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HETEROBIFUNCTIONAL COMPOSITIONS FOR TARGETED PROTEIN DEGRADATION AND METHODS FOR THEIR USE

Abstract

Compositions and methods for control and/or modification of endogenous protein degradation are described. The compositions are directed to heterobifunctional molecules having protein and enzyme system binding moieties linked together by an organic linker group. The compositions are selective for binding to certain endogenous proteins and function to recruit endogenous decomposition systems such as the polyubiquitin system for peptide cleavage and reassimilation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The subject patent application claims the benefit of priority to U.S. Provisional Patent Application Nos. 63/156,593 (filed Mar. 4, 2021; now pending). The full disclosure of the priority application is incorporated herein by reference in its entirety and for all purposes.

BACKGROUND
[0002] A powerful approach was recently developed that integrates fragment-based ligand discovery with chemical proteomics, called fragment-based ligand mapping in cells, to globally survey ligandable proteins and their ligandable sites (FbLMiC, FIG. **1**A) (11). In FbLMiC, small-molecule probes, called fully functionalized fragment (FFF) probes are designed to contain: 1) a structurally minimized “constant” region bearing a photoactivatable diazirine group and alkyne handle, which together enable UV light-induced covalent modification and detection, enrichment, and identification of compound-bound protein targets; and 2) a “variable” recognition region consisting of structurally diverse small-molecule fragments (MW<300 Da) to promote interactions with a subset of the proteome. See FIGS. **1**B and **1**C.
[0003] Notable strengths of FbLMiC are: 1) fragments can be optimized into higher affinity ligands through FbLMiC-guided medicinal chemistry, resulting in lead chemical probes that can selectively modulate protein function (as has been shown with first-in-class chemical probes for PTGR2, SLC25A20 and PGRMC2, (11, 12)); 2) FFFs can map ligand binding site on endogenous protein targets, revealing fragments that interact at a variety of protein sites (e.g. active sites, cofactor binding sites, allosteric sites); 3) FFF probes engage proteins reversibly, overcoming the limitations of other chemical proteomic profiling techniques, such as activity-based protein profiling (ABPP), which require covalent reactions with amino acid side chains. Recently this platform has been expanded and has demonstrated that enantiomerically matched FFF's can be used to expedite the discovery of selective fragment-protein interactions (13). See enantioprobes; FIG. **1**C.
[0004] To streamline the identification of high affinity and selectivity chemical probes against proteins with established FFF leads, a strategy has recently been developed and is termed competitive FbLMiC (FIGS. **2**A and **2**B). Here, the potency and selectivity of candidate small-molecule ligands is assessed across 100s-1000s of proteins. This method integrates targeting and tandem mass tagging (TMT) methods (14) to enable the optimization of chemical probes for protein targets directly within their endogenous biological environments. In competitive FbLMiC, up to nine different competitor analogs (and a DMSO control) are evaluated per experiment for blockade of FFF interactions with endogenous targets. Competitor libraries consist of members that share parent fragment core structures, either purchased or synthesized in-house. By performing targeted MS experiments, MS runtimes can be dramatically shortened, enabling the screening of ˜150 compounds/day. Analogs that show the strongest blockade are optimized for potency and proteomic selectivity by an iterative cycle of MS-based FbLMiC-guided medicinal chemistry to furnish lead ‘binders.’ The original pilot FFF library consisted of only 13 members for proof of principle studies (11). For this work, a specialized library of ˜150 “scout” FFF probes was synthesized and was based upon fragment (<250 Da) cores commonly found in bioactive compounds (e.g. drugs, natural products, human metabolites) (15) and possessed structures that are

synthetically accessible for derivatization. Initial profiling of this library has demonstrated outstanding coverage with an unprecedented ligandability map of 5000+ proteins, including those that fall out of traditional “druggable” classes (e.g. adaptor proteins, transcription factors). However, the protein binding activity accomplished with members of the FFF library does not enable refined selection of the proteins of interest (POIs), modification of the POIs and/or their selective, programmed degradation.

[0005] A rudimentary solution involves reliance upon rationally designed heterobifunctional small molecules. These heterobifunctional molecules facilitate the study of an increasingly wide range of biological phenomena and have proven enabling in drugging challenging therapeutic targets and processes. Such molecules typically consist of two ligands or binders that are connected via a covalent linker, yielding a chimeric compound that can mediate the formation of ternary complexes between two unique proteins (6, 7). The ability to “recruit” enzymes to proteins of interest (POIs) endows these molecules with great potential as a generalizable strategy to influence PTMs.

[0006] Recently, it has been shown that one class of PTM enzymes, E3 ubiquitin ligases, can be recruited to target POIs via heterobifunctional molecules (8-10). These molecules, often referred to as proteolysis targeting chimeras (PROTACs), induce molecular proximity between an E3 ligase and a POI, leading to ubiquitination and targeted protein degradation (TPD). Despite the tremendous therapeutic potential of TPD, two major challenges overshadow the generalization of this approach, i) the small number of E3 ligase recruiters that have been identified, despite the excess of 600 predicted E3 ligases; and ii) the undetermined fraction of the proteome accessible to TPD, and correspondingly, the availability of small molecule ligands capable of binding to target POIs that can be coopted into such modalities.

[0007] Nearly all PROTACs utilize established small molecule ligands (e.g. inhibitors), limiting the scope of proteins that can be targeted and requiring synthetic ‘retrofitting’ that may be disruptive to binding to the POI. Further, due to this dearth of available ligands, the fraction of the proteome that can be targeted with PROTAC-type strategies is unknown. Therefore, a goal of the present invention is the development of PROTAC's that are based upon surveys of the entire proteome for proteins that may be tractable to targeted protein degradation in parallel to the identification of bifunctional degrader leads instead of choosing singular targets with a priori established ligands. Therefore, a goal of the present invention is the development of heterobifunctional compositions that functions to bind endogenous proteins and to modify them and/or to enable programmed selective degradation.

SUMMARY

[0008] An aspect of the present invention is directed to heterobifunctional fragment based degrader molecules, FragTACS, having protein binding target moieties that are selected from unbiased whole proteome affinity interactions. Another aspect of the present invention is directed to methods for in vitro and/or in vivo endogenous protein degradation through the agency of heterobifunctional FragTACs.

[0009] Embodiments of the heterobifunctional FragTACS incorporate small molecule fragments that are preferably endogenous protein binding target moieties, small molecule recruiter moieties for preferably endogenous degradation enzymes and an organic group linking these two moieties together. The FragTACS may be depicted by a generic Formula I:

PBF-L-RBF   Formula I

[0010] The acronym PBF is a protein binding fragment of a small organic molecule moiety that enables selection of the protein of interest from a milieu of proteins, preferably endogenous proteins. The acronym RBF is a recruiter binding fragment of moiety of a small organic molecule moiety that is capable of recruiting in a cytoplasm context an enzyme that degrades, fragments and/or divided endogenous proteins into fragments for re-assimilation. The acronym L is an organic linker group having combinable, reactive functions at its termini that enable covalent attachment to

the PBF and RBF. Exemplary PBF fragments include but are not limited to:
##STR00001##
Exemplary RBF fragments include but are not limited to thalidomide derivatives for the CRBN ligase and the N-(4-thiazol-1-yl phenethyl) 4-hydroxyprolinamide for the VHL ligase
##STR00002##
These examples of RBF fragments enable recruitment of E3 ubiquitin ligase activity. Exemplary linker groups include short and oligomeric polyol (PEG) and alkylenyl, moieties having amine and/or carboxyl termini for binding to the PBF and RBF fragments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIGS. **1**A, **1**B and **1**C depict Fragment-based Ligand-ability Mapping in Cells. FIG. **1**A depicts Fully functionalized fragment (FFF) probes are composed of a drug-like fragment as well as a retrieval tag, enabling the covalent capture of fragment-bound protein targets directly in cells upon UV irradiation. Fragment targets, as well as the site of fragment interaction, can be identified and quantified by mass spectrometry- and gel-based methods. FIG. **1**B depicts the general structure of FFF showing the constant affinity tag region (red), which consists of a photoreactive group (diazirine) and a latent affinity (alkyne) group, as well as the variable region (blue), which contains fragment recognition elements for binding to proteins in cells. FIG. **1**C depicts example structures of fragment scout- and enantioprobes.

[0012] FIGS. **2**A and **2**B depict targeted competitive FbLMiC workflow for chemical probe development for prioritized proteins. FIG. **2**A depicts optimization of scout probe to lead binder through iterative competitive FbLMiC. FIG. **2**B depicts targeted MS-FbLMiC accelerates chemical probe development by enabling rapid, multiplexed analyses of a defined set of prioritized targets in native biological systems. Shown is a targeted MS-FbLMiC workflow that we have developed for assessing small molecule interactions in prioritized targets, where a cell preparation is treated with up to nine compounds (and a DMSO control) and a corresponding FFF scout probe, click conjugation to a desthiobiotin-azide tag, streptavidin enrichment, elution, and labeling of enriched peptides with tandem-mass tags (TMT). Samples are then combined and analyzed in a single LC-MS/MS/MS experiment, where peptides of interest in prioritized proteins are selected for sequential MS1, MS2, and MS3 analysis. MS3 signals are used to quantify peptides, where reductions in signal (e.g., >80%) mark compound-sensitive sites.

[0013] FIGS. **3**A, **3**B and **3**C depict fragment-based discovery of degradable proteins. FIG. **3**A depicts chemical structures of a preliminary set of fragment-based PROTACs (FragTACs), which consist of a small molecule fragment chemically linked to established E3 recruiting ligands. FIG. **3**B shows representative data for a FragTAC (1, 100 μM, 6 hrs) incubated with HCC1806 cells. A relative abundance of ˜7000 proteins was determined using quantitative multiplexed proteomics. Vertical dashed lines mark 3-fold changes relative to DMSO and horizontal dashed lines mark p value=0.05. Red box (inset) shows targets with >3-fold decreased abundance. FIG. **3**C depicts western blots confirming dose-dependent downregulation of 3 example targets (TDP-43, FAM136A, and CCAR2).

[0014] FIGS. **4**A, **4**B, **4**C and **4**D depict expanding the druggable landscape of for targeted protein degradation. FIG. **4**A provides the strategy to expand the number of targets and E3 proteins available for TPD approaches. FIG. **4**B shows a synthesizable library of ˜300 FragTACs and examine their ability to modulate protein levels via unbiased proteome-wide proteomic analyses in primary human immune cells. FIG. **4**C shows a representative subset of E3 ligases for which FbLDisC has identified hit ligands. Note that hit ligands have been discovered for members of most E3 subfamilies. FIG. **4**D shows a model system for testing whether druggable sites on E3 ligases support TPD. Ligands for a given E3 are coupled to a compound AP11867 that binds to the FKBP12-F36V protein and assayed for inducing degradation of recombinant FKBP12-F36V in cells that endogenously express the E3 ligase.

[0015] FIG. **5** depicts in schematic style how a FragTAC binds to a target protein and recruits a fragmentation ligase, which in this example is the ubiquitin ligase complex. The ubiquitin ligase ubiquitinizes the protein to add ubiquitin peptide chains to the protein. This ubiquitination marks the protein for proteolysis through proteasome enzymatic degradation to yield amino acid fragments for re-assimilation.

[0016] FIG. **6** shows results of Western blot studies of proteasome/neddylation inhibition of several target proteins in HCC1806 cells.

[0017] FIG. **7** shows results of Western blot studies of dose-dependent response of several target proteins.

[0018] FIG. **8** shows results of Western blot studies of time course of the responses of several target proteins.

[0019] FIG. **9** shows results of cell viability assays of representative examples.

DETAILED DESCRIPTION

[0020] In general, the invention provides heterobifunctional FragTACs that contain a protein binding fragment (PBF) and a recruiter binding fragment (RBF) that are connected via a linker moiety (L). As exemplified herein, the PBF moiety can be any small molecule that targets one or more endogenous proteins, the RBF moiety can be any compound that recruits one or more endogenous degradation enzymes, and the L moiety can be any organic group that optimally links the two moieties with no or minimum impact on their biological functions.

[0021] In various embodiments, the PBF moiety can be selected from any suitable commercially available fragments or synthetically accessible fragments, as described herein. Exemplary commercially available and synthetically derivatizable PBF fragments include but are not limited to:

##STR00003## ##STR00004## ##STR00005## ##STR00006## ##STR00007## ##STR00008## ##STR00009## ##STR00010## ##STR00011## ##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020##

[0022] In these compound structures, X is F, Cl, Br or I; Y is COOH or NH.sub.2; Z is O, NH or S.

[0023] In some embodiments, the PBF moiety is a synthetically accessible fragment. Exemplary synthetically accessible PBF fragments include but are not limited to:

TABLE-US-00001 [00021]embedded image Suzuki-Miyaura Coupling Organoboranes General Examples [00022]embedded image X = F, Cl, Br, I [00023]embedded image Y = NH, S, O [00024]embedded image R = F, Cl, Br, I, alkyl, acyl [00025]embedded image Y = NH, S, O Specific Examples [00026]embedded image [00027]embedded image [00028]embedded image [00029]embedded image [00030]embedded image [00031]embedded image [00032]embedded image Organohalides Aliphatic [00033]embedded image [00034]embedded image [00035]embedded image [00036]embedded image [00037]embedded image [00038]embedded image [00039]embedded image [00040]embedded image Aromatic [00041]embedded image X = Cl, Br, I [00042]embedded image [00043]embedded image [00044]embedded image [00045]embedded image [00046]embedded image [00047]embedded image [00048]embedded image [00049]embedded image Buchwald-Hartwig Amination Aryl Halides Bromides [00050]embedded image [00051]embedded image [00052]embedded image [00053]embedded image [00054]embedded image [00055]embedded image [00056]embedded image [00057]embedded image Iodides [00058]embedded image [00059]embedded image [00060]embedded image [00061]embedded image [00062]embedded image [00063]embedded image [00064]embedded image [00065]embedded image Secondary Amines [00066]embedded image [00067]embedded image

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embedded image [00098]embedded image

[0024] In some embodiments, the PBF moiety is a synthetically accessible benzhydrylpiperazine derived fragment with a structure shown in Formula XL VII.

##STR00099##

In the structure, R is alkyl group, aryl group, COOEt or H; U is CH or N; V is CH or N; W is CH or N; X is H, F, Cl, Br, I, alkyl group or aryl group; Y is CH or N; Z is CH or N. Exemplary synthetically accessible benzhydrylpiperazine derivative fragments include but are not limited to:

##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105##

[0025] In some other embodiments, the employed PBF moiety is a synthetically accessible natural product-derived fragment. Exemplary synthetically accessible natural product-derived PBF fragments include but are not limited to:

##STR00106## ##STR00107##

[0026] Like the PBF moiety, the RBF moiety in the FragTACs of the invention can also employ a number of suitable compounds. These include, e.g., CRBN ligands, VHL ligands, IAP ligands, MDM2 ligands, RNF ligands, DCAF ligands, KEAP1 ligands and FEMIB ligands. In some embodiments, the RBF moiety can be a CRBN ligand with a structure shown in any one of Formulae II-V below:

CRBN Ligands (IMiDs):

##STR00108##

[0027] In these structures, X is CH.sub.2 or CO; Y is NH, O, alkyne or CH.sub.2. Exemplary CRBN-derived RBFs include but are not limited to:

##STR00109## ##STR00110##

[0028] In some embodiments, the RBF moiety can be a VHL ligand with a structure shown in any one of Formulae VI-IX.

VHL Ligands (Von-Hippel Lindau):

##STR00111##

In these structures, stereochemistry of C1 is I or(S); R is H, CH.sub.2 or CH.sub.3; X is F or CN; Y is S or H; Z is O or H. Specific examples of VHL-derived RBFs include but are not limited to:

##STR00112## ##STR00113##

[0029] In some embodiments, the RBF moiety can be an IAP ligand with a structure shown in any one of Formulae X-XIII.

IAP Ligands (Inhibitor of Apoptosis Proteins):

##STR00114##

[0030] In some other embodiments, the RBF moiety can be a MDM2 ligand with a structure shown in any one of Formulae XIV-XVI.

MDM2 Ligands:

##STR00115##

[0031] In still some other embodiments, the RBF moiety can be a RNF ligand with a structure shown in any one of Formulae XVII-XIX.

RNF Ligands (Ring Finger Proteins):

##STR00116##
[0032] In some embodiments, the RBF moiety can be a DCAF ligand with a structure shown in any one of Formulae XX-XXI.
DCAF Ligands:
##STR00117##
[0033] In some other embodiments, the RBF moiety can be a KEAP1 ligand or a FEM1B ligand with a structure shown in Formula XXII or XXIII, respectively.
KEAP1 Ligand:
##STR00118##
[0034] The linker (L) moiety in the FragTACs of the invention can employ any suitable compound or moiety that is capable of conjugating the PBF and RBF moieties without significant impact on their interactions with their cognate protein partners. In some embodiments, the L moiety can be a polyethylene glycol (PEG) linker with a structure shown in any one of Formulae XXIV-XXVI. In some other embodiments, the L moiety can be an aliphatic linker with a structure shown in any one of Formulae XXVII-XXX. In still some other embodiments, the L moiety can be a hybrid linker with a structure shown in any one of Formulae XXXI-XXXIX. In still some other embodiments, the L moiety can be an aryl-based linker with a structure shown in any one of Formulae XL and XLI. In some other embodiments, the L moiety can be a heterocycle-based linker with a structure shown in any one of Formulae XLII and XLIII. In some other embodiments, the L moiety can be a click chemistry-generated linker with a structure shown in any one of Formulae XLIV-XL VI.
Polyethylene glycol (PEG) linkers:
##STR00119##
In these structures, n is an integer between 0 and 10.
Aliphatic Linkers:
##STR00120##
In these structures, n is an integer between 0 and 10.
Hybrid Linkers:
##STR00121##
In these structures, n and m are each independently integers between 0 and 10.
Aryl-Based Linkers:
##STR00122##
In these structures, n and m are each independently integers between 0 and 10.
Heterocycle-Based Linkers:
##STR00123##
In these structures, n and m are each independently integers between 0 and 10; X is CH or N.
Click Chemistry-Generated Linkers:
##STR00124##
In these structures, n and m are each independently integers between 0 and 10. Formulae XLIV and XLV are generated through the cycloaddition of an azide with an alkyne. Formula XLVI is generated through the cycloaddition of tetrazine and trans-cyclooctene.
[0035] One aspect of the present invention is directed to heterobifunctional compositions that use small molecule fragments appended to established E3 ligands. As illustrated in FIG. **5**, these heterobifunctional compositions identify degradable targets and synthetically progress-able ligands via unbiased whole proteome MS-based proteomics. This aspect of the invention has established that low target affinity interactions can lead to potent protein degradation.
[0036] Embodiments of this aspect of the heterobifunctional compositions are directed to a small library which was designed and synthesized to incorporate of bifunctional, fragment-based degrader molecules (‘FragTACs’). Generic Formula I, depicted above in the Summary, incorporates the protein binding terminus (PBT), the linker (L) and the recruiter binding terminus (RBT) as an embodiments of FragTACs. Exemplary embodiments of the PBT moiety include

heterocyclic multicyclic organic molecular fragments listed below.

[0037] Exemplary embodiments of the L moiety include the linear organic chain fragments listed below. Exemplary embodiments of the RBT moieties include the organic molecular fragments listed below. These RBT moieties bind with the ligases as shown in the list including VHL, Cereblon, RNF-4, MDMR, RNF114, SNIPER and KEAP. Exemplary Cereblon and VHL ligand based FragTACs are listed below. The doses and percentages given in the list indicate the percent fragmentation at the given doses when the example is administered to a corresponding cell culture. See also FIG. **3**A, eight members.

[0038] Focused embodiments of the heterobifunctional compositional FragTACs incorporate established von-Hippel Lindau (VHL) or Cereblon (CRBN) E3 ligands (18-20) which are chemically linked to any one of four fragment scaffolds. The fragment scaffolds were previously demonstrated to exhibit broad proteomic interactions (11). The ability of the FragTAC library to induce proteomic changes in breast cancer cells was profiled and identified a combined ˜120 downregulated proteins that span a broad range of classes (e.g. enzymes, transcription factors, chaperone proteins) as well as several members of protein complexes. FragTAC-1, in particular, substantially downregulated 43 proteins, including transcriptional repressor TDP-43, regulator protein CCAR2 and functionally uncharacterized protein FAM136A, which was confirmed to occur in a dose-dependent fashion (FIGS. **3**B and **3**C). Protein loss was blocked with the proteosome inhibitor MG132, suggesting that FragTAC-1 promotes proteasomal degradation (not shown). Notably, there are no known small molecule ligand for these proteins.

Examples of heterocyclic multicyclic organic molecular PBT fragments:

##STR00125## ##STR00126## ##STR00127## ##STR00128##

Examples of linear organic chain L moieties:

Peg Linkers

##STR00129##

Aliphatic Linkers

##STR00130##

Examples of organic molecular RBT moieties:

##STR00131## ##STR00132## ##STR00133##

Cereblon Ligand-Based FragTAC Probes:

##STR00134##

VHL Ligand-Based FragTAC Probes:

##STR00135## ##STR00136##

De Novo Discovery of Degradable Protein Targets Using Fragment-Based PROTACs (FragTACs).

[0039] With the success in identifying degradable proteins using a small library of FragTACs, the strategy enveloping embodiments of the invention is expanded by synthesizing a larger FragTAC library (˜300) and applying them to identify degradable targets in therapeutically relevant human immune model systems using multiplex proteomic workflows (FIGS. **4**A-**4**D). The goal of this strategy enables 1) identification of characteristic chemical features of PROTACs that lead to successful TPD; 2) a gain of an understanding of the impact of ligand affinity and promiscuity in selectivity and efficiency; 3) comparatively assessment of routinely employed E3 ligands for their ability to induce TPD; 4) broad annotation of human immune targets that may be tractable to TPD; 5) establishment of a template to transition and optimize promiscuous fragment-based degraders to selective PROTACs for targets with compelling therapeutic or biological value.

[0040] Towards this end, a library may be prepared using ˜30 small molecule fragments chemically linked to either CRBN or VHL E3 ligands (FIG. **4**B). Fragments may be chosen based on their chemical diversity and non-overlapping proteomic interactions determined via the FbLDiSC workflows (FIGS. **1**A-**1**C) to maximize proteomic coverage. In initial studies, 4-5 unique linkers (e.g. PEG, aliphatic) may be utilized, enabling the generation of mini-libraries around each fragment that explore variables such as E3s and linker composition. This library may be screened in pooled primary peripheral blood mononuclear cells (PBMCs) from de-identified donors for their ability to induce TPD via unbiased quantitative TMT proteomics as established in our preliminary studies (FIG. **3**B). Human PBMCs may be chosen as an initial model system, as they are composed of a diverse cell population (T cells, B cells, dendritic cells, etc.) that contain uniquely expressed, immune-relevant targets, thus increasing the probability to uncover degradable, therapeutically translatable targets that serve critical roles in inflammation, infection, and cancer progression, for example. The variables discussed above may be encoded in the library to assess their independent contributions as well as establish a baseline of proteins susceptible to this TPD system. The FbLDiSC workflows may be used to assess and optimize potency and selectivity of lead FragTACs, as needed, for prioritized targets. See FIGS. **2**A-**2**B and FIGS. **3**A-**3**C.

Chemoproteomic-Enabled Discovery of E3 Ligase-Binding Compounds that Support TPD in Human Cells.

[0041] Multiple E3 ligase components, including CRBN (31), VHL (32), RNF114 (33), and DCAF16 (34) have been shown to engage in tripartite complexes where bridging small molecules can direct specific protein substrates to ubiquitination and degradation. These findings, combined with the large number of E3 ligases (>600 (35)) in humans, have stimulated interest in the broader draggability potential of TPD (36). The recent proteome-wide ligandability maps using FbLDisC have identified hit ligands (or 'binders') for >40 E3 ligases (11, 13) (FIG. **4**C). The capacity of these E3-binders to support TPD may be tested using a recently described dTAG model system (37) (FIG. **4**D). E3 binders that induce degradation may be optimized into lead chemical probes using FbLDisC-guided medicinal chemistry as described in FIGS. **2**A-**2**B. The chemical probes may be investigated for their ability to degrade targets with established ligands. Enzyme ligase E3 systems may be prioritized with restricted tissue expression, (e.g. in cancer or immune cells), as they may offer safer paths for drugs compared to broadly expressed E3 systems. To this end several fragment-based ligands for several E3 ligases have been validated. See Table 1.

[0042] Additional studies were performed to validate the activities of the FragTAC probes in HCC1806 cells via Western blot analyses. Results from these studies are shown in FIGS. **6**-**9**.

TABLE-US-00002 TABLE 1 Examples E3s with newly validated ligands Gene Type Localization Tissue CUL1 Cullin Lung fibroblasts CUL4A Cullin CUL4B Cullin Nucleus DDB2 DDB1-Nucleus; Accumulates at sites of DNAUbiquitously expressed; corneal Adaptor damage following UV irradiation. endothelium EDD1 HECT Nucleus Testis, brain, pituitary and kidney. FBXO3 FBOX Nucleus; Colocalizes with PML FBXO7 FBOX Cytoplasm, Nucleus, Mitochondrion, Cytoplasm, Cytosol FBXO9 FBOX Cytoplasm HUWE1 HECT Cytoplasm, Nucleus Heart, brain and placenta PWP1 DDB1- Nucleus; Nucleolus; Chromosome Placenta, skeletal muscle, kidney Adaptor and pancreas SYVN1 VBM ER membrane Ubiquitously expressed TRIM21 RING Cytoplasm; Cytoplasmic vesicle; Heart and fetal lung. Autophagosome; Nucleus; P-body UBR1 RING Cytoplasm, Cytosol Broadly, skeletal muscle, kidney and pancreas UBR4 UIM Membrane, Cytoplasm, Cytoplasm, Cytoskeleton, Nucleus UHRF1 RING Nucleus; Localizes to replication foci Thymus, bone marrow, testis, lung and heart. Overexpressed in breast cancer.

Results of Proteomics Studies

[0043] Examples of downregulated proteins targeted by FragTACs are listed in Table 2. The results show targets with 2 unique peptides or more and depleted 2-fold or more relative to DMSO control (P value <0.05). Compound structures can be found in the examples of FragTAC Probes described herein (Table 3).

TABLE-US-00003 TABLE 2 Types of proteins targeted by FragTAC probes Accession Proteins Number Probe Code A2M P01023 IF-051 AATF Q9NY61 LPC020 ABCC5 O15440 LPC020 ABHD10 Q9NUJ1 LPC020 ABLIM1 O14639 IF-051, CMC-6-123 ACAA2 P42765 LPC020 ACADM P11310 LPC020 ACADSB P45954 LPC020 ACAT1 P24752 LPC020 ACO2 Q99798 LPC020 ACOT1 Q86TX2 LPC020 ADAM15 Q13444 CMC-6-109, CMC-6-111, IF-072 ADIRF Q15847 IF-051, IF-085 AGR2 O95994 LPC020 AHCTF1 Q8WYP5 LPC020 AHNAK Q09666 IF-

051 AHNAK2 Q8IVF2 IF-051 AHSG P02765 LPC020, IF-085 AIDA Q96BJ3 IF-051 AK2 P54819 LPC022, LPC020, IF-051, CMC-6-123, IF-049 AK3 Q9UIJ7 LPC020 AK4 P27144 LPC020 AKAP8 O43823 LPC020 ALAS1 P13196 CMC-6-123 ALCAM Q13740 IF-051, LPC020, IF-049 ALDH1B1 P30837 LPC020 ALYREF Q86V81 IF-051, LPC022, LPC020 ANAPC5 Q9UJX4 LPC020, LPC011 ANLN Q9NQW6 CMC-6-109, CMC-6-111, IF-072, IF-080 ANPEP P15144 IF-072 API5 Q9BZZ5 IF-051 APOB P04114 LPC020 ARID1B Q8NFD5 IF-051 ARMC10 Q8N2F6-2 IF-051, IF-049 ARPC5L Q9BPX5 IF-051 ATAD2 Q6PL18 LPC011 ATP1A2 P50993 LPC020 ATP2B1 P20020 IF-051 ATP2B4 P23634 IF-051 ATP5F1A P25705 LPC020 ATP5F1B P06576 LPC020 ATP5F1C P36542 LPC022, LPC020 ATP5F1D P30049 LPC022, LPC020 ATP5PD O75947 LPC020 ATP5PO P48047 LPC020 ATPAF1 Q5TC12 LPC020 B4GALT1 P15291 LPC020 BACH1 O14867 IF-051, IF-049 BAZ1B Q9UIG0 CMC-6-109, CMC-6-111, IF-072 BBX Q8WY36 IF-051 BCAM P50895 IF-051, CMC-6-109 BCKDK O14874 IF-051 BCL2L12 Q9HB09 LPC020 BCLAF1 Q9NYF8 IF-051, LPC020 BICRA Q9NZM4 IF-051 BOD1L1 Q8NFC6 IF-051, IF-049 BOLA3 Q53S33 IF-051 BPTF Q12830 LPC020 BRD2 P25440 IF-051, IF-065 BRD8 Q9H0E9 IF-051 BRD9 Q9H8M2 IF-051 BTF3L4 Q96K17 IF-051 BUD13 Q9BRD0 IF-051, LPC020, IF-049 BUD31 P41223 IF-051 C2orf49 Q9BVC5 IF-051 C9orf78 Q9NZ63 IF-051 CACTIN Q8WUQ7 IF-051 CAMLG P49069 IF-051 CARD19 Q96LW7-2 IF-051, CMC-6-123 CASK O14936 IF-051 CBLL1 Q75N03 IF-051 CBR4 Q8N4T8 LPC020 CBX1 P83916 IF-051, CMC-6-109, IF-072 CBX3 Q13185 IF-051, CMC-6-109, IF-072 CBX5 P45973 IF-051 CCAR2 Q8N163 LPC020 CCDC12 Q8WUD4 IF-051, LPC022, LPC020, CMC-6-123 CCDC124 Q96CT7 IF-051 CCDC137 Q6PK04 IF-051 CCDC58 Q4VC31 IF-051, LPC022, LPC020, CMC-6-123 CCDC90B Q9GZT6 LPC011 CCNE2 O96020 LPC011 CCNL1 Q9UK58 LPC011 CCNT1 O60563 IF-051 CD109 Q6YHK3 IF-051, LPC020 CD151 P48509 IF-072 CD3EAP O15446 IF-051 CD46 P15529 IF-051 CD58 P19256 IF-051, CMC-6-123 CD59 P13987 IF-051 CD74 P04233-2 CMC-6-109, CMC-6-111, IF-072 CD9 P21926 IF-049 CDC40 O60508 IF-051, LPC022, LPC020, IF-049 CDC5L Q99459 IF-051, LPC022, LPC020 CDH1 P12830 IF-051, LPC022, LPC020, IF-049 CDH13 P55290 CMC-6-109, CMC-6-111, IF-072 CDH2 P19022 IF-051 CDH3 P22223 CMC-6-109, CMC-6-111, IF-072 CDK12 Q9NYV4 IF-051, IF-049 CDK2AP1 O14519 IF-051 CDK2AP2 O75956 IF-051, IF-049 CDK5RAP1 Q96SZ6 CMC-6-109, CMC-6-111, IF-072, LPC020 CDK5RAP1 Q96SZ6-3 LPC022 CENPC Q03188 IF-051 CFL2 Q9Y281 IF-051 CHAF1B Q13112 IF-051 CHAMP1 Q96JM3 CMC-6-109, CMC-6-111, IF-072 CHCHD1 Q96BP2 IF-051 CHCHD2 Q9Y6H1 IF-051, LPC020, IF-072, CMC-6-123 CHCHD2P9 Q5T1J5 IF-051 CHCHD4 Q8N4Q1 IF-051 CHD6 Q8TD26 IF-051 CHD7 Q9P2D1 LPC020 CHTOP Q9Y3Y2 IF-051 CIRBP Q14011 IF-051 CKAP2 Q8WWK9 LPC020 CKMT1A P12532 IF-051, LPC022, LPC020 CLCC1 Q96S66 LPC022, LPC020 CLIC4 Q9Y696 IF-051 CLPB Q9H078 IF-051, LPC020, CMC-6-123 CLPB Q9H078-2 IF-051 CNPY2 Q9Y2B0 LPC020 CNST Q6PJW8 IF-051, IF-049 COA7 Q96BR5 LPC022, LPC020, IF-072, CMC-6-123, IF-085 COL17A1 Q9UMD9 IF-051 COQ9 O75208 LPC020 COX20 Q5RI15 LPC020 COX6B1 P14854 LPC020 COX7A2 P14406 LPC020 COX7A2L O14548 LPC022, LPC020 COX7C P15954 CMC-6-123, LPC020 CPOX P36551 LPC022, LPC020, IF-072, IF-051, IF-049 CPSF7 Q8N684 IF-051 CRYZ Q08257 LPC020 CSRP1 P21291 IF-051 CSTF2 P33240 IF-051, LPC022, LPC020 CSTF2T Q9H0L4 LPC022, LPC020 CSTF3 Q12996 IF-051, LPC022, LPC020, IF-049 CT45A10 P0DMU9 IF-051, IF-049 CTNNA1 P35221 IF-051 CTNNB1 P35222 IF-051 CTNND1 O60716-3 LPC020 CTSB P07858 IF-051, LPC020, IF-072 CTSD P07339 CMC-6-123, LPC020, IF-072 CTSH P09668 IF-072, IF-051, CMC-6-123, CMC-6-111, CMC-6-109, IF-085 CTSS P25774 IF-072 CTSV O60911 IF-051 CTSZ Q9UBR2 IF-051, IF-080, IF-085 CTTN Q14247 IF-051 CUSTOS Q96C57 IF-051 CUX1 P39880 IF-051, IF-049 CWC15 Q9P013 IF-051, LPC020 CYCS P99999 IF-051, LPC022, LPC020, CMC-6-123 CYP1B1 Q16678 IF-073 DAP3 P51398 LPC020 DAZAP1 Q96EP5 IF-051, LPC020 DBN1 Q16643 IF-051 DDX21 Q9NR30 IF-051 DDX52 Q9Y2R4 LPC011 DFFA O00273 IF-051 DHTKD1 Q96HY7 LPC020 DHX9 Q08211 IF-051 DIABLO

Q9NR28 LPC022, LPC020, IF-051, CMC-6-123, IF-049 DIDO1 Q9BTC0 IF-051, LPC022, LPC020 DNAJA3 Q96EY1 LPC020 DNAJB14 Q8TBM8 LPC020 DNAJC8 O75937 IF-051 DNASE2 O00115 LPC020, IF-072 DOLPP1 Q86YN1 LPC011 DPP7 Q9UHL4 CMC-6-123, IF-072 DSG2 Q14126 IF-051, IF-049 ECHS1 P30084 LPC020 ECI2 O75521 LPC020 ECSIT Q9BQ95 LPC020 EDF1 O60869 IF-051 EFNB1 P98172 IF-051, IF-049 EFNB2 P52799 IF-051, CMC-6-123 EFR3A Q14156 LPC020 EIF1 P41567 IF-051 ELAVL1 Q15717 LPC022, LPC020 ELOA Q14241 IF-051 EMG1 Q92979 IF-051 EPN1 Q9Y613 IF-051, LPC022, LPC020 EPN2 O95208 IF-051 EPPK1 P58107 LPC020 ERAL1 O75616 LPC020 ERBIN Q96RT1 LPC020 ERH P84090 IF-051, LPC020 ERP29 P30040 IF-051, LPC020 ETFA P13804 LPC020 ETFDH Q16134 LPC020 EZR P15311 IF-051 FAHD1 Q6P587 IF-051, LPC020, IF-049 FAM136A Q96C01 IF-051, LPC022, LPC020, IF-049 FAM169A Q9Y6X4 IF-051, IF-049 FAM177A1 Q8N128 IF-073 FAM49B Q9NUQ9 IF-051 FAM83B Q5T0W9 LPC020 FAM83H Q6ZRV2 LPC020 FASTKD3 Q14CZ7 LPC020 FCF1 Q9Y324 IF-049 FDX1 P10109 LPC020 FDXR P22570 LPC020 FECH P22830 LPC020 FIP1L1 Q6UN15 IF-051, IF-049 FKBP1A P62942 IF-051 FNBP4 Q8N3X1 IF-051 FSTL1 Q12841 IF-051 FTL P02792 CMC-6-123, IF-085 FUBP3 Q96I24 LPC020 FUCA2 Q9BTY2 LPC020 FUS P35637 IF-051, LPC022, LPC020 FYTTD1 Q96QD9 IF-051, LPC020, IF-049, CMC-6-123 FYTTD1 Q96QD9-2 LPC022, LPC020 GADD45GIP1 Q8TAE8 LPC020 GATAD2B Q8WXI9 IF-051 GATB O75879 LPC022, LPC020 GCSH P23434 CMC-6-123, LPC020 GFER P55789 IF-051, LPC020, IF-072, CMC-6-123 GFM2 Q969S9 LPC020 GGH Q92820 IF-072 GHITM Q9H3K2 LPC022, LPC020, IF-072, IF-051, CMC-6-123, IF-060, IF-073, IF-062, IF-064, IF-085 GIPC2 Q8TF65 LPC011 GLA P06280 IF-051 GLB1L2 Q8IW92 LPC020 GLO1 Q04760 IF-051 GLRX5 Q86SX6 LPC020 GLS O94925-3 LPC022, LPC020 GLUD1 P00367 LPC020 GM2A P17900 IF-051, CMC-6-123, IF-085 GMFB P60983 IF-051 GNG12 Q9UBI6 LPC020 GPATCH4 Q5T3I0 IF-051 GPC1 P35052 IF-051, CMC-6-111, IF-072, LPC020 GPKOW Q92917 CMC-6-109, IF-051, IF-072 GPT2 Q8TD30 LPC022, LPC020 GRPEL2 Q8TAA5 LPC022, LPC020 GRSF1 Q12849 LPC020 GSTK1 Q9Y2Q3 LPC020 GTF2B Q00403 IF-051, IF-049 GTF2I P78347 CMC-6-109, IF-072 H1FX Q92522 CMC-6-109, CMC-6-111, IF-072 H3F3A P84243 LPC011 HADHB P55084 LPC020 HARS2 P49590 LPC020 HAX1 O00165 CMC-6-109, IF-072 HCCS P53701 LPC020 HCLS1 P14317 IF-051 HEBP1 Q9NRV9 IF-051, LPC020 HEXB P07686 CMC-6-123 HIBCH Q6NVY1 LPC020 HINT1 P49773 IF-051 HINT2 Q9BX68 IF-051, LPC020 HIST1H1C P16403 CMC-6-123 HIST1H1T P22492 IF-049 HIST2H2AB Q8IUE6 IF-051 HIST2H2BF Q5QNW6 IF-051 HIST2H3A Q71DI3 LPC011 HMGA1 P17096-2 IF-051 HMGB2 P26583 IF-051 HMGCL P35914 LPC020 HMGN1 P05114 IF-051 HMGN2 P05204 IF-051 HMGN4 O00479 IF-051 HMGN5 P82970 IF-051 HNRNPA0 Q13151 IF-051, LPC022, LPC020 HNRNPA1 P09651 IF-051, LPC020 HNRNPA1 P09651-2 LPC020 HNRNPA2B1 P22626 LPC022, LPC020 HNRNPA3 P51991 LPC022, LPC020 HNRNPAB Q99729-2 IF-051 HNRNPAB Q99729-3 IF-051, LPC020 HNRNPCL1 O60812 LPC020 HNRNPD Q14103 IF-051 HNRNPDL O14979 IF-051, LPC022, LPC020 HNRNPH1 P31943 LPC022, LPC020 HNRNPH2 P55795 LPC022, LPC020 HNRNPH3 P31942 LPC020 HNRNPK P61978-2 IF-051, LPC020, LPC011 HNRNPL P14866 IF-051, LPC022, LPC020 HNRNPM P52272 LPC020 HNRNPR O43390 IF-051, LPC020 HNRNPR O43390-2 LPC020 HNRNPUL2 Q1KMD3 LPC020 HPCAL1 P37235 IF-051 HPDL Q96IR7 LPC020 HSD17B10 Q99714 LPC020 HSDL2 Q6YN16 LPC020 HSPD1 P10809 LPC020 HSPE1 P61604 CMC-6-123 HTRA2 O43464 IF-051, LPC022, LPC020, IF-049 IBA57 Q5T440 LPC020 ICE1 Q9Y2F5 CMC-6-111, IF-072 IDH3G P51553 IF-051, LPC020, IF-049 IFI30 P13284 IF-072, IF-051, CMC-6-111, CMC-6-109, IF-049 IGFBP7 Q16270 IF-051 IK Q13123 IF-051, LPC020, IF-049 IL6ST P40189 IF-051, LPC020, CMC-6-123 ILF2 Q12905 IF-051 ILF3 Q12906-7 IF-051 IMMT Q16891 LPC020 INCENP Q9NQS7 IF-080 INF2 Q27J81 IF-051, LPC020, IF-049, CMC-6-123 ISCA1 Q9BUE6 LPC020 ISCU Q9H1K1 IF-051, CMC-6-123 ISG20L2 Q9H9L3 IF-051, CMC-6-123 ISOC2 Q96AB3 LPC020 ITGA3 P26006 CMC-6-109, CMC-6-111, IF-072 ITIH4 Q14624 IF-051

ITM2B Q9Y287 LPC020 JUP P14923 IF-051 KCTD5 Q9NXV2 LPC011 KDELC2 Q7Z4H8 LPC020 KDM2A Q9Y2K7 LPC020, LPC011 KHSRP Q92945 IF-051 KIAA0391 O15091 LPC020 KIAA1217 Q5T5P2 IF-051 KIF20A O95235 IF-080 KIF22 Q14807 CMC-6-109, CMC-6-111, IF-072, IF-080 KIF23 Q02241 CMC-6-109, CMC-6-111, IF-072 KIFC1 Q9BW19 CMC-6-109, CMC-6-111, IF-072 KLF5 Q13887 LPC020 KRT1 P04264 CMC-6-111 KRT10 P13645 CMC-6-111, LPC034 KRT13 P13646 LPC020 KRT14 P02533 LPC020 KRT17 Q04695 LPC020 KRT18 P05783 LPC020 KRT19 P08727 IF-051, LPC020, IF-049 KRT2 P35908 CMC-6-111, LPC034 KRT5 P13647 LPC020 KRT6A P02538 LPC020 KRT7 P08729 LPC020 KRT79 Q5XKE5 LPC020 KRT8 P05787 LPC020 KRT9 P35527 CMC-6-111 KTN1 Q86UP2 IF-051 L1RE1 Q9UN81 LPC011 L2HGDH Q9H9P8 LPC020 LAMA3 Q16787-1 IF-051 LAMTOR2 Q9Y2Q5 LPC020 LARP1 Q6PKG0 IF-051 LASP1 Q14847 IF-051 LCOR Q96JN0-3 IF-051, IF-049 LGALS1 P09382 IF-051 LGMN Q99538 LPC022, LPC020, IF-072, IF-051, CMC-6-123, IF-049, IF-073, IF-085 LMNA P02545 IF-051 LMNB1 P20700 IF-051 LMNB2 Q03252 IF-051 LMO7 Q8WWI1 IF-051, IF-072, CMC-6-123 LMTK2 Q8IWU2 IF-051 LNPEP Q9UIQ6 IF-051 LPXN O60711 CMC-6-109, CMC-6-111, IF-072 LRCH3 Q96II8 LPC011 LRIF1 Q5T3J3 LPC022, LPC020 LSM14A Q8ND56 IF-051 LSM2 Q9Y333 IF-051 LSM6 P62312 IF-051 LSM7 Q9UK45 IF-051 LSM8 O95777 IF-051 LVRN Q6Q4G3 CMC-6-109, CMC-6-111, IF-072 LYPD3 O95274 IF-051, IF-049 LYRM7 Q5U5X0 IF-049 MAGOH P61326 IF-051 MALSU1 Q96EH3 LPC020 MAN1A2 O60476 IF-051 MAN2B1 O00754 IF-072 MANF P55145 IF-051 MAP7 Q14244 LPC011 MARCH5 Q9NX47 LPC020 MATR3 P43243 IF-051, LPC022, LPC020, IF-049 MAVS Q7Z434 IF-051, IF-049 MBP P02686 IF-051 MCAT Q8IVS2 LPC020 MCM3AP O60318 LPC020 MDC1 Q14676 IF-051, LPC020 ME2 P23368 LPC020 MED1 Q15648 LPC020 MED11 Q9P086 LPC011 MED21 Q13503 IF-051 MED4 Q9NPJ6 LPC020 MEN1 O00255 CMC-6-109, CMC-6-111, IF-072 MESD Q14696 IF-051 MFSD6 Q6ZSS7 IF-049 MGA Q8IWI9 LPC020 MGME1 Q9BQP7 IF-049 MIC13 Q5XKP0 LPC022, LPC020 MICA Q29983 IF-051 MICU1 Q9BPX6 LPC020 MIF P14174 IF-051 MKI67 P46013 IF-072, IF-051, CMC-6-111, CMC-6-109, IF-049 MMP1 P03956 IF-051 MMTAG2 Q9BU76 LPC020 MNAT1 P51948 LPC020 MOB1B Q7L9L4 IF-051 MPRIP Q6WCQ1 IF-051 MRPL1 Q9BYD6 LPC020 MRPL10 Q7Z7H8 LPC020 MRPL12 P52815 IF-051, CMC-6-123 MRPL13 Q9BYD1 LPC022, LPC020 MRPL14 Q6P1L8 LPC022, LPC020 MRPL15 Q9P015 LPC020 MRPL16 Q9NX20 LPC022, LPC020 MRPL17 Q9NRX2 LPC022, LPC020 MRPL20 Q9BYC9 LPC020 MRPL21 Q7Z2W9 LPC020 MRPL23 Q16540 LPC022, LPC020 MRPL24 Q96A35 LPC020 MRPL3 P09001 LPC020 MRPL33 O75394 CMC-6-123, LPC020 MRPL34 Q9BQ48 CMC-6-123 MRPL37 Q9BZE1 LPC022, LPC020 MRPL38 Q96DV4 LPC020 MRPL39 Q9NYK5 LPC020 MRPL4 Q9BYD3 LPC020 MRPL40 Q9NQ50 LPC020 MRPL41 Q8IXM3 LPC022, LPC020, IF-049, LPC011 MRPL42 Q9Y6G3 LPC020 MRPL45 Q9BRJ2 LPC022, LPC020 MRPL46 Q9H2W6 LPC020 MRPL47 Q9HD33 LPC020 MRPL49 Q13405 LPC020 MRPL50 Q8N5N7 LPC020 MRPL52 Q86TS9 LPC020 MRPL54 Q6P161 IF-049 MRPL9 Q9BYD2 LPC020 MRPS10 P82664 LPC020 MRPS11 P82912 LPC020 MRPS12 O15235 LPC022 MRPS15 P82914 LPC020 MRPS16 Q9Y3D3 LPC020 MRPS17 Q9Y2R5 LPC020 MRPS18B Q9Y676 LPC020 MRPS18C Q9Y3D5 LPC022, LPC020 MRPS2 Q9Y399 LPC020 MRPS22 P82650 LPC020 MRPS23 Q9Y3D9 LPC022, LPC020 MRPS25 P82663 LPC020 MRPS26 Q9BYN8 LPC020 MRPS28 Q9Y2Q9 LPC020 MRPS30 Q9NP92 LPC022, LPC020 MRPS31 Q92665 LPC020 MRPS34 P82930 LPC020 MRPS35 P82673 LPC020 MRPS6 P82932 LPC020 MRPS7 Q9Y2R9 LPC020 MRPS9 P82933 LPC020 MSLN Q13421 IF-051, LPC020, CMC-6-123 MSLN Q13421-2 IF-051, IF-049 MSN P26038 IF-051 MT-CO2 P00403 LPC020 MT-ND1 P03886 LPC020 MT1X P80297 IF-051 MTDH Q86UE4 IF-051 MTERF3 Q96E29 LPC020 MTFR1 Q15390 IF-051, CMC-6-123, IF-085 MTG1 Q9BT17 LPC022, LPC020 MTHFD1L Q6UB35 LPC020 MTHFD2 P13995 LPC020 MTREX P42285 IF-051 MXRA7 P84157 IF-051, IF-049 MYDGF Q969H8 IF-051 MYO19 Q96H55 IF-072 NAGA P17050 IF-072 NCBP3 Q53F19 IF-051 NCL P19338 IF-051 NCOA4 Q13772 LPC020 NCOA5

Q9HCD5 IF-051, IF-049 NDUFA10 O95299 LPC020 NDUFA13 Q9P0J0 LPC020 NDUFA2 O43678 LPC020 NDUFA4 O00483 LPC022, LPC020, IF-049 NDUFA5 Q16718 LPC020 NDUFA7 O95182 IF-049 NDUFA8 P51970 LPC020 NDUFA9 Q16795 LPC020 NDUFAB1 O14561 IF-051, IF-049 NDUFAF2 Q8N183 LPC020 NDUFAF4 Q9P032 LPC020 NDUFB10 O96000 LPC020 NDUFB4 O95168 LPC020 NDUFS1 P28331 LPC020 NDUFS2 O75306 LPC020 NDUFS3 O75489 LPC020 NDUFS6 O75380 LPC020 NDUFS8 O00217 LPC020 NDUFV1 P49821 LPC020 NDUFV2 P19404 LPC020 NECTIN1 Q15223 IF-051, LPC020, IF-049, CMC-6-123 NECTIN2 Q92692 IF-051, LPC020 NENF Q9UMX5 IF-051 NFS1 Q9Y697 LPC020 NFU1 Q9UMS0 IF-051, IF-049 NGRN Q9NPE2 LPC020 NOC4L Q9BVI4 LPC020 NOL12 Q9UGY1 IF-051, IF-049 NOL7 Q9UMY1 IF-051 NOLC1 Q14978-2 IF-051 NOLC1 Q14978-3 LPC011 NONO Q15233 IF-051, LPC020 NOP53 Q9NZM5 LPC020 NPC2 P61916 IF-072, IF-051, CMC-6-123, IF-073, IF-085 NPM3 O75607 LPC020 NRAS P01111 LPC020 NSD2 O96028 IF-051 NSUN5 Q96P11 LPC020 NT5E P21589 IF-051 NUCB1 Q02818 CMC-6-123, IF-085 NUMA1 Q14980 IF-051, LPC020, IF-049 NUP37 Q8NFH4 LPC020 NUP50 Q9UKX7 IF-051 NUP54 Q7Z3B4 LPC020 NUP62 P37198 LPC020 NUSAP1 Q9BXS6 IF-051 NVL O15381 IF-051, CMC-6-109, CMC-6-111, IF-072 NXF1 Q9UBU9 IF-051 NXT1 Q9UKK6 LPC020 OAS1 P00973 IF-051 OAT P04181 LPC020 OCLN Q16625 IF-051 OGDH Q02218 LPC020 OVOL1 O14753 IF-051 OXA1L Q15070 LPC020 P4HB P07237 IF-051 PABPN1 Q86U42 IF-051, LPC020, IF-049 PAM16 Q9Y3D7 LPC020 PANX1 Q96RD7 LPC020 PARL Q9H300 IF-051, LPC020 PATZ1 Q9HBE1 CMC-6-111, IF-072 PAWR Q96IZ0 IF-051 PBRM1 Q86U86 LPC020 PCF11 O94913 IF-051 PCLAF Q15004 IF-073 PDAP1 Q13442 IF-051 PDE6C P51160 IF-051 PDHB P11177 LPC020 PDIA3 P30101 IF-051 PDIA6 Q15084 LPC020 PEBP1 P30086 IF-051 PHB P35232 LPC020 PHF14 O94880 IF-072, LPC011 PHF3 Q92576 IF-051, LPC022, LPC020 PHIP Q8WWQ0 IF-072 PICALM Q13492 IF-051 PKP1 Q13835 CMC-6-111, IF-072 PKP2 Q99959-2 IF-051 PLEC Q15149-3 LPC020 PLEC Q15149-4 IF-051 PLEKHA6 Q9Y2H5 IF-072 PLRG1 O43660 IF-051 PLSCR1 O15162 LPC020 PLSCR3 Q9NRY6 LPC020 POFUT1 Q9H488 IF-051, LPC020 POFUT2 Q9Y2G5-1 LPC020 POGZ Q7Z3K3 IF-072 POLDIP2 Q9Y2S7 LPC020 POLR2D O15514 LPC020 POLR2F P61218 IF-051, CMC-6-123 POLR2G P62487 IF-051 POLRMT O00411 LPC020 POM121C A8CG34 IF-051, LPC020 POSTN Q15063 IF-051 PPIB P23284 IF-051 PPIF P30405 IF-051, IF-049 PPP1R9B Q96SB3 IF-051, IF-049 PPT1 P50897 IF-051, IF-049, IF-072 PQBP1 O60828 IF-051, LPC020, IF-049 PRC1 O43663 IF-080 PRKN O60260-5 LPC020 PRKRIP1 Q9H875 IF-051 PRNP P04156 IF-051, LPC022, LPC020 PRPF19 Q9UMS4 IF-051 PRSS37 A4D1T9 LPC020 PSAP P07602 IF-051, LPC020, IF-049 PSEN1 P49768 IF-051 PSIP1 O75475 IF-051 PSPC1 Q8WXF1 IF-051 PTBP1 P26599 IF-051, LPC020 PTBP3 O95758-1 LPC020 PTCD1 O75127 LPC020 PTK7 Q13308 IF-051, IF-049 PTMA P06454 IF-051, IF-085 PTPMT1 Q8WUK0 LPC020 PTRH1 Q86Y79 LPC020, LPC011 PYM1 Q9BRP8 IF-051, CMC-6-123 QRSL1 Q9HOR6 LPC022 RACGAP1 Q9H0H5 IF-072, IF-080, IF-051, CMC-6-111, CMC-6-109, IF-049 RAD21 O60216 IF-051 RALY Q9UKM9 LPC020 RAVER1 Q8IY67 LPC020 RBBP6 Q7Z6E9 IF-051 RBBP7 Q16576 IF-051 RBM14 Q96PK6 LPC022, LPC020 RBM15 Q96T37 LPC020 RBM19 Q9Y4C8 IF-051, IF-049 RBM22 Q9NW64 IF-051 RBM27 Q9P2N5 LPC020 RBM3 P98179 CMC-6-109, IF-051 RBM4 Q9BWF3 LPC020 RBM6 P78332 IF-051 RBM8A Q9Y5S9 IF-051 RBMX P38159 IF-051, LPC020, IF-049 RBMXL1 Q96E39 IF-051 RDX P35241 IF-051 REEP5 Q00765 IF-051, LPC022, LPC020, IF-049 REXO4 Q9GZR2 IF-051, LPC020, IF-049 RGS10 O43665 IF-051, IF-049 RIF1 Q5UIP0 IF-051, LPC020, IF-049 RMDN1 Q96DB5 IF-051, LPC020, CMC-6-123 RNASET2 O00584 CMC-6-123 RNF121 Q9H920 IF-051 RNMT O43148 IF-051 RP9 Q8TA86 IF-051 RPF1 Q9H9Y2 LPC020, LPC011 RPLP1 P05386 IF-073 RPS21 P63220 IF-051 RPUSD3 Q6P087 LPC020 RREB1 Q92766 IF-051 RRN3 Q9NYV6 LPC011 RRP36 Q96EU6 IF-051 RSBN1 Q5VWQ0 IF-051 RSF1 Q96T23 IF-051 RSRC1 Q96IZ7 IF-051 S100A10 P60903 IF-051 S100A14 Q9HCY8 IF-051, LPC020, IF-049 SAFB Q15424 IF-051, LPC020, IF-049 SAFB2 Q14151 IF-051, LPC022,

LPC020, IF-049 SAMD1 Q6SPF0 IF-051, IF-049 SAMM50 Q9Y512 LPC020 SAP130 Q9H0E3 IF-051 SARNP P82979 IF-051 SCAF11 Q99590 IF-051 SCAF8 Q9UPN6 IF-051 SCFD1 Q8WVM8 IF-051 SCO1 O75880 IF-051 SDC4 P31431 IF-051 SDF2L1 Q9HCN8 IF-051 SDHA P31040 LPC020 SDHAF2 Q9NX18 LPC020 SDHB P21912 LPC020 SELENOI Q9C0D9 LPC011 SERPINH1 P50454 IF-051 SETX Q7Z333 IF-073 SF1 Q15637-6 LPC020 SF3A1 Q15459 IF-051 SF3A3 Q12874 IF-051 SF3B2 Q13435 IF-051 SF3B4 Q15427 LPC020 SFPQ P23246 IF-051 SFSWAP Q12872 IF-051 SFXN1 Q9H9B4 LPC020 SFXN3 Q9BWM7 LPC020 SHMT2 P34897 LPC020 SLC16A1 P53985 IF-051 SLC1A4 P43007 IF-051 SLC1A5 Q15758 IF-051 SLC25A10 Q9UBX3 LPC020 SLC25A6 P12236 LPC020 SLC30A1 Q9Y6M5 LPC020, IF-072 SLC3A2 P08195 IF-051 SLC43A3 Q8NBI5 IF-051 SLC48A1 Q6P1K1 LPC020 SLC6A6 P31641 LPC020 SLC9A1 P19634 IF-051 SLC9A3R1 O14745 IF-051 SLC9A3R2 Q15599 IF-051 SLIRP Q9GZT3 LPC020 SLMAP Q14BN4 IF-051, IF-049 SLU7 O95391 LPC022, LPC020 SMARCB1 Q12824 IF-051, LPC020 SMARCD1 Q96GM5 IF-051 SMARCD2 Q92925-2 LPC020 SMIM12 Q96EX1 IF-051 SMU1 Q2TAY7 IF-051 SNIP1 Q8TAD8 LPC020 SNRNP70 P08621 IF-051 SNRPA1 P09661 IF-051 SNRPC P09234 IF-051 SNRPD1 P62314 LPC011 SNW1 Q13573 IF-051 SON P18583-5 LPC020 SON P18583-6 LPC020 SORL1 Q92673 IF-049 SORT1 Q99523 CMC-6-109, IF-072 SOX15 O60248 IF-051 SPECC1 Q5M775 IF-051 SPIN1 Q9Y657 IF-051 SPINT1 O43278 IF-051, IF-049 SPRY4 Q9C004 IF-051 SPRYD4 Q8WW59 IF-049 SRCAP Q6ZRS2 IF-051 SREK1 Q8WXA9-2 IF-051 SRP9 P49458 IF-051 SRRT Q9BXP5 IF-051 SRSF9 Q13242 LPC020 SSBP1 Q04837 LPC020 ST14 Q9Y5Y6 IF-051, LPC020, IF-049, IF-072 STK24 Q9Y6E0 IF-051 STOML2 Q9UJZ1 LPC020 STX12 Q86Y82 IF-051 STX18 Q9P2W9 IF-051 STX5 Q13190 IF-051 STX7 O15400 IF-051 SUGP1 Q8IWZ8 IF-051 SUMF2 Q8NBJ7 LPC020 SYAP1 Q96A49 IF-051 SYF2 O95926 IF-051 SYMPK Q92797 IF-049 SYNE2 Q8WXH0 IF-051 TACC1 O75410 IF-051, IF-049 TACO1 Q9BSH4 LPC020 TAF11 Q15544 IF-049 TAF15 Q92804 IF-051, LPC020 TAGLN2 P37802 IF-051 TARDBP Q13148 LPC022, LPC020 TBCA O75347 IF-051 TBL1XR1 Q9BZK7 IF-051 TBP P20226 LPC020 TCF20 Q9UGU0 IF-051 TCOF1 Q13428-3 IF-051 TCOF1 Q13428-7 LPC020 TEX264 Q9Y619 IF-051 TGFBI Q15582 IF-051, CMC-6-123 THOC7 Q619Y2 LPC020 THRAP3 Q9Y2W1 LPC020 TIMM13 Q9Y5L4 IF-051, LPC022, LPC020 TIMM17A Q99595 LPC020 TIMM23B Q5SRD1 LPC022, LPC020 TIMM29 Q9BSF4 IF-051, IF-049 TIMM50 Q3ZCQ8 LPC020 TK1 P04183 CMC-6-109, CMC-6-111, IF-072 TMBIM6 P55061 LPC049, LPC011, IF-065, IF-063 TMEM106B Q9NUM4 IF-051, CMC-6-123 TMEM126A Q9H061 LPC020 TMEM214 Q6NUQ4 IF-072 TMEM237 Q96Q45 IF-049 TMEM30A Q9NV96 IF-051 TMEM40 Q8WWA1 LPC020 TMPO P42166 LPC020 TMSB4X P62328 IF-051 TNFRSF12A Q9NP84 LPC020 TOMM6 Q96B49 LPC020 TOMM7 Q9P0U1 LPC020 TOP2A P11388 CMC-6-109, CMC-6-111, IF-072, LPC011 TOR1AIP2 Q8NFQ8 IF-051 TOR4A Q9NXH8 LPC049 TP53BP1 Q12888 IF-051, LPC020 TPP1 O14773 LPC020, IF-072 TPST1 O60507 LPC020 TPX2 Q9ULW0 CMC-6-109, CMC-6-111, IF-072, IF-080 TRIM28 Q13263 CMC-6-109, CMC-6-111, IF-072 TRIM33 Q9UPN9 IF-072 TRMT10C Q7L0Y3 LPC020 TSFM P43897 LPC020 TSNAX Q99598 LPC011 TST Q16762 LPC020 TTC19 Q6DKK2 IF-051, IF-049, CMC-6-123 TUFM P49411 LPC020 TXN2 Q99757 IF-051, LPC020 TXNDC12 O95881 IF-051 TXNDC5 Q8NBS9 IF-051 TXNIP Q9H3M7 IF-073, IF-080, IF-085 TXNL4A P83876 IF-051 TYW1 Q9NV66 LPC011 U2AF2 P26368 IF-051 U2SURP O15042 LPC020 U2SURP O15042-2 IF-051 UBE2G2 P60604 IF-051 UBE2J2 Q8N2K1 IF-051 UHRF1 Q96T88 CMC-6-109, CMC-6-111, IF-072, IF-080 URB1 O60287 LPC020 USP36 Q9P275 IF-051, IF-049 UTP15 Q8TED0 LPC011 UTP18 Q9Y5J1 LPC011 VAMP7 P51809 IF-051, IF-049 VDAC1 P21796 LPC020 VDAC2 P45880 LPC020 VIM P08670 IF-051, CMC-6-123 VIRMA Q69YN4 LPC020 VSIR Q9H7M9 CMC-6-109, CMC-6-111, IF-072 VTI1B Q9UEU0 IF-051 WDFY1 Q8IWB7 IF-072 WDR33 Q9C0J8 LPC020 WDR76 Q9H967 CMC-6-109, CMC-6-111, IF-072 WTAP Q15007 IF-051, IF-049 XPNPEP3 Q9NQH7 LPC020 XRN2 Q9H0D6 IF-051 YAP1 P46937 IF-051 YARS2 Q9Y2Z4 LPC020 YIPF3 Q9GZM5 IF-051, LPC020, IF-049 YLPM1 P49750 IF-051, IF-049

ZBTB1 Q9Y2K1 IF-051, CMC-6-123, IF-085 ZBTB11 O95625 IF-080 ZC3H13 Q5T200 IF-051, LPC020 ZC3H14 Q6PJT7 IF-051, IF-072 ZC3H18 Q86VM9 IF-051 ZC3HC1 Q86WB0 LPC020 ZCCHC8 Q6NZY4 IF-051 ZFR Q96KR1 LPC022, LPC020 ZMYM4 Q5VZL5 LPC022, LPC020 ZNF148 Q9UQR1 IF-051 ZNF185 O15231 IF-051, IF-049 ZNF217 O75362 IF-051, CMC-6-123, IF-085 ZNF512B Q96KM6 IF-051 ZNF638 Q14966 IF-051, LPC020 ZNF800 Q2TB10 IF-072 ZNF830 Q96NB3 IF-051, LPC020

Biological Materials and Methods

[0044] Materials: Rabbit anti-SAFB1 (PA5-2135P, 1:3,000 dilution) and rabbit anti-FAM136A (PA5-56345, 1:1,000 dilution) were ordered from Life Technologies. Rabbit anti-SAFB2 (11642-1-AP, 1:3,000 dilution), rabbit anti-TARDBP (10782-2-AP, 1:1,000 dilution) were ordered from Proteintech. Rabbit anti-PTMA (PA5-71580, 1:1,000 dilution) was ordered from Invitrogen. Rabbit anti-LGMN (93627S, 1:1,000), rabbit anti-PPP1R9B (14136S, 1:1,000 dilution), rabbit anti-DBC1 (5693S, 1:1,000 dilution), rabbit anti-MAVS (3993S, 1:1,000 dilution), rabbit anti-Beta-actin (4967S, 1:1,000 dilution), rabbit anti-histone H3 (9715S, 1:1,000 dilution), mouse anti-HA (2367S, 1:2,000 dilution), rabbit anti-HA (3724S, 1:2,000 dilution), rabbit anti-FLAG (14793, 1:1,000 dilution), mouse anti-FLAG (8146S, 1:1,000 dilution), anti-rabbit HRP (7074P2, 1:10,000 dilution) were ordered from Cell Signaling Technology. Epoxomicin (A2606, proteasome inhibitor) was ordered from APEx Bio. MG132 (HY-13259, proteasome inhibitor) and MLN4924 (HY-70062, neddylation inhibitor) were ordered from MedChem Express.

[0045] Mammalian cell culture: MDA-MB-231 cells (ATCC) were maintained in DMEM supplemented with 10% (v/v) fetal bovine serum (FBS), 1% (v/v) penicillin/streptomycin, and 2 mM glutamine. HCC1806 cells (ATCC) were maintained in RPMI supplemented with 10% (v/v) fetal bovine serum (FBS), 1% (v/v) penicillin/streptomycin, and 2 mM glutamine. All cell lines were grown at 37° C. in a humidified 5% CO_2 atmosphere.

[0046] Treatment of Live Cells with FragTAC Probes: Separate 6 cm (for gel-based analyses) or 10 cm (for MS-based analyses) dishes of MDA-MB-231 cells or HCC1806 cells were grown to 80-95% confluency with DMEM media and RPMI media respectively. The growth medium was aspirated, and the cells were washed with Dulbecco's phosphate buffered saline (DPBS). The cells were incubated with serum-free media containing fragment probes (200 μM) or control probes (200 μM) for 6 hours at 37° C. under a 5% CO_2 atmosphere. The cultures were scraped, washed with cold DPBS, and collected into 15 mL centrifuge tubes, then transferred to 1.5 mL Eppendorf tubes. The cell suspensions were centrifuged (1,400 g, 3 min) and the pellets were stored at −80° C. until the next stage of processing.

[0047] Sample preparation for TMT MS Quantitative Proteomics: For treatment of cell lysates, cells were collected, cell pellets were resuspended in 100 μL DPBS containing 1× Halt protease inhibitor cocktail, and lysed by sonication (15 ms on, 40 ms off, 15% amplitude, 1 s total). Protein concentrations were normalized (2 mg/mL in 100 μL with cold DPBS) using the Lowry Protein Assay (Pierce). Urea (48 mg) was weighed for each sample in 2 mL LoBind Eppendorf tubes and cell lysates (100 μL=200 μg) were added (final urea concentration=8 M). Pellets were resuspended in a freshly prepared 1:1 solution (50 μL) TCEP (200 mM in DPBS) and K_2CO_3 (600 mM in DPBS) and incubated (30 min, 37° C.) while shaking. After reaction, a solution of freshly prepared iodoacetamide (70 μL, 400 mM in DPBS) was added and incubated for 30 min at room temperature while protected from light. After reaction, 1.8 mL of cold 3:1:2 MeOH/$CHCl_3$/H_2O solution was added to each tube, and the samples were centrifuged (10,000×g, 10 min, 4° C.), forming a disc. The supernatant was carefully removed, MeOH (600 μL) was added, samples were centrifuged as previously described and the supernatant was removed. Cell pellets were resuspended in TEAB (160 μL, 100 mM, pH 8.5) and sonicated as described previously. Endoproteinase LysC (20 μL, ½ vial, 10 μg dissolved in 220 μL of 100 mM TEAB pH 8.5) was added to each and incubated at 37° C. for 2 h with shaking. Sequencing-grade modified porcine trypsin (20 μL, 1 vial, 20 μg dissolved in 220 μL of 100 mM TEAB pH 8.5), a solution of Protease Max (2 μL, 1% (w/v) in 100 mM TEAB pH 8.5) and a solution of $CaCl_2$) (2 μL, 100 mM) were added to the samples and incubated at 37° C. overnight (14 h) with shaking. The digest was separated by centrifugation (12,000×g, 10 min, 4° C.). Peptide concentration was determined using Pierce Quantitative Fluorometric Peptide Assay Kit (Thermo Fisher Scientific, 23290) according to manufacturer's instructions. MS-grade acetonitrile (12.5 μL) was added to each and samples were labeled with respective TMT 10 plex isotope (8 μL, 20 μg/μL) for 1 h with occasional vortexing at RT. To quench, hydroxylamine (3 μL, 5% v/v) was added to each sample, vortexed, and incubated for 15 min at RT. Formic acid (5 μL) was added to each tube to acidify and the samples were dried under vacuum centrifugation. The samples were combined by redissolving the contents of one tube in a solution of trifluoroacetic acid (TFA, 400 μL, 0.1% in water) and transferred into each sample until all samples were redissolved. The stepwise process was repeated with formic acid (Buffer A, 200 μL, 0.1% in water) for a final volume of 600 μL. The samples were fractionated using a fractionation kit (Pierce high pH Reversed-Phase Fractionation Kit, Thermo Fisher Scientific 84868) according to manufacturer's instructions. The peptide fractions were eluted from the spin column with consecutive solutions of 0.1% triethylamine combined with MeCN (5-75% MeCN). The fractions were combined pairwise (fraction 1 and fraction 10, fraction 2 and fraction 11, etc.), dried via vacuum centrifugation, and stored at −80° C. until ready for mass spectrometer injection.

[0048] LC-MS analysis of TMT samples: TMT labeled samples were redissolved in MS buffer A (20 μL, 0.1% formic acid in water). 3 μL of each sample was loaded onto an Acclaim PepMap 100 precolumn (75 μm×2 mm) and eluted on an Acclaim PepMap RSLC analytical column (75 μm×15 cm) using the UltiMate 3000 RSLCnano system (Thermo Fisher Scientific). Buffer A was prepared as described above and buffer B (0.1% formic acid in MeCN) were used in a 220 min gradient (flow rate 0.3 mL min, 35° C.) of 2% buffer B for 10 min, followed by an incremental increase to 30% buffer B over 192 min, 60% buffer B for 5 min, 60-95% buffer B for 1 min, hold at 95% buffer B for 5 min, followed by descent to 2% buffer B for 1 min followed by re-equilibration at 2% for 6 min. The elutions were analyzed with a Thermo Fisher Scientific Orbitrap Fusion Lumos mass spectrometer with a cycle time of 3 s and nano-LC electrospray ionization source applied voltage of 2.0 kV. MS1 spectra were recorded at a resolution of 120,000 with an automatic gain control (AGC) value of 1×10^6 ions, maximum injection time of 50 ms (dynamic exclusion enabled, repeat count 1, duration 20 s). The scan range was specified from 375 to 1,500 m/z. Peptide fragmentation MS2 spectra was recorded via collision-induced diffusion (CID) and quadrupole ion trap analysis (AGC 1.8×10^4, 30% collision energy, maximum inject time 120 ms, isolation window 1.6). MS3 spectra was generated by high-energy collision-induced dissociation (HCD) with collision energy of 65%. Precursor selection included up to 10 MS^2 ions for the MS^3 spectrum.

[0049] TMT proteomics data analysis: Proteomic analysis was performed with the processing software Proteome Discoverer 2.4 (Thermo Fisher Scientific). Peptide sequences were identified by matching proteome databases with experimental fragmentation patterns via the SEQUEST HT algorithm. Fragment tolerances were set to 0.6 Da, and precursor mass tolerances set to 10 ppm with one missed cleavage site allowed. Spectra were searched against the *Homo Sapiens* proteome database (42,358 sequences) using a false discovery rate of 1% (Percolator). MS^3 peptide quantitation was performed with a mass tolerance of 20 ppm. Identified proteins were required to have at least two unique peptides. TMT ratios obtained by Proteome Discoverer were transformed with $\log_2(x)$, and p-values were calculated via Student's two-tailed t-tests with two biological replicates.

[0050] Cell viability assays: Cells were seeded in white-opaque 96-well plates in full growth media at a density of 2,000 cells/well (50 μL) and were allowed to grow for 14 h at 37° C. in a humidified 5% CO_2 atmosphere. The cells were then treated with compounds in triplicate and incubated at 37° C. in a humidified 5% CO_2 atmosphere for 6 hours. Cell viability was determined using the luciferase-based Cell Titer-Glo Luminescent Cell Viability Assay (Promega) following manufacturer's guidelines. Data represents the average and standard deviation of triplicates in measured luminescence.

[0051] Immunoblots: After cells were harvested, cell pellets were resuspended in 100 μL DPBS containing 1× Halt protease inhibitor cocktail, and lysed by sonication (15 ms on, 40 ms off, 15% amplitude, 1 s total). Protein concentrations were normalized (2 mg/mL in 100 μL with cold DPBS) as previously described. 4X SDS gel loading buffer (33 μL) was added to the solution of protein lysate and the resulting mixture was heated at 95° C. for 15 min. Proteins (15 μg total protein loaded per gel lane) were resolved by SDS-PAGE (10% acrylamide) made in-house, and transferred to PVDF membrane (0.2 μM, 1620177, Bio-Rad). The membrane was blocked with 5% BSA in Tris-buffered saline with Tween (TBST) buffer (0.1% Tween 20, 20 mM Tris-HCl 7.6, 150 mM NaCl) at room temperature for 1 h. The antibody was diluted with fresh 5% BSA in TBST buffer (dilutions were performed following manufacturer's guidelines) and incubated with membrane overnight (14 h) at 4° C. Membrane was washed three times with TBST buffer, left 5 minutes between each wash on a rocker and then incubated with secondary antibody in 5% dry milk in TBST at room temperature for 2 h on a rocker. Membrane was washed three times with TBST buffer and visualized by in-gel fluorescence on a Bio-Rad ChemiDoc MP Imaging System. The images were processed using Image Lab (version 5.2.1) software.

[0052] Quantification and Statistical Analysis: All data fitting and statistical analysis were performed using GraphPad Prism version 9.00 for Windows and Mac. Statistical significance was defined as $p<0.05$ and determined by 2-tailed Student's/tests.

Compound Synthesis and Characterization

[0053] General Synthetic Information: All commercial reagents acquired from Sigma-Aldrich, Fisher Scientific, Combi-Blocks, MedChemExpress, AstaTech, Matrix Scientific and BroadPharm were used without further purification. Distilled water was used for all water necessities in synthetic procedures (e.g., reagent, solvent, work-up). All reactions were run with anhydrous solvents. TLC analyses were completed with EMD Millipore silica gel coated (250 μM) F254 glass plates and visualized with UV light (254 nm) or by employing diverse stains (e.g., iodine, CAM, PMA, Ninhydrin, DNP, KMnO.sub.4), followed by gentle heating. Flash column chromatography was performed with silica gel 60. Automated purifications were performed on Biotage Isolera One purification system using manually packed columns of 5G, 10G or 25G of silica. Preparative Thin Layer Chromatography (pTLC) was carried out using EMD Millipore silica gel coated (250 μM) F254 glass plates or glass backed plates 1000-2000 μm thickness (Analtech). .sup.1H- and .sup.13C-NMR spectra were recorded on a Bruker AVANCE NEO 400 MHz instrument with SampleXpress, Bruker AVANCE NEO 500 MHz NMR instrument equipped with 5 mm BBFO SP probe, and SampleCase-24 sample changer, and Bruker AVANCE III HD 600, instrument equipped with a 5 mm CPDCH CryoProbe. Data was collected at ambient temperature unless otherwise stated using standard pulse methods as supplied by Bruker software. Coupling constants are quoted to the nearest 0.1 Hz and multiplicities are given by the following abbreviations and combinations: m (multiplet), s (singlet), d (doublet), dd (doublet of doublets), ddd (doublet of doublet of doublets), t (triplet), td (triplet of doublets), tt (triplet of triplets), q (quartet), br (broad). All NMR data was processed in MestReNova v12.0.2. Chemical shifts for proton and carbon resonances are reported in parts per million (ppm) on the 8 scale relative to the residual protons of the deuterated solvent of relevance. Mass spectrometry data were collected on a Thermo Scientific ISQ single-quadrupole instrument (ESI; low resolution), Agilent 6125 and 6135 single-quadrupole instruments (ESI; low resolution), and Agilent 6230 single-quadrupole TOF (ESI-TOF; high resolution).

##STR00137##

[0054] General Procedure 1: Synthesis of amides using HATU as coupling reagent To a solution of amine (1.1 eq.), carboxylic acid (1.0 eq.) and HATU (1.5 eq.) in dry DMF at 0° C. was added DIPEA (3.0 eq.). The reaction mixture was stirred for 2-18 h and then quenched with DI H.sub.2O and diluted with Ethyl acetate (EtOAc). The aqueous layer was extracted with EtOAc (3 times), and the combined organic layers were washed with H_2O (2 times), a saturated aqueous solution of NH_4Cl (2 times) and a saturated aqueous solution of NaCl (2 times) before being dried over anhydrous Na_2SO_4 and filtered. Volatiles were removed by rotary evaporation and the crude product was purified by automated column chromatography, pTLC, or organic solvent washes (i.e., the organic solvent was added into the vial and mixed with the solid decanting).

##STR00138##

[0055] General Procedure 2: Synthesis of amides using EDC as coupling reagent. To a solution of amine (1.1 eq.), carboxylic acid (1.0 eq.), EDC (1.5 eq.) and HOBt (1.5 eq.) in dry DCM at 0° C. was added DIPEA (3.0 eq.). The reaction mixture was stirred overnight, then quenched with DI H_2O. The aqueous layer was extracted with CH_2Cl_2 (3 times), and the combined organic layers were washed with H_2O (2 times), a saturated aqueous solution of NH_4Cl (2 times) and a saturated aqueous solution of NaCl (2 times) before being dried over anhydrous Na_2SO_4 and filtered. Volatiles were removed by rotary evaporation and the crude product was purified by automated column chromatography, pTLC, or organic solvent washes (i.e., the organic solvent was added into the vial and mixed with the solid decanting).

##STR00139##

[0056] General Procedure 3: Tert-butyl ester deprotection of carboxylic acids using TFA. To a solution of t-Butyl protected acid (1.0 eq.) in dry CH_2Cl_2 was added a solution of TFA (20-50%.). The reaction mixture was stirred at RT for 0.5-12 h. The crude product was concentrated through a nitrogen flow, diluted in CH_2Cl_2 and evaporated again. No purification was needed.

##STR00140##

[0057] General Procedure 4: Boc deprotection of amines using TFA. To a solution of Boc protected amine (1.0 eq.) in dry CH_2Cl_2 was added a solution of TFA (20-50%). The reaction mixture was stirred at RT for 0.5-12 h. The crude product was concentrated through a nitrogen flow, diluted in CH_2Cl_2 and evaporated again. No purification was needed.

##STR00141##

[0058] General Procedure 5: Ester hydrolysis. The ester (1.0 eq.) was dissolved in a mixture of THF/MeOH/H_2O (3:1:2) and LiOH, H_2O (7.0 eq.) was added. The reaction mixture was stirred at RT for 3-12 h. The solvents were evaporated, and the residue was acidified with IN aqueous until pH=1, before being extracted with EtOAc, dried over anhydrous Na_2SO_4 and filtered. The organic solvent was evaporated, and no purification was needed.

Starting Materials:

##STR00142##

Characterization Data:

##STR00143##

Tert-butyl 3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4yl)oxy) acetamido) ethoxy) propanoate (1)

[0059] General Procedure 1. Reaction scale: 10.5 mg (31.60 μmol, 1.0 eq.) of SM0 and 6.4 μL (33.55 μmol, 1.1 eq.) of L1. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 1 as a white foam (14.7 mg, 97%). R_f=0.63 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{24}H_{29}N_3O_9$ 503.2, found $[M+H]^+$ 504.0. 1H NMR (400 MHZ, $CDCl_3$) δ 8.69 (s, 1H), 7.77-7.69 (t, 1H), 7.65 (d, J=5.9 Hz, 1H), 7.54 (d, J=7.3 Hz, 1H), 7.18 (d, J=8.4 Hz, 1H), 5.04-4.96 (m, 1H), 4.64 (s, 2H), 3.71 (s, 2H), 3.67-3.54 (m, 4H), 3.50 (ddt, J=14.6, 7.8, 3.1 Hz, 1H), 2.89 (t, J=11.3 Hz, 2H), 2.56 (dd, J=7.9, 6.1 Hz, 2H), 2.21-2.09 (m, 2H), 1.42 (d, J=1.7 Hz, 9H). ^{13}C NMR (101 MHZ, $CDCl_3$) δ 171.05, 168.21, 166.82, 166.68, 154.43, 136.96, 133.69, 119.30, 118.11, 117.29, 80.99, 77.36, 77.04, 76.94, 76.73, 69.11, 67.84, 66.68, 49.24, 38.98, 38.63, 36.09, 31.34, 29.72, 28.08, 22.78.

##STR00144##

Tert-butyl 3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) ethoxy) ethoxy) propanoate (2)

[0060] General Procedure 1. Reaction scale: 20 mg (60.0 μmol, 1.0 eq.) of SM0 and 15 μL (65 μmol. 1.1 eq.) of L3. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 2 as a yellow foam (19.9 mg, 59%). R/=0.57 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{26}H_{33}N_3O_{10}$ 547.2, found $[M+H]^+$ 548.4. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.70 (s, 1H), 7.73 (dd, J=8.4, 7.4 Hz, 1H), 7.59 (br. t, 1H), 7.55-7.48 (m, 1H), 7.18 (d, J=8.5 Hz, 1H), 5.04-4.92 (m, 1H), 4.65 (s, 2H) 3.73-3.52 (m, 10H), 2.89-2.70 (m, 4H), 1.42 (s, 11H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 171.72, 171.45, 168.54, 167.24, 167.13, 166.17, 154.88, 137.40, 134.14, 119.69, 118.47, 117.71, 81.24, 77.79, 77.68, 77.48, 77.16, 70.67, 70.59, 69.93, 68.30, 67.31, 49.71, 39.47, 39.06, 36.62, 31.85, 30.15, 28.53, 23.12.

##STR00145##

Tert-butyl 1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azapentadecan-15-oate (3)

[0061] General Procedure 1. Reaction scale: 20 mg (60.2 μmol, 1.0 eq.) of SM0 and 15 μL (66.2 μmol. 1.1 eq.) of L5. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 3 as a yellow foam (21.5 mg, 61%). R_f=0.76 (8% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI−) calc'd for $C_{28}H_{37}N_3O_{11}$ 591.2, found $[M-H]^-$ 590.0. 1H NMR (400 MHZ, $CDCl_3$) δ 8.92 (s, 1H), 7.72 (dd, J=8.4, 7.3 Hz, 1H), 7.63 (br. t, 1H), 7.53 (d, J=7.3 Hz, 1H), 7.18 (d, J=8.3 Hz, 1H), 4.99-4.88 (m, 1H), 4.64 (s, 2H), 3.70-3.57 (m, 14H), 2.92-2.80 (m, 2H), 2.48 (t, J=6.6 Hz, 2H), 1.42 (s, 11H). ^{13}C NMR (101 MHZ, $CDCl_3$) δ 171.10, 170.97, 168.15, 166.88, 166.67, 165.80, 154.49, 136.94, 133.70, 119.39, 118.09, 117.28, 80.62, 77.37, 77.25, 77.05, 76.93, 76.73, 70.39, 70.35, 70.32, 70.24, 69.43, 67.96, 66.87, 49.32, 39.13, 38.62, 36.15, 31.42, 29.69, 28.09, 22.71.

##STR00146##

Tert-butyl (1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl) carbamate (4)

[0062] Reaction scale: 100 mg (301.0 μmol, 1.0 eq.) of SM0 and 96.8 mg (331.1 μmol. 1.1 eq.) of L7. Purified by Biotage Isolera (2-14% MeOH/CH_2Cl_2 gradient, 16 CV) to afford 4 as a yellow foam (113 mg, 62%). R_f=0.63 (8% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{28}H_{38}N_3O_{11}$ 606.3, found $[M+H]^+$ 607.2. ^1H NMR (500 MHZ, $CDCl_3$) δ 9.23 (s, 1H), 7.69 (t, J=7.9 Hz, 1H), 7.58 (t, J=5.6 Hz, 1H), 7.49 (d, J=7.3 Hz, 1H), 7.15 (s, 1H), 4.96-4.87 (m, 1H), 4.61 (s, 2H), 3.62-3.48 (m, 17H), 2.87-2.76 (m, 2H), 1.38 (s, 11H).

##STR00147##

Tert-butyl 1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12,15-tetraoxa-3-azaoctadecan-18-oate (5)

[0063] General Procedure 1. Reaction scale: 10 mg (32.5 μmol, 1.0 eq.) of SM0 and 10.18 μL (35.8 μmol. 1.1 eq.) of L9. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 5 as a white foam (15.7 mg, 76%). R/=0.53 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{30}H_{41}N_3O_{12}$ 635.3, found $[M+H]^+$ 636.0. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.99 (s, 1H), 7.72 (dd, J=8.4, 7.4 Hz, 1H), 7.63 (t, J=5.2 Hz, 1H), 7.53 (d, J=7.3 Hz, 1H), 7.18 (d, J=8.4 Hz, 1H), 5.03-4.86 (m, 1H), 4.64 (s, 2H), 3.69-3.59 (m, 18H), 2.88-2.78 (m, 2H), 2.47 (t, J=6.5 Hz, 3H), 1.42 (s, 11H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 171.34, 168.34, 167.10, 166.70, 154.44, 136.97, 133.65, 119.42, 117.25, 80.61, 77.42, 77.10, 76.79, 70.55, 70.38, 70.28, 69.50, 67.87, 66.79, 60.40, 49.31, 39.08, 38.63, 36.19, 31.46, 29.70, 28.08, 22.68, 21.05, 14.19.

##STR00148##

Tert-butyl 6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) hexanoate

(6)

[0064] General Procedure 1. Reaction scale: 15 mg (45.1 μmol, 1.0 eq.) of SM0 and 10.0 μL (49.7 μmol. 1.1 eq.) of L13. Purified by pTLC (8% MeOH/CH_2Cl_2) to afford 6 as an off-white solid (19.6 mg, 87%). R_f=0.79 (8% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{25}H_{31}N_3O_8$ 501.2, found $[M+H]^+$ 502.0. ^1H NMR (400 MHZ, $CDCl_3$) δ 9.66 (s, 1H), 7.75-7.71 (m, 1H), 7.61 (s, 1H), 7.55 (d, J=7.3 Hz, 1H), 7.39 (d, J=1.4 Hz, 2H), 7.28 (d, J=7.3 Hz, 2H), 7.20 (d, J=1.4 Hz, 1H), 4.98-4.91 (m, 1H), 4.72-4.55 (m, 3H), 3.63 (q, J=3.8 Hz, 3H), 3.45 (t, J=5.1 Hz, 2H), 2.35 (t, J=4.9 Hz, 4H), 1.25 (s, 9H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 171.77, 171.13, 168.15, 166.66, 166.41, 166.08, 154.62, 141.96, 136.86, 133.40, 128.43, 127.68, 126.98, 120.06, 118.32, 117.43, 77.16, 77.05, 76.84, 76.62, 76.52, 75.77, 68.48, 51.92, 51.44, 49.17, 45.70, 41.64, 39.03, 32.85, 31.22, 29.53, 28.69, 26.59, 25.16, 22.67.

##STR00149##

Tert-butyl (6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) hexyl) carbamate (7)

[0065] ^1H NMR (400 MHZ, $CDCl_3$) δ 7.74 (m, 1H), 7.55 (d, J=7.3 Hz, 1H), 7.46 (m, 1H), 7.19 (d, J=8.3 Hz, 1H), 4.99 (m, 1H), 4.71-4.60 (m, 3H), 3.46-3.28 (m, 2H), 3.17-3.04 (m, 2H), 2.93-2.76 (m, 3H), 2.14 (m, 1H), 1.64-1.56 (m, 2H), 1.48-1.33 (m, 16H).

##STR00150##

Tert-butyl 3-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) propanoate (8)

[0066] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1 and 11.2 mg (51.1 μmol. 1.1 eq.) of L2. Purified by pTLC (8% MeOH/CH_2Cl_2) to afford 8 as a clear oil (22.6 mg, 77%). R_f=0.59 (8% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{32}H_{46}N_4O_7S$ 630.3, found $[M+H]^+$ 631.1. ^1H NMR (600 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.43 (t, J=6.0 Hz, 1H), 7.36-7.30 (m, 4H), 6.95 (d, J=8.3 Hz, 1H), 4.72 (t, J=7.9 Hz, 1H), 4.55 (dd, J=15.0, 6.7 Hz, 1H), 4.50 (dt, J=4.3, 2.1 Hz, 1H), 4.44 (d, J=8.2 Hz, 1H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.09-4.07 (m, 1H), 3.71-3.64 (m, 4H), 3.59 (dd, J=11.3, 3.7 Hz, 1H), 2.54-2.44 (m, 7H), 2.13-2.07 (m, 1H), 1.42 (s, 9H), 0.92 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.08, 171.82, 171.19, 170.79, 170.63, 150.34, 148.45, 138.16, 131.63, 130.92, 129.52, 128.12, 128.09, 80.85, 77.26, 77.05, 76.84, 70.10, 66.82, 66.79, 66.66, 66.29, 60.42, 58.39, 57.77, 56.62, 55.99, 43.22, 36.69, 36.08, 35.85, 34.79, 31.93, 30.17, 29.71, 29.67, 29.37, 29.33, 28.93, 28.10, 28.08, 26.64, 26.41, 22.70, 21.27, 21.07, 17.45, 16.04, 14.14.

##STR00151##

Tert-butyl 3-(3-(((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) propanoate (9)

[0067] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1′ and 11.3 mg (51.1 μmol. 1.1 eq.) of L2. Purified by pTLC (8% MeOH/CH_2Cl_2) to afford 9 as a white foam (0.6 mg, 77%). R/=0.57 (8% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{32}H_{46}N_4O_7S$ 630.3, found $[M+H]^+$ 631.3. ^1H NMR (600 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.43 (t, J=6.0 Hz, 1H), 7.36-7.30 (m, 4H), 6.95 (d, J=8.3 Hz, 1H), 4.72 (t, J=7.9 Hz, 1H), 4.55 (dd, J=15.0, 6.7 Hz, 1H), 4.50 (dt, J=4.3, 2.1 Hz, 1H), 4.44 (d, J=8.2 Hz, 1H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.09-4.07 (m, 1H), 3.71-3.64 (m, 4H), 3.59 (dd, J=11.3, 3.7 Hz, 1H), 2.54-2.44 (m, 7H), 2.13-2.07 (m, 1H), 1.42 (s, 9H), 0.92 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.08, 171.82, 171.19, 170.79, 170.63, 150.34, 148.45, 138.16, 131.63, 130.92, 129.52, 128.12, 128.09, 80.85, 77.26, 77.05, 76.84, 70.10, 66.82, 66.79, 66.66, 66.29, 60.42, 58.39, 57.77, 56.62, 55.99, 43.22, 36.69, 36.08, 35.85, 34.79, 31.93, 30.17, 29.71, 29.67, 29.37, 29.33, 28.93, 28.10, 28.08, 26.64, 26.41, 22.70, 21.27, 21.07, 17.45, 16.04, 14.14.

##STR00152##

Tert-butyl 3-(2-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) ethoxy) propanoate (10)

[0068] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1 and 13.5 mg (51.1 μmol. 1.1 eq.) of L4. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 10 as a yellow oil (14.9 mg, 51%). R/=0.54 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{34}H_{50}N_4O_8S$ 674.3, found $[M+H]^+$ 675.2. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.48 (t, J=6.0 Hz, 1H), 7.38-7.30 (m, 4H), 7.05 (d, J=8.2 Hz, 1H), 4.73 (t, J=8.0 Hz, 1H), 4.56 (dd, J=15.0, 6.6 Hz, 1H), 4.50 (dt, J=4.4, 2.1 Hz, 1H), 4.44 (d, J=8.2 Hz, 1H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.09 (dt, J=11.5, 1.8 Hz, 1H), 3.72-3.65 (m, 4H), 3.61-3.56 (m, 5H), 2.52-2.45 (m, 8H), 2.13-2.06 (m, 1H), 1.42 (s, 9H), 0.93 (s, 9H).

##STR00153##

Tert-butyl 3-(2-(3-(((S)-1-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy) ethoxy) propanoate (11)

[0069] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1′ and 13.4 mg (51.1 μmol. 1.1 eq.) of L4. Purified by pTLC (8% MeOH/CH_2Cl_2) to afford 11 as a clear oil (19.3 mg, 62%). R$=0.55 (10% MeOH/$CH_2Cl_2$, UV-active); LC-MS (ESI+) calc'd for $C_{34}H_{50}N_4O_8S$ 674.3, found $[M+H]^+$ 675.1. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.49 (t, J=6.0 Hz, 1H), 7.39-7.29 (m, 4H), 7.07 (d, J=8.1 Hz, 1H), 4.72 (t, J=8.0 Hz, 1H), 4.60-4.41 (m, 3H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.14-4.05 (m, 1H), 3.68 (q, J=6.1 Hz, 4H), 3.60 (s, 5H), 2.55-2.42 (m, 8H), 2.11 (dd, J=13.4, 8.1 Hz, 1H), 1.42 (s, 9H), 0.93 (s, 9H). ^{13}C NMR (101 MHZ, $CDCl_3$) δ 172.23, 171.79, 170.98, 170.88, 150.32, 148.44, 138.23, 131.64, 130.88, 129.49, 128.11, 80.70, 77.37, 77.25, 77.05, 76.73, 70.43, 70.17, 70.10, 67.14, 66.88, 58.40, 57.82, 56.63, 43.19, 36.66, 36.15, 35.92, 34.76, 29.71, 28.10, 26.41, 16.05.

##STR00154##

Tert-butyl(S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoate (12)

[0070] General Procedure 1. Reaction scale: 70 mg (162.6 μmol, 1.0 eq.) of SM1 and 54.8 mg (178.8 μmol. 1.1 eq.) of L6. Purified by Biotage Isolera (2-15% MeOH/CH_2Cl_2 gradient, 16 CV) to afford 12 as a white foam (81 mg, 70%). R/=0.56 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI−) calc'd for $C_{36}H_{54}N_4O_9S$ 718.4, found $[M+HCOO]^-$ 763.2. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.60 (s, 1H), 7.48 (t, J=6.0 Hz, 1H), 7.26 (s, 4H), 7.02 (d, J=8.5 Hz, 1H), 4.60 (t, J=8.0 Hz, 1H), 4.45 (dd, J=15.4, 7.2 Hz, 3H), 4.25 (dd, J=15.1, 5.4 Hz, 1H), 3.94 (d, J=11.2 Hz, 1H), 3.60 (td, J=6.1, 3.8 Hz, 4H), 3.57-3.49 (m, 9H), 2.44-2.34 (m, 7H), 2.31 (td, J=8.3, 4.1 Hz, 1H), 2.04 (ddt, J=13.0, 8.1, 1.9 Hz, 1H), 1.35 (s, 9H), 0.87 (s, 9H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.00, 171.50, 171.20, 170.94, 150.35, 148.36, 138.30, 131.65, 130.76, 129.41, 128.21, 128.01, 80.60, 77.43, 77.31, 77.11, 76.79, 70.43, 70.39, 70.36, 70.26, 69.98, 67.10, 66.84, 58.69, 57.64, 56.73, 50.39, 45.73, 43.09, 36.67, 36.33, 36.20, 35.11, 29.66, 28.08, 26.38, 16.00, 8.54.

##STR00155##

Tert-butyl(S)-15-((2S,4S)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoate (13)

[0071] General Procedure 1. Reaction scale: 15 mg (34.8 μmol, 1.0 eq.) of SM1′ and 11.7 mg (38.3 μmol. 1.1 eq.) of L6. Purified by Biotage Isolera (2-15% MeOH/CH_2Cl_2 gradient, 16 CV) to afford 13 as a white foam (81 mg, 70%). R/=0.62 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{36}H_{54}N_4O_9S$ 718.4, found $[M+H]^+$ 719.2. ^1H NMR (400 MHz, $CDCl_3$) δ 8.69 (s, 1H), 7.67-7.59 (m, 1H), 7.42-7.32 (m, 4H), 6.95 (d, J=8.7 Hz, 1H), 5.59 (s, 1H), 4.73 (d, J=9.0 Hz, 1H), 4.65 (dd, J=15.0, 7.0 Hz, 1H), 4.49 (d, J=8.7 Hz, 2H), 4.31 (dd, J=15.0, 5.1 Hz, 1H), 3.96 (dd, J=11.0, 4.2 Hz, 1H), 3.80 (dd, J=11.1, 1.5 Hz, 1H), 3.74-3.69 (m, 3H), 3.66 (d, J=12.6 Hz, 5H), 3.60 (q, J=1.7 Hz, 4H), 2.55-2.45 (m, 7H), 2.35 (d, J=14.2 Hz, 1H), 2.19 (ddd, J=14.1, 8.9, 4.8 Hz, 1H), 1.44 (s, 9H), 0.93 (s, 9H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.72, 172.18, 171.57, 170.95, 150.40, 137.45, 131.18, 129.61, 128.17, 80.62, 77.35, 77.23, 77.03, 76.72, 71.15, 70.51, 70.47, 70.43, 70.32, 67.20, 66.88, 59.87, 58.62, 57.12, 43.48, 36.80, 36.23, 35.10, 34.89, 29.70, 28.10, 26.32, 16.03.

##STR00156##

Tert-butyl ((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl) carbamate (14)

[0072] General Procedure 1. Reaction scale: 30 mg (69.9 μmol, 1.0 eq.) of SM1 and 24.6 mg (76.6 μmol. 1.1 eq.) of L8. Purified by Biotage Isolera (2-15% MeOH/CH.sub.2Cl.sub.2 gradient, 16 CV) to afford 14 as a yellow oil (40 mg, 78%). R$=0.52 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI−) calc'd for C.sub.36H.sub.55N.sub.5O.sub.9S 733.4, found [M+HCOO].sup.− 778.2. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.43 (br. s, 1H), 7.36-7.29 (m, 4H), 7.02 (d, J=8.4 Hz, 1H), 5.11 (d, J=7.5 Hz, 1H), 4.70 (t, J=8.0 Hz, 1H), 4.57-4.45 (m, 3H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.07 (br. s, 1H), 3.70 (t, J=5.7 Hz, 2H), 3.60 (d, J=15.4 Hz, 10H), 3.50 (t, J=5.2 Hz, 2H), 2.52-2.41 (m, 7H), 2.15-2.07 (m, 1H), 1.42 (s, 9H), 0.93 (s, 9H). .sup.13C NMR (101 MHz, CDCl.sub.3) δ 172.10, 171.73, 170.93, 156.08, 150.35, 148.42, 138.23, 131.65, 130.86, 129.47, 128.09, 79.26, 77.37, 77.26, 77.06, 76.74, 70.46, 70.23, 70.12, 70.06, 67.15, 60.40, 58.51, 57.71, 56.69, 43.18, 40.34, 36.64, 36.04, 34.92, 29.70, 28.44, 26.41, 21.06, 16.03, 14.20.

##STR00157##

Tert-butyl(S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaicosanoate (15)

[0073] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1 and 12.7 mg (51.1 μmol. 1.1 eq.) of L9. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 15 as a yellow foam (16.3 mg, 51%). R.sub.f=0.54 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.38H.sub.58N.sub.4O.sub.10S 762.4, found [M+H].sup.+ 763.3.

##STR00158##

Tert-butyl 4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoate (16)

[0074] General Procedure 1. Reaction scale: 10 mg (23.2 μmol, 1.0 eq.) of SM1 and 4.5 mg (25.6 μmol. 1.1 eq.) of L10. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 16 as a white solid (7.5 mg, 53%). R.sub.f=0.54 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.30H.sub.42N.sub.4O.sub.6S 586.3, found [M+H].sup.+ 587.1. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.44 (t, J=6.0 Hz, 1H), 7.37-7.27 (m, 4H), 6.56 (d, J=8.6 Hz, 1H), 4.73 (t, J=8.0 Hz, 1H), 4.47 (d, J=8.6 Hz, 3H), 4.35 (d, J=5.3 Hz, 1H), 4.05 (d, J=11.6 Hz, 1H), 3.58 (dd, J=11.4, 3.5 Hz, 1H), 2.60-2.42 (m, 10H), 1.42 (s, 9H), 0.93 (s, 9H).

##STR00159##

Tert-butyl 6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoate (17)

[0075] General Procedure 1. Reaction scale: 20 mg (46.5 μmol, 1.0 eq.) of SM1 and 10.3 mg (51.1 μmol. 1.1 eq.) of L11. Purified by pTLC (8% MeOH/CH.sub.2Cl.sub.2) to afford 17 as a white solid (24.6 mg, 86%). R.sub.f=0.55 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.32H.sub.46N.sub.4O.sub.6S 614.3, found [M+H].sup.+ 615.1. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.60 (s, 1H), 7.27 (t, J=4.6 Hz, 5H), 6.30 (d, J=8.7 Hz, 1H), 4.63 (t, J=8.0 Hz, 1H), 4.51-4.40 (m, 3H), 4.29-4.20 (m, 1H), 4.00 (d, J=11.4 Hz, 1H), 3.60-3.48 (m, 2H), 2.43 (s, 4H), 2.14-2.08 (m, 4H), 1.51 (m, 4H), 1.34 (s, 9H), 0.85 (s, 9H).

##STR00160##

Tert-butyl (7-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl) carbamate (18)

[0076] The protocols are the same as that described in Dolle et al., *J. Med. Chem.* 2021, 64, 15, 10682-10710.

##STR00161##

Methyl 8-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-

yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoate (19)

[0077] General Procedure 1. Reaction scale: 10.0 mg (23.2 μmol, 1.0 eq.) of SM1 and 4.6 μL (25.6 μmol. 1.1 eq.) of L13. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 19 as a white solid (10.0 mg, 72%). R.sub.f=0.61 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.31H.sub.4N.sub.4O.sub.6S 600.3, found [M+H].sup.+ 601.1. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.68 (s, 1H), 7.40-7.29 (m, 5H), 6.15 (d, J=8.7 Hz, 1H), 4.71 (t, J=7.9 Hz, 1H), 4.61-4.46 (m, 3H), 4.33 (dd, J=14.9, 5.2 Hz, 1H), 4.15-4.03 (m, 1H), 3.64 (s, 3H), 2.51 (s, 3H), 2.27 (d, J=7.5 Hz, 2H), 2.19-2.15 (m, 2H), 1.63-1.56 (m, 4H), 1.31-1.25 (m, 7H), 0.93 (s, 9H).

##STR00162##

Methyl 10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoate (20)

[0078] General Procedure 1. Reaction scale: 10.0 mg (23.2 μmol, 1.0 eq.) of SM1 and 5.53 mg (25.6 μmol. 1.1 eq.) of L16. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 20 as a white solid (10.3 mg, 71%). R.sub.f=0.64 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.33H.sub.48N.sub.4O.sub.6S 628.3, found [M+H].sup.+ 629.1. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.40-7.28 (m, 5H), 6.14 (d, J=8.7 Hz, 1H), 4.71 (t, J=7.9 Hz, 1H), 4.60-4.45 (m, 3H), 4.33 (dd, J=15.0, 5.2 Hz, 1H), 4.15-4.06 (m, 1H), 3.65 (d, J=4.8 Hz, 4H), 2.51 (s, 3H), 2.30-2.23 (m, 2H), 2.20-2.12 (m, 3H), 1.59 (d, J=9.7 Hz, 4H), 1.29-1.26 (m, 9H), 0.92 (s, 9H).

##STR00163##

N-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamide (21)

[0079] General Procedure 3 then 2. Reaction scale: 13.0 mg (29.1 μmol) of 1 and 8.1 mg (32.0 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 21 as a white foam (7.5 mg, 38%). R/=0.63 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.37H.sub.39N.sub.5O.sub.8 681.3, found [M+H].sup.+ 682.0. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.10 (s, 1H), 7.72 (d, J=1.1 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.38 (s, 3H), 7.26 (d, J=1.4 Hz, 4H), 7.20-7.16 (m, 4H), 4.99-4.92 (m, 2H), 4.63 (d, J=2.9 Hz, 2H), 3.78 (p, J=1.5 Hz, 2H), 3.62 (t, J=5.1 Hz, 6H), 3.46 (d, J=4.8 Hz, 3H), 2.85-2.66 (m, 7H), 1.60 (m 3H). .sup.13C NMR (101 MHZ, CDCl.sub.3) δ 171.14, 169.63, 168.26, 166.95, 166.63, 165.95, 154.60, 142.10, 136.96, 133.65, 128.63, 127.86, 127.18, 119.69, 118.25, 117.39, 77.34, 77.23, 77.02, 77.02, 76.71, 75.90, 68.90, 68.19, 67.36, 60.41, 51.97, 51.51, 49.24, 45.85, 41.78, 39.37, 33.38, 31.29, 29.71, 29.37, 22.91, 14.21, 14.12.

##STR00164##

N-(2-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamide (22)

[0080] General Procedure 3 then 2. Reaction scale: 19.9 mg (36.3 μmol, 1 eq) of 2 and 10.1 mg (39.9 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 22 as a white foam (7.5 mg, 38%). R.sub.f=0.61 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.39H.sub.43N.sub.5O.sub.9 725.3, found [M+H].sup.+ 726.2. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 9.10 (s, 1H), 7.72 (dd, J=8.4, 7.4 Hz, 1H), 7.63 (t, J=4.9 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.38 (m, 3H), 7.26 (m, 4H), 7.20-7.16 (m, 4H), 4.99-4.92 (m, 2H), 4.63 (s, 2H), 3.80 (td, J=6.9, 1.6 Hz, 3H), 3.78-3.62 (m, 13H), 3.46 (d, J=4.8 Hz, 3H), 2.85-2.66 (m, 5H). .sup.13C NMR (101 MHZ, CDCl.sub.3) δ 171.14, 169.63, 168.26, 166.95, 166.63, 165.95, 154.60, 142.10, 136.96, 133.65, 128.63, 127.86, 127.18, 119.69, 118.25, 117.39, 77.34, 77.23, 77.02, 77.02, 76.71, 75.90, 68.90, 68.19, 67.36, 60.41, 51.97, 51.51, 49.24, 45.85, 41.78, 39.37, 33.38, 31.29, 29.71, 29.37, 22.91, 14.21, 14.12.

##STR00165##

N-(2-(2-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy) ethoxy)ethyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamide (23)

[0081] General Procedure 3 then 2. Reaction scale: 16.11 mg (30.08 μmol, 1.0 eq) of 3 and 8.35 mg (33.09 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (8% MeOH/DCM) to afford 23 as a white foam (17.3 mg, 75%). R.sub.f=0.58 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.41H.sub.47N.sub.5O.sub.10 769.3, found [M+H].sup.+ 770.1. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.08 (s, 1H), 7.73 (dd, J=8.4, 7.3 Hz, 1H), 7.62 (t, J=5.2 Hz, 1H), 7.54 (d, J=7.3 Hz, 1H), 7.40 (d, J=7.0 Hz, 4H), 7.30-7.24 (m, 4H), 7.23-7.14 (m, 3H), 4.98-4.89 (m, 1H), 4.64 (s, 2H), 4.22 (s, 1H), 3.78 (t, J=6.8 Hz, 2H), 3.67-3.51 (m, 14H), 3.49-3.42 (m, 2H), 2.91-2.66 (m, 4H), 2.60 (t, J=6.8 Hz, 2H), 2.36 (q, J=5.4 Hz, 4H). .sup.13C NMR (101 MHz, CDCl.sub.3) δ 169.3, 168.1, 166.8, 166.7, 165.8 (2C), 154.5, 142.2 (2C), 136.9, 128.6 (4C), 127.9 (4C), 127.2 (2C), 119.3, 117.3, 77.4, 77.0, 76.7, 75.9, 70.4, 70.3, 70.3, 69.5, 67.9, 67.2, 52.0, 51.5, 49.3, 41.8, 33.3, 31.5, 29.7, 29.4, 22.7.

##STR00166##

N-(15-(4-benzhydrylpiperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamide (24)

[0082] General Procedure 3 then 2. Reaction scale: 15.0 mg (25.9 μmol, 1.0 eq) of 5 and 7.2 mg (28.5 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 24 as a white foam (12.7 mg, 61%). R/=0.73 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.43H.sub.51N.sub.5O.sub.11 813.4, found [M+H].sup.+ 814.1. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.08 (s, 1H), 7.73 (dd, J=8.4, 7.3 Hz, 1H), 7.62 (t, J=5.2 Hz, 1H), 7.54 (d, J=7.3 Hz, 1H), 7.40 (d, J=7.0 Hz, 4H), 7.30-7.24 (m, 4H), 7.23-7.14 (m, 3H), 4.98-4.89 (m, 1H), 4.64 (s, 2H), 4.22 (s, 1H), 3.78 (t, J=6.8 Hz, 2H), 3.67-3.51 (m, 18H), 3.49-3.42 (m, 2H), 2.91-2.66 (m, 4H), 2.60 (t, J=6.8 Hz, 2H), 2.36 (q, J=5.4 Hz, 4H). .sup.13C NMR (101 MHz, CDCl.sub.3) δ 171.11, 169.32, 168.16, 166.82, 154.46, 142.20, 136.93, 133.72, 128.61, 127.87, 127.15, 119.28, 117.27, 77.34, 77.23, 77.03, 76.71, 75.94, 70.59, 70.45, 70.35, 70.30, 69.47, 67.88, 67.26, 60.41, 52.01, 51.55, 49.34, 45.76, 41.73, 39.12, 33.41, 31.47, 29.71, 22.70, 14.21.

##STR00167##

N-(6-(4-benzhydrylpiperazin-1-yl)-6-oxohexyl)-2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamide (25)

[0083] General Procedure 3 then 2. Reaction scale: 17 mg (38.17 μmol, 1.0 eq) of 6 and 10.4 mg (41.21 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (8% MeOH/DCM) to afford 25 as a white foam (25 mg, 99%). R.sub.f=0.67 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.38H.sub.41N.sub.5O.sub.7 679.3, found [M+H].sup.+ 680.0. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.08 (br. s, 1H), 7.73 (dd, J=8.4, 7.4 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.39 (d, J=7.3 Hz, 4H), 7.26 (s, 4H), 7.20-7.17 (m, 3H), 4.98-4.93 (m, 1H), 4.63 (q, J=3.1 Hz, 2H), 4.21 (s, 1H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.48-3.27 (m, 7H), 2.90-2.76 (m, 1H), 2.73-2.58 (m, 3H), 2.36 (s, 3H), 1.60-1.30 (m, 6H). .sup.13C NMR (101 MHZ, CDCl.sub.3) δ 171.96, 171.31, 168.33, 166.84, 166.60, 166.27, 154.81, 142.14, 137.04, 133.58, 128.62, 127.87, 127.17, 120.24, 118.51, 117.61, 77.35, 77.23, 77.03, 76.81, 76.71, 75.96, 68.67, 52.11, 51.63, 49.35, 45.89, 41.83, 39.21, 33.04, 31.41, 29.71, 28.87, 26.78, 25.35, 22.86.

##STR00168##

(2S,4R)-1-((S)-2-(3-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (26)

[0084] General Procedure 3 then 2. Reaction scale: 10.2 mg (17.8 μmol, 1.0 eq) of 8 and 4.93 mg (19.52 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 26 as a pale yellow solid (6 mg, 42%). R.sub.f=0.46 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.45H.sub.56N.sub.6O.sub.6S 808.4, found [M+H].sup.+ 809.3. .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.68 (s, 1H), 7.47-7.30 (m, 11H), 7.20 (d, J=7.4 Hz, 2H), 6.95 (d, J=8.3 Hz, 1H), 4.73-4.65 (m, 1H), 4.59-4.52 (m, 1H), 4.45 (d, J=8.6 Hz, 1H), 4.29 (dd, J=15.1, 5.1 Hz, 1H), 3.79-3.49 (m, 10H), 2.52 (s, 7H), 2.03 (d, J=15.4 Hz, 1H), 1.61 (m, 9H), 0.90 (s, 9H).

##STR00169##

(2S,4S)-1-((S)-2-(3-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (27)

[0085] General Procedure 3 then 2. Reaction scale: 11.4 mg (19.9 μmol, 1 eq.) of 9 and 5.6 mg (22 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 27 as a white solid (6 mg, 42%). R.sub.f=0.46 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.45H.sub.56N.sub.6O.sub.6S 808.4, found [M+H].sup.+ 809.2. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.41-7.26 (m, 12H), 7.20 (d, J=7.4 Hz, 2H), 4.68-4.56 (m, 2H), 4.48 (d, J=8.9, 2H), 4.25 (dd, J=15.1, 5.1 Hz, 1H), 3.92 (dd, J=10.6, 4.1 Hz, 1H), 3.81-3.65 (m, 7H), 3.59 (s, 1H), 3.45 (d, J=5.3 Hz, 1H), 2.66-2.55 (m, 2H), 2.52 (s, 4H), 2.18 (dd, J=8.9, 5.4 Hz, 2H), 1.56 (s, 9H), 0.89 (s, 9H).

##STR00170##

(2S,4R)-1-((S)-2-(3-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (28)

[0086] General Procedure 3 then 2. Reaction scale: 7.5 mg (11.1 μmol, 1.0 eq) of 10 and 3.1 mg (12.2 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 28 as a white foam (6.4 mg, 25%). R/=0.46 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.47H.sub.60N.sub.6O.sub.7S 852.4, found [M+H].sup.+ 853.3. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.52 (t, J=6.0 Hz, 1H), 7.41-7.35 (m, 5H), 7.34 (s, 3H), 7.28 (d, J=7.1 Hz, 3H), 7.21-7.16 (m, 2H), 7.03 (d, J=8.3 Hz, 1H), 4.71 (t, J=8.1 Hz, 1H), 4.57 (dd, J=15.0, 6.6 Hz, 1H), 4.50-4.42 (m, 2H), 4.32 (dd, J=15.0, 5.2 Hz, 1H), 4.20 (s, 1H), 4.12-4.06 (m, 1H), 3.80-3.65 (m, 5H), 3.59 (ddd, J=10.3, 6.3, 4.3 Hz, 7H), 3.49-3.41 (m, 3H), 2.56 (td, J=6.8, 1.3 Hz, 2H), 2.51 (s, 4H), 2.46 (q, J=4.6 Hz, 3H), 2.34 (dt, J=10.7, 5.2 Hz, 4H), 0.93 (s, 9H). .sup.13C NMR (101 MHz, CDCl.sub.3) δ 172.14, 171.70, 170.88, 169.32, 150.27, 148.45, 142.08, 138.26, 131.63, 130.86, 129.47, 128.63, 128.10, 127.83, 127.19, 77.34, 77.22, 77.02, 76.70, 75.92, 70.48, 70.38, 70.13, 67.38, 67.16, 58.34, 57.70, 56.67, 51.97, 51.50, 45.75, 43.18, 41.72, 36.77, 35.95, 34.85, 33.38, 26.41, 16.08.

##STR00171##

(2S,4S)-1-((S)-2-(3-(2-(3-(4-benzhydrylpiperazin-1-yl)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (29)

[0087] General Procedure 3 then 1. Reaction scale: 16.2 mg (26.2 μmol, 1.0 eq) of 11 and 7.3 mg (28.8 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 29 as a white foam (19.6 mg, 88%). R/=0.51 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.47H.sub.60N.sub.6O.sub.7S 852.4, found [M+H].sup.+ 853.3. .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.59 (s, 1H), 7.44 (t, J=6.0 Hz, 1H), 7.32-7.28 (m, 4H), 7.26 (d, J=1.2 Hz, 4H), 7.21-7.17 (m, 5H), 7.12-7.08 (m, 2H), 6.98 (d, J=8.4 Hz, 1H), 4.63 (t, J=8.1 Hz, 1H), 4.48 (dd, J=15.0, 6.7 Hz, 1H), 4.43-4.38 (m, 2H), 4.23 (dd, J=15.0, 5.3 Hz, 1H), 4.12 (s, 1H), 4.06-3.99 (m, 1H), 3.69-3.57 (m, 5H), 3.51 (dddd, J=15.6, 12.6, 8.7, 5.3 Hz, 7H), 3.36 (dd, J=6.2, 4.0 Hz, 2H), 2.48 (td, J=6.9, 1.8 Hz, 2H), 2.43 (s, 3H), 2.38 (t, J=5.6 Hz, 2H), 2.26 (dt, J=17.0, 5.2 Hz, 4H), 0.85 (s, 9H). .sup.13C NMR (151 MHz, CDCl.sub.3) δ 172.16, 171.69, 171.20, 170.96, 169.38, 150.32, 148.43, 142.09, 138.31, 131.67, 130.83, 130.08, 129.47, 129.41, 128.64, 128.09, 127.84, 127.21, 77.26, 77.04, 76.83, 75.93, 70.47, 70.38, 70.14, 67.37, 67.18, 60.42, 60.37, 58.45, 57.73, 56.71, 56.00, 51.98, 51.51, 45.76, 43.18, 41.74, 36.74, 36.06, 34.94, 33.38, 31.94, 29.72, 29.38, 26.42, 22.71, 21.08, 20.80, 16.07, 14.22, 14.14.

##STR00172##

(2S,4R)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (30)

[0088] General Procedure 3 then 1. Reaction scale: 21.9 mg (33.0 μmol, 1.0 eq) of 12 and 9.2 mg (36.35 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 30 as a white foam (10.5 mg, 35%). R/=0.52 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.49H.sub.64N.sub.6O.sub.8S 896.5, found [M+H].sup.+ 897.3. .sup.1H

NMR (600 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.47 (t, J=6.0 Hz, 1H), 7.40-7.38 (m, 4H), 7.36-7.33 (m, 4H), 7.29-7.25 (m, 5H), 7.21-7.16 (m, 2H), 7.02 (d, J=8.4 Hz, 1H), 4.72 (t, J=8.1 Hz, 1H), 4.56 (dd, J=15.0, 6.7 Hz, 1H), 4.50-4.46 (m, 2H), 4.32 (dd, J=15.0, 5.3 Hz, 1H), 4.21 (s, 1H), 4.10 (d, J=11.0 Hz, 1H), 3.74 (tt, J=6.1, 3.1 Hz, 3H), 3.69 (dddd, J=9.8, 8.5, 5.0, 2.7 Hz, 3H), 3.60 (dd, J=13.1, 2.0 Hz, 9H), 3.45 (t, J=5.1 Hz, 2H), 2.57 (td, J=6.8, 4.1 Hz, 2H), 2.51 (s, 3H), 2.47 (ddd, J=7.2, 4.7, 3.1 Hz, 2H), 2.35 (dt, J=16.0, 4.1 Hz, 4H), 0.93 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.19, 171.75, 170.91, 169.39, 150.32, 148.45, 142.12, 138.26, 131.65, 130.90, 130.88, 129.49, 128.64, 128.12, 127.85, 127.20, 77.26, 77.14, 77.04, 76.83, 75.94, 70.45, 70.37, 70.30, 70.19, 70.12, 67.17, 67.14, 58.39, 57.76, 56.70, 51.96, 51.52, 50.82, 45.73, 43.20, 41.74, 36.61, 35.98, 34.84, 33.71, 33.38, 30.37, 30.18, 29.72, 26.42, 16.08.
##STR00173##
(2S,4S)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (31)
[0089] General Procedure 3 then 1. Reaction scale: 10 mg (15.1 μmol, 1.0 eq) of 13 and 4.2 mg (16.6 μmol, 1.1 eq) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 30 as a white foam (10.5 mg, 35%). R_f=0.57 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI−) calc'd for $C_{49}H_{64}N_6O_8S$ 896.5, found $[M+HCOO]^-$ 941.1. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.68 (s, 1H), 7.46 (d, J=6.2 Hz, 1H), 7.42-7.37 (m, 4H), 7.34 (d, J=1.7 Hz, 4H), 7.29-7.24 (m, 5H), 7.21-7.16 (m, 2H), 7.05 (d, J=8.3 Hz, 1H), 4.72 (t, J=8.0 Hz, 1H), 4.57 (dd, J=15.0, 6.6 Hz, 1H), 4.47 (d, J=8.8 Hz, 2H), 4.32 (dd, J=15.0, 5.2 Hz, 1H), 4.21 (s, 1H), 4.14-4.09 (m, 1H), 3.72 (dt, J=17.1, 4.5 Hz, 5H), 3.60 (d, J=8.8 Hz, 9H), 3.45 (t, J=5.0 Hz, 2H), 2.61-2.54 (m, 2H), 2.51 (s, 3H), 2.46 (d, J=5.1 Hz, 2H), 2.35 (dt, J=10.0, 5.0 Hz, 4H), 0.93 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.85, 172.00, 171.51, 169.28, 150.36, 148.50, 142.08, 129.73, 129.55, 128.63, 128.24, 128.11, 127.92, 127.83, 127.57, 127.19, 77.22, 77.01, 76.80, 75.94, 71.16, 70.51, 70.44, 70.32, 70.23, 67.25, 67.21, 59.82, 58.57, 57.01, 55.98, 51.97, 51.52, 45.72, 43.42, 41.70, 36.79, 35.22, 34.97, 33.69, 33.43, 31.92, 30.16, 30.03, 29.70, 29.65, 29.60, 29.47, 29.36, 29.32, 27.21, 26.70, 26.31, 25.53, 23.17, 22.69, 21.49, 16.06, 14.18, 14.12.
##STR00174##
(2S,4R)-1-((S)-19-(4-benzhydrylpiperazin-1-yl)-2-(tert-butyl)-4,19-dioxo-7,10,13,16-tetraoxa-3-azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (32)
[0090] General Procedure 3 then 2. Reaction scale: 15.3 mg (21.65 μmol) of 15 and 6.01 mg (23.81 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 32 as a white solid (7.0 mg, 35%). R/=0.47 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{51}H_{68}N_6O_9S$ 940.5, found $[M+H]^+$ 941.4. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.68 (s, 1H), 7.43-7.31 (m, 9H), 7.27 (d, J=5.3 Hz, 5H), 7.23-7.15 (m, 2H), 7.03 (d, J=8.2 Hz, 1H), 4.73 (t, J=8.0 Hz, 1H), 4.57 (dd, J=14.9, 6.6 Hz, 1H), 4.52-4.43 (m, 2H), 4.33 (dd, J=14.9, 5.2 Hz, 1H), 4.22 (s, 1H), 4.14 (d, J=7.1 Hz, 1H), 3.78-3.70 (m, 5H), 3.66-3.58 (m, 15H), 3.46 (dd, J=11.2, 6.2 Hz, 2H), 2.61-2.57 (m, 2H), 2.52 (d, J=1.7 Hz, 3H), 2.50-2.46 (m, 2H), 2.39-2.30 (m, 4H), 0.93 (s, 9H).
##STR00175##
(2S,4R)-1-((S)-2-(4-(4-benzhydrylpiperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (33)
[0091] General Procedure 3 then 2. Reaction scale: 6.7 mg (12.63 μmol) of 16 and 3.5 mg (13.89 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 33 as a pale yellow solid (4.2 mg, 44%). R_f=0.58 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI+) calc'd for $C_{43}H_{52}N_6O_5S$ 764.4, found $[M+H]^+$ 766.1. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.42-7.32 (m, 12H), 7.27 (s, 3H), 7.20 (d, J=7.3 Hz, 2H), 4.75 (t, J=8.1 Hz, 2H), 4.57 (dd, J=14.9, 6.6 Hz, 1H), 4.49 (d, J=8.3 Hz, 2H), 4.33 (dd, J=14.9, 5.1 Hz, 1H), 4.22 (s, 1H), 4.12 (d, J=6.8 Hz, 1H), 3.65 (d, J=7.6 Hz, 2H), 3.56 (d, J=13.7 Hz, 4H), 3.49 (s, 1H), 3.41 (d, J=5.5 Hz, 3H), 2.51 (s, 3H), 2.35 (d, J=4.9 Hz, 4H), 0.92 (s, 9H).

##STR00176##
(2S,4R)-1-((S)-2-(6-(4-benzhydrylpiperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (34)

[0092] General Procedure 3 then 2. Reaction scale: 9 mg (16.11 μmol) of 17 and 4.47 mg (17.72 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 21 as a pale yellow solid (3.3 mg, 26%). R.sub.f=0.54 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.45H.sub.56N.sub.6O.sub.5S 792.4, found [M+H].sup.+ 793.9. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.73 (dd, J=8.4, 7.4 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, J=7.3 Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, J=7 Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.60-2.35 (m, 13H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

##STR00177##
(2S,4R)-1-((S)-2-(8-(4-benzhydrylpiperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (35)

[0093] General Procedure 5 then 2. Reaction scale: 8.5 mg (14.49 μmol) of 19 and 4.02 mg (15.94 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 23 as a pale yellow solid (6.4 mg, 54%). R.sub.f=0.53 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.47H.sub.60N.sub.6O.sub.5S 820.4, found [M+H].sup.+ 821.0. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.73 (dd, J=8.4, 7.4 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, J=7.3 Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, J=7 Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.60-2.35 (m, 17H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

##STR00178##
(2S,4R)-1-((S)-2-(10-(4-benzhydrylpiperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (36)

[0094] General Procedure 5 then 2. Reaction scale: 10 mg (16.27 μmol) of 20 and 4.52 mg (17.89 μmol) of benzhydrylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 36 as a pale yellow solid (8.5 mg, 62%). R/=0.62 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.49H.sub.64N.sub.6O.sub.5S 848.5, found [M+H].sup.+ 849.3. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.73 (dd, J=8.4, 7.4 Hz, 1H), 7.54 (dd, J=7.4, 0.6 Hz, 1H), 7.50 (br. t, 2H), 7.39 (d, J=7.3 Hz, 5H), 7.30 (s, 1H), 7.25 (br. t, 2H), 7.20-7.17 (m, 3H), 4.72 (t, 2H), 4.58 (s, 1H) 4.50 (s, 1H), 4.60-4.30 (dd, J=7 Hz, 1H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.60-2.35 (m, 21H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

##STR00179##
(2S,4R)-1-((S)-16-((R)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (37)

[0095] General Procedure 3 then 1. Reaction scale: 15 mg (22.6 μmol, 1.0 eq.) of 12 and 8.2 mg (24.9 μmol, 1.1 eq.) of (4R)-benzhydrylpiperazine. Purified by pTLC (8% MeOH/CH.sub.2Cl.sub.2) to afford 37 as an off-white solid (8.3 mg, 38%). R.sub.f=0.55 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.55H.sub.70N.sub.6O.sub.8S [M+H].sup.+: 972.5, found 974.0; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.34 (s, 13H), 7.19 (s, 6H), 7.01 (d, J=8.3 Hz, 1H), 4.73 (s, 1H), 4.47 (d, J=8.3 Hz, 2H), 4.28-4.02 (m, 3H), 3.79-3.03 (m, 19H), 2.50 (s, 8H), 2.17-2.03 (m, 4H), 0.91 (s, 9H). Note: presence of rotamers.

##STR00180##
(2S,4R)-1-((S)-16-((S)-4-benzhydryl-2-phenylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-

carboxamide (38)

[0096] General Procedure 1 and 3. Reaction scale: 21.6 mg (30.0 μmol, 1.00 equiv) of 12 and 14.9 mg (35.0 μmol, 1.17 equiv) of (4S)-phenylbenzhydrylpiperazine derivative. Purified by PTLC (10% MeOH/EtOAc) to afford 38 as a clear oil (5.5 mg, 19%). R.sub.f=0.47 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for C.sub.55H.sub.70N.sub.6O.sub.8S [M+2H].sup.2+: 487.2, found 487.1; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.41-7.27 (m, 10H), 7.23-7.13 (m, 9H), 6.34 (d, J=8.5 Hz, 1H), 5.76 (br s, 1H), 4.59 (m, 1H), 4.51-4.46 (m, 2H), 4.27-4.18 (m, 3H), 3.83-3.30 (m, 19H), 2.91 (m, 1H), 2.60-2.46 (m, 6H), 2.41 (m, 2H), 2.31-2.19 (m, 2H), 2.07 (m, 1H), 0.87-0.76 (m, 9H). Note: presence of rotamers.

##STR00181##

(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-(4-(phenyl(pyridin-2-yl)methyl) piperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (39)

[0097] General Procedure 3 then 1. Reaction scale: 20 mg (30.2 μmol, 1.0 eq.) of 12 and 8.4 mg (33.2 μmol, 1.1 eq.) of phenylpyridin-methylpiperazine. Purified by pTLC (8% MeOH/CH.sub.2Cl.sub.2) to afford 39 as white foam (18.2 mg, 67%). R.sub.f=0.62 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.48H.sub.63N.sub.7O.sub.8S [M+H].sup.+: 897.5, found 898.7; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.59 (s, 1H), 8.44-8.40 (m, 1H), 7.55 (td, J=7.7, 1.8 Hz, 1H), 7.45 (dd, J=7.9, 1.0 Hz, 1H), 7.43-7.40 (m, 1H), 7.38 (dt, J=8.1, 1.5 Hz, 2H), 7.28-7.24 (m, 4H), 7.20 (dd, J=8.3, 6.9 Hz, 2H), 7.18 (s, 1H), 7.14-7.10 (m, 1H), 7.03 (ddt, J=7.3, 4.9, 1.2 Hz, 1H), 6.95 (dd, J=8.3, 1.6 Hz, 1H), 4.63 (td, J=8.0, 1.4 Hz, 1H), 4.47 (dd, J=15.0, 6.7 Hz, 1H), 4.43-4.38 (m, 2H), 4.34 (s, 1H), 4.24 (dd, J=15.0, 5.3 Hz, 1H), 4.06-3.99 (m, 2H), 3.68-3.59 (m, 4H), 3.55-3.48 (m, 12H), 3.41-3.37 (m, 2H), 2.49 (qd, J=6.8, 1.6 Hz, 2H), 2.42 (s, 3H), 2.39 (ddd, J=8.7, 6.4, 3.9 Hz, 3H), 2.28-2.23 (m, 2H), 0.85 (s, 9H). .sup.13C NMR (151 MHz, CDCl.sub.3) δ 172.20, 171.74, 171.19, 170.95, 169.49, 161.47, 150.33, 149.30, 148.43, 140.48, 138.28, 136.93, 131.66, 130.85, 129.48, 128.71, 128.23, 128.11, 127.60, 122.29, 122.25, 77.54, 77.30, 77.26, 77.05, 76.92, 76.84, 70.41, 70.32, 70.27, 70.25, 70.15, 70.10, 67.16, 67.14, 60.42, 58.44, 57.75, 56.86, 56.70, 56.00, 51.97, 51.55, 45.59, 43.18, 41.59, 36.58, 36.06, 36.03, 34.87, 33.34, 31.94, 29.71, 29.37, 29.33, 26.66, 26.42, 22.70, 21.07, 19.25, 18.17, 16.06, 14.21, 14.14.

##STR00182##

(2S,4R)-1-((S)-16-((S)-4-benzhydryl-3-methylpiperazin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (40)

[0098] General Procedure 1 and 3. Reaction scale: 18.4 mg (26.0 μmol, 1.00 equiv) of 12 and 23.6 mg (64.0 μmol, 2.46 equiv) of (4S)-methyl benzhydrylpiperazine. Purified by PTLC (10% MeOH/EtOAc) to afford 40 as a clear oil (12.5 mg, 53%). R.sub.f=0.39 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for C.sub.50H.sub.68N.sub.6O.sub.8S [M+2H].sup.2+: 456.2, found 456.4; .sup.1H NMR (600 MHz, CDCl.sub.3) δ 8.67 (s, 1H), 7.44-7.41 (m, 2H), 7.39-7.32 (m, 7H), 7.28-7.24 (m, 5H), 7.21-7.13 (m, 3H), 4.72 (m, 1H), 4.55-4.46 (m, 3H), 4.35-4.31 (m, 1H), 4.16 (m, 1H), 4.10 (d, J=11.6 Hz, 1H), 3.76-3.54 (m, 16H), 3.46-3.37 (m, 1H), 3.00-2.91 (m, 2H), 2.63-2.39 (m, 12H), 2.16 (m, 1H), 0.94-0.91 (m, 9H).

##STR00183##

2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-N-(2-(2-(2-(3-oxo-3-(4-tosylpiperazin-1-yl) propoxy) ethoxy) ethoxy)ethyl) acetamide (41)

[0099] General Procedure 3 then 1. Reaction scale: 12 mg (22.4 μmol, 1.0 eq.) of 3 and 5.9 mg (24.7 μmol, 1.1 eq.) of tosylpiperazine. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 41 as a clear oil (4.8 mg, 28%). R.sub.f=0.65 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.35H.sub.43N.sub.5O.sub.12S.sub.2 757.3, found [M+H].sup.+ 758.2. .sup.1H NMR (600 MHz, CDCl.sub.3) δ 8.95 (s, 1H), 7.74 (dd, J=8.4, 7.4 Hz, 1H), 7.63 (s, 1H), 7.62-7.60 (m, 3H), 7.55 (d, J=7.3 Hz, 1H), 7.33 (d, J=8.0 Hz, 3H), 7.19 (d, J=8.4 Hz, 1H), 4.94 (dd, J=12.4, 5.4 Hz, 1H), 3.73 (td, J=6.6, 1.4 Hz, 3H), 3.68-3.66 (m, 3H), 3.64 (dd, J=4.7, 2.7 Hz, 2H), 3.01-2.95 (m, 5H), 2.86-2.68 (m, 4H), 2.43 (s, 4H), 1.27 (s, 9H). .sup.13C NMR (151 MHZ, CDCl.sub.3) δ 171.09, 169.63, 168.16, 166.83, 166.67, 165.83, 154.47, 144.10, 137.01, 133.69, 132.33, 129.86, 127.79, 119.35, 118.06, 117.33, 77.24, 77.03, 76.82, 70.41, 70.28, 70.26, 70.24, 69.49, 67.91, 67.12, 60.42, 56.00, 53.45, 49.32, 46.16, 45.82, 45.10, 41.25, 40.90, 39.11, 38.63, 35.95, 33.71, 33.39, 31.94, 31.92, 31.47, 30.19, 29.79, 29.72, 29.54, 29.49, 29.38, 29.34, 29.25, 27.23, 25.55, 23.19, 22.70, 21.57, 21.08, 14.22, 14.14, 14.07.

##STR00184##

2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-N-(6-oxo-6-(4-tosylpiperazin-1-yl) hexyl) acetamide (42)

[0100] General Procedure 3 then 1. Reaction scale: 8 mg (18.0 μmol, 1.0 eq.) of 3 and 4.8 mg (19.8 μmol, 1.1 eq.) of tosylpiperazine. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 42 as an off-white solid (7.1 mg, 28%). R/=0.75 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.32H.sub.37N.sub.5O.sub.9S.sub.2 667.2, found [M+H].sup.+ 668.3. .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 9.38 (s, 1H), 7.75 (d, J=1.1 Hz, 1H), 7.63-7.61 (m, 2H), 7.59 (s, 1H), 7.56 (d, J=7.4 Hz, 1H), 7.34 (s, 2H), 7.20 (d, J=8.3 Hz, 1H), 4.94 (dt, J=8.0, 4.0 Hz, 1H), 4.79-4.55 (m, 4H), 3.74-3.65 (m, 5H), 3.53 (t, J=5.2 Hz, 3H), 3.41-3.32 (m, 5H), 2.97 (dd, J=12.1, 5.3 Hz, 5H), 2.87 (d, J=3.1 Hz, 1H), 2.79-2.68 (m, 3H), 2.44 (s, 4H). .sup.13C NMR (151 MHz, CDCl.sub.3) δ 171.94, 171.21, 168.30, 166.82, 166.54, 166.29, 154.79, 144.17, 137.11, 133.55, 132.23, 129.90, 127.78, 120.27, 118.47, 117.67, 77.24, 77.03, 76.82, 68.66, 56.00, 49.31, 46.19, 45.87, 45.05, 40.93, 39.13, 33.71, 32.95, 31.95, 31.39, 30.18, 29.72, 29.38, 29.34, 28.84, 27.23, 26.65, 25.12, 23.19, 22.84, 22.75, 22.71, 21.58, 14.21, 14.14.

##STR00185##

(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(4-tosylpiperazin-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (43)

[0101] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μmol) of 12 and 2.39 mg (9.96 μmol) of tosylpiperazine. Purified by pTLC (10% MeOH/DCM) to afford 43 as a white solid (3.3 mg, 41%). R.sub.f=0.65 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.43H.sub.60N.sub.6O.sub.10S.sub.2 884.4, found [M+HCOO].sup.− 931.2. .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.64-7.60 (m, 2H), 7.46-7.41 (m, 1H), 7.36-7.30 (m, 6H), 6.95 (d, J=8.3 Hz, 1H), 4.73 (t, J=8.1 Hz, 1H), 4.62-4.56 (m, 1H), 4.51-4.45 (m, 2H), 4.35-4.29 (m, 1H), 3.77-3.64 (m, 7H), 3.65-3.51 (m, 13H), 3.06-2.88 (m, 4H), 2.54-2.50 (m, 6H), 2.15-2.11 (m, 1H), 0.94 (s, 12H). .sup.13C NMR (151 MHZ, CDCl.sub.3) δ 172.15, 171.80, 170.90, 169.80, 150.31, 148.47, 144.15, 138.25, 132.46, 132.28, 131.65, 130.89, 129.88, 129.49, 128.82, 128.11, 127.79, 77.24, 77.03, 76.82, 70.39, 70.23, 70.17, 70.13, 70.09, 68.17, 67.20, 67.06, 58.42, 57.78, 56.77, 56.00, 46.13, 45.81, 45.04, 43.21, 40.82, 38.74, 37.11, 36.54, 35.95, 34.76, 33.71, 33.45, 33.41, 32.77, 31.95, 30.37, 30.18, 30.05, 29.72, 29.50, 29.38, 29.34, 28.94, 27.23, 26.72, 26.43, 23.75, 23.19, 23.00, 22.71, 21.58, 16.08, 14.20, 14.14, 14.07, 10.97, 7.49.

##STR00186##

(2S,4R)-1-((S)-3,3-dimethyl-2-(6-oxo-6-(4-tosylpiperazin-1-yl) hexanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (44)

[0102] General Procedure 3 then 1. Reaction scale: 10 mg (17.9 μmol, 1.0 eq.) of 3 and 4.7 mg (19.7 μmol, 1.1 eq.) of tosylpiperazine. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 41 as a white solid (4.0 mg, 29%). R.sub.f=0.65 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.39H.sub.52N.sub.6O.sub.7S.sub.2 780.3, found [M+H].sup.+ 781.8. .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.69 (s, 1H), 7.61 (d, J=8.4 Hz, 2H), 7.37-7.31 (m, 7H), 6.25 (d, J=8.5 Hz, 1H), 4.76-4.71 (m, 2H), 4.59 (dd, J=14.9, 6.7 Hz, 3H), 4.52-4.47 (m, 4H), 4.40-4.26 (m, 5H), 4.15-4.08 (m, 3H), 3.35 (s, 13H), 2.97 (dd, J=7.0, 3.9 Hz, 1H), 0.92 (s, 9H). Note: presence of rotamers.

##STR00187##

N-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-(4,5-diphenyloxazol-2-yl) propenamide (45)

[0103] General Procedure 1 and 3. Reaction scale: 13.4 mg (22.0 μmol, 1.00 equiv) of 4 and 7.6 mg (26.0 μmol, 1.20 equiv) of oxaprozin. Purified by PTLC (10% MeOH/EtOAc) to afford 45 as a clear oil (9.7 mg, 57%). R_f=0.31 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $C_{41}H_{44}N_5O_{11}$ $[M+H]^+$: 782.3, found 782.1; ^1H NMR (600 MHz, $CDCl_3$) δ 9.27 (br s, 1H), 7.71-7.68 (m, 2H), 7.61-7.59 (m, 2H), 7.55-7.51 (m, 3H), 7.36-7.29 (m, 6H), 7.14 (d, J=7.8 Hz, 1H), 6.93 (m, 1H), 4.92 (dd, J=12.4, 5.5 Hz, 1H), 4.61 (s, 2H), 3.66-3.50 (m, 16H), 3.19 (m, 2H), 2.85-2.81 (m, 1H), 2.78-2.65 (m, 4H), 2.11-2.07 (m, 1H).

##STR00188##

N-(6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) hexyl)-3-(4,5-diphenyloxazol-2-yl) propenamide (46)

[0104] General Procedure 1 and 3. Reaction scale: 8.7 mg (16.0 μmol, 1.00 equiv) of 7 and 7.0 mg (24.0 μmol, 1.50 equiv) of oxaprozin. Purified by PTLC (10% MeOH/CH_2Cl_2) to afford 46 as a white foam (5.7 mg, 50%). LC-MS (ESI−) calc'd for $C_{39}H_{38}N_5O_8$ $[M-H]^-$: 704.3, found 704.0; ^1H NMR (600 MHZ, $CDCl_3$) δ 9.34 (br s, 1H), 7.71 (m, 1H), 7.60-7.58 (m, 2H), 7.55-7.52 (m, 3H), 7.46 (m, 1H), 7.36-7.29 (m, 5H), 7.17 (d, J=8.3 Hz, 1H), 6.36 (m, 1H), 4.95 (m, 1H), 4.62 (m, 2H), 3.37 (m, 1H), 3.29 (m, 2H), 3.21-3.16 (m, 3H), 2.85 (m, 1H), 2.78-2.70 (m, 4H), 2.13-2.09 (m, 1H), 1.55-1.45 (m, 4H), 1.37-1.29 (m, 4H).

##STR00189##

(2S,4R)-1-((S)-2-(tert-butyl)-19-(4,5-diphenyloxazol-2-yl)-4,17-dioxo-7,10,13-trioxa-3,16-diazanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (47)

[0105] General Procedure 3 then 1. Reaction scale: 8 mg (12.6 μmol, 1.0 eq.) of 14 and 4.1 mg (13.9 μmol, 1.1 eq.) of oxaprozin. Purified by pTLC (10% MeOH/CH_2Cl_2) to afford 47 as white powder (6.2 mg, 54%). R_f=0.58 (10% MeOH/CH_2Cl_2, UV-active); LC-MS (ESI−) calc'd for $C_{49}H_{60}N_6O_9S$ 908.4, found $[M+HCOO]^-$ 954.2. ^1H NMR (400 MHZ, $CDCl_3$) δ 8.67 (s, 1H), 7.55 (ddt, J=14.2, 6.2, 1.9 Hz, 5H), 7.37-7.30 (m, 10H), 7.15 (s, 1H), 7.09 (d, J=8.4 Hz, 1H), 4.67 (t, J=8.2 Hz, 1H), 4.56-4.48 (m, 2H), 4.42 (s, 1H), 4.32 (dd, J=15.0, 5.4 Hz, 1H), 4.13-4.07 (m, 1H), 3.74 (q, J=5.2 Hz, 1H), 3.68-3.64 (m, 1H), 3.62-3.53 (m, 11H), 3.43 (tt, J=8.8, 4.7 Hz, 2H), 3.18 (t, J=7.6 Hz, 2H), 2.77 (t, J=7.7 Hz, 2H), 2.46 (t, J=5.6 Hz, 2H), 2.36 (ddd, J=13.1, 8.5, 4.3 Hz, 2H), 2.06 (d, J=14.3 Hz, 3H), 0.94 (s, 9H). ^{13}C NMR (151 MHz, $CDCl_3$) δ 172.16, 171.66, 171.62, 171.11, 162.85, 150.35, 148.44, 145.47, 138.26, 134.85, 133.32, 132.40, 130.85, 129.55, 129.47, 128.87, 128.73, 128.70, 128.66, 128.63, 128.50, 128.43, 128.22, 128.17, 128.12, 128.06, 127.90, 126.51, 126.33, 77.24, 77.03, 76.82, 70.51, 70.40, 70.30, 70.08, 69.92, 68.86, 67.23, 58.55, 57.82, 57.72, 56.87, 56.00, 45.33, 43.28, 43.13, 39.78, 39.46, 36.54, 36.40, 35.82, 34.91, 33.71, 32.74, 31.95, 30.18, 30.06, 29.72, 29.68, 29.63, 29.38, 29.34, 27.69, 27.23, 26.72, 26.43, 26.38, 23.95, 23.20, 22.72, 21.51, 16.05, 14.21, 14.14.

##STR00190##

(2S,4R)-1-((S)-2-(7-(3-(4,5-diphenyloxazol-2-yl) propanamido) heptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (48)

[0106] General Procedure 1 and 3. Reaction scale: 15.2 mg (23.0 μmol, 1.00 equiv) of 18 and 9.9 mg (34.0 μmol, 1.20 equiv) of oxaprozin. Purified by PTLC (10% MeOH/EtOAc) to afford 48 as a clear oil (14.0 mg, 73%). R_f=0.23 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for $C_{47}H_{57}N_6O_6S$ $[M+H]^+$: 833.4, found 833.4; ^1H NMR (600 MHz, $CDCl_3$) δ 8.65 (s, 1H), 7.58-7.49 (m, 5H), 7.36-7.30 (m, 10H), 6.49 (m, 1H), 6.28 (d, J=8.7 Hz, 1H), 4.67 (t, J=8.1 Hz, 1H), 4.55-4.50 (m, 2H), 4.45 (m, 1H), 4.31 (dd, J=15.0, 5.4 Hz, 1H), 4.05 (d, J=11.4 Hz, 1H), 3.58 (dd, J=11.3, 3.6 Hz, 1H), 3.22-3.12 (m, 4H), 2.71 (t, J=7.4 Hz, 2H), 2.49 (s, 3H), 2.38-2.34 (m, 1H), 2.21-2.16 (m, 1H), 2.13-2.07 (m, 2H), 1.60-1.42 (m, 4H), 1.29-

1.20 (m, 5H), 0.93 (s, 9H).
##STR00191##
5-chloro-N-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3-azatetradecan-14-yl)-3-phenyl-1H-indole-2-carboxamide (49)
[0107] General Procedure 1 and 3. Reaction scale: 9.5 mg (18.0 μmol, 1.00 equiv) of 4 and 6.0 mg (22.0 μmol, 1.20 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 49 as a yellow foam (9.2 mg, 75%). R.sub.f=0.55 (10% MeOH/EtOAc, UV-active); LC-MS (ESI−) calc'd for C.sub.36H.sub.33N.sub.5O.sub.7Cl [M−H].sup.−: 682.2, found 682.0; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 10.22 (s, 1H), 9.76 (br s, 1H), 7.71 (m, 1H), 7.54-7.45 (m, 7H), 7.40-7.37 (m, 2H), 7.22-7.17 (m, 2H), 6.02 (m, 1H), 5.01-4.98 (m, 1H), 4.62 (m, 2H), 3.41-3.17 (m, 4H), 2.90-2.73 (m, 3H), 2.13 (m, 1H), 1.52 (m, 2H), 1.43-1.30 (m, 4H), 1.22-1.18 (m, 2H).
##STR00192##
5-chloro-N-(6-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) hexyl)-3-phenyl-1H-indole-2-carboxamide (50)
[0108] General Procedure 1 and 3. Reaction scale: 9.5 mg (18.0 μmol, 1.00 equiv) of 7 and 6.0 mg (22.0 μmol, 1.20 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 50 as a yellow foam (9.2 mg, 75%). R.sub.f=0.55 (10% MeOH/EtOAc, UV-active); LC-MS (ESI−) calc'd for C.sub.36H.sub.33N.sub.5O.sub.7Cl [M−H].sup.−: 682.2, found 682.0; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 10.22 (s, 1H), 9.76 (br s, 1H), 7.71 (m, 1H), 7.54-7.45 (m, 7H), 7.40-7.37 (m, 2H), 7.22-7.17 (m, 2H), 6.02 (m, 1H), 5.01-4.98 (m, 1H), 4.62 (m, 2H), 3.41-3.17 (m, 4H), 2.90-2.73 (m, 3H), 2.13 (m, 1H), 1.52 (m, 2H), 1.43-1.30 (m, 4H), 1.22-1.18 (m, 2H).
##STR00193##
5-chloro-N-((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl)-3-phenyl-1H-indole-2-carboxamide (51)
[0109] General Procedure 3 then 1. Reaction scale: 8 mg (12.6 μmol, 1.0 eq.) of 14 and 4.9 mg (13.9 μmol, 1.1 eq.) of chlorophenylindole. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 51 as yellow powder (5.7 mg, 51%). R.sub.f=0.57 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI−) calc'd for C.sub.46H.sub.55ClN.sub.6O.sub.8S 886.4, found [M−H].sup.− 885.5. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 10.22 (s, 1H), 8.67 (s, 1H), 7.55-7.47 (m, 4H), 7.46-7.37 (m, 3H), 7.33 (s, 4H), 7.21 (dd, J=8.7, 2.0 Hz, 1H), 7.11 (d, J=8.6 Hz, 1H), 6.37 (s, 1H), 4.81 (t, J=8.2 Hz, 1H), 4.60-4.54 (m, 2H), 4.50 (s, 1H), 4.37 (dd, J=15.0, 5.3 Hz, 1H), 4.17 (d, J=11.4 Hz, 1H), 3.65-3.56 (m, 5H), 3.47 (d, J=6.5 Hz, 10H), 2.48 (s, 3H), 2.38 (dd, J=14.4, 8.3 Hz, 2H), 2.17 (s, 3H), 0.97 (s, 9H).
##STR00194##
5-chloro-N-(7-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptyl)-3-phenyl-1H-indole-2-carboxamide (52)
[0110] General Procedure 1 and 3. Reaction scale: 18.3 mg (28.0 μmol, 1.00 equiv) of 18 and 11.6 mg (43.0 μmol, 1.54 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 52 as a yellow solid (17.9 mg, 79%). R.sub.f=0.38 (10% MeOH/EtOAc, UV-active); LC-MS (ESI+) calc'd for C.sub.44H.sub.52N.sub.6O.sub.5SCl [M+H].sup.+: 811.3, found 811.1; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 10.40 (m, 1H), 8.65 (s, 1H), 7.49-7.28 (m, 12H), 7.16 (m, 1H), 6.57 (m, 1H), 5.91 (m, 1H), 4.74 (m, 1H), 4.59 (d, J=9.0 Hz, 1H), 4.51-4.47 (m, 2H), 4.35 (dd, J=15.0, 5.6 Hz, 1H), 4.09 (d, J=11.3 Hz, 1H), 3.62 (m, 1H), 3.16 (m, 2H), 2.46 (s, 3H), 2.38 (m, 1H), 2.19-2.06 (m, 3H), 1.53-1.46 (m, 2H), 1.27-1.22 (m, 3H), 1.17 (m, 2H), 1.06 (m, 2H), 0.96 (s, 9H).
##STR00195##
(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(4-oxo-3-(3,4,5-trifluorobenzyl)-3,5,7,8-tetrahydropyrido[4,3-d]pyrimidin-6 (4H)-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (53)

[0111] General Procedure 1 and 3. Reaction scale: 14.2 mg (20.0 μmol, 1.00 equiv) of 12 and 15.2 mg (38.0 μmol, 1.90 equiv) of SM2. Purified by PTLC (100% EtOAc) then plate was dried and run again (10% MeOH/CH.sub.2Cl.sub.2) to afford 53 as a white foam (4.3 mg, 23%). .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (s, 1H), 8.09-8.05 (m, 1H), 7.53-7.45 (m, 1H), 7.36 (m, 4H), 7.07-6.96 (m, 3H), 5.03-4.95 (m, 2H), 4.74 (m, 1H), 4.62-4.42 (m, 5H), 4.33 (dd, J=15.0, 5.2 Hz, 1H), 4.12 (m, 1H), 3.85-3.76 (m, 3H), 3.74-3.68 (m, 3H), 3.63-3.55 (m, 10H), 2.77 (m, 1H), 2.72-2.69 (m, 3H), 2.54-2.51 (m, 4H), 2.49-2.44 (m, 2H), 2.19-2.10 (m, 1H), 0.93 (s, 9H).
##STR00196##
(2S,4R)-1-((S)-1-(4-(benzyloxy)phenyl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (54)
[0112] General Procedure 1 and 3. Reaction scale: 8.6 mg (12.0 μmol, 1.00 equiv) of 14 and 7.5 mg (31.0 μmol, 2.58 equiv) of SM2. Purified by PTLC (10% MeOH/EtOAc) to afford 54 as a white foam (8.5 mg, 84%). 1H NMR (400 MHZ, CDCl.sub.3) δ 8.69 (s, 1H), 7.43-7.30 (m, 11H), 7.17 (d, J=8.58 Hz, 2H), 6.92 (d, J=8.58 Hz, 2H), 5.04 (s, 2H), 4.70 (m, 1H), 4.58-4.48 (m, 3H), 4.33 (dd, J=14.8, 5.2 Hz, 1H), 4.11 (d, J=11.3 Hz, 1H), 3.74-3.35 (m, 19H), 2.50-2.44 (m, 6H), 1.42 (d, J=5.7 Hz, 1H), 0.93 (s, 9H).
##STR00197##
(2S,4R)-1-((S)-19-(tert-butyl)-4,17-dioxo-1-(4-oxo-3,5,6,7-tetrahydro-4H-cyclopenta[4,5]thieno[2,3-d]pyrimidin-2-yl)-8,11,14-trioxa-2-thia-5,18-diazaicosan-20-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (55)
[0113] General Procedure 1 and 3. Reaction scale: 10.4 mg (14.0 μmol, 1.00 equiv) of 14 and 4.2 mg (18.0 μmol, 1.29 equiv) of SM2. Purified by PTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 55 as a yellow oil (9.0 mg, 77%). LC-MS (ESI+) calc'd for C.sub.42H.sub.57N.sub.6O.sub.9S.sub.2 [M+H].sup.+: 853.4, found 853.7.
##STR00198##
1-benzyl-N-((S)-14-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl) carbamoyl) pyrrolidine-1-carbonyl)-15,15-dimethyl-12-oxo-3,6,9-trioxa-13-azahexadecyl)-6-oxo-1,6-dihydropyridazine-3-carboxamide (56)
[0114] General Procedure 1 and 3. Reaction scale: 10.3 mg (14.0 μmol, 1.00 equiv) of 14 and 5.0 mg (21.0 μmol, 1.50 equiv) of SM2. Purified by PTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 56 as a white solid (8.4 mg, 71%). .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.89 (d, J=9.6 Hz, 1H), 7.52 (m, 1H), 7.41 (m, 2H), 7.36-7.30 (m, 7H), 6.97 (d, J=9.6 Hz, 1H), 6.81 (d, J=8.4 Hz, 1H), 5.35 (s, 2H), 4.68 (m, 1H), 4.54-4.49 (m, 3H), 4.35 (dd, J=14.9, 5.6 Hz, 1H), 4.15-4.09 (m, 2H), 3.67-3.57 (m, 16H), 2.52-2.50 (m, 6H), 2.42 (m, 1H), 0.94 (s, 9H).
##STR00199##
(2S,4R)-1-((2S)-2-(tert-butyl)-16-(3-(2-methoxy-5-(3-methylisoxazol-5-yl)pyrimidin-4-yl) piperidin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (57)
[0115] General Procedure 1 and 3. Reaction scale: 7.0 mg (10.0 μmol, 1.00 equiv) of 12 and 6.0 mg (15.0 μmol, 1.50 equiv) of SM2. Purified by PTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 57 as a white foam (6.8 mg, 74%). R.sub.f=0.68 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI−) calc'd for C.sub.47H.sub.63N.sub.8O.sub.12S [M+HCO.sub.2].sup.−: 963.4, found 963.0; .sup.1H NMR (600 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 8.66 (d, J=2.7 Hz), 7.54-7.44 (m, 1H), 7.36-7.32 (m, 4H), 7.06-7.00 (s, 1H), 6.43 (s, 1H), 4.73-4.69 (m, 1H), 4.59-4.54 (m, 1H), 4.51-4.45 (m, 2H), 4.36-4.29 (m, 1H), 4.10 (br d, J=11.5 Hz, 1H), 4.07-4.05 (m, 3H), 4.00-3.90 (m, 1H), 3.80-3.66 (m, 5H), 3.64-3.55 (m, 11H), 3.11-3.05 (m, 1H), 2.67-2.63 (m, 1H), 2.62-2.57 (m, 1H), 2.51 (s, 3H), 2.49-2.44 (m, 3H), 2.38-2.36 (m, 3H), 2.16-2.10 (m, 1H), 1.97-1.76 (m, 5H), 0.93 (s, 9H).
##STR00200##
(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(4-(4-(pyrimidin-2-ylmethyl) piperidine-1-

carbonyl)phenyl)-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (58)

[0116] General Procedure 1 and 3. Reaction scale: 8.0 mg (11.0 μmol, 1.00 equiv) of 12 and 9.2 mg (22.0 μmol, 2.00 equiv) of SM2. Purified by PTLC (100% EtOAc) then plate was dried and run again (10% MeOH/CH.sub.2Cl.sub.2) to afford 58 as a yellow oil (1.3 mg, 13%). LC-MS (ESI+) calc'd for C.sub.50H.sub.67N.sub.8O.sub.9S [M+H].sup.+: 956.0, found 956.4; .sup.1H NMR (600 MHz, CDCl.sub.3) δ 8.66 (m, 3H), 7.38-7.27 (m, 9H), 7.14 (t, J=4.9 Hz, 1H), 6.98 (d, J=8.4 Hz, 1H), 4.63 (m, 1H), 4.54-4.45 (m, 4H), 4.34 (td, J=15.7, 5.4 Hz, 2H), 3.98 (d, J=11.4 Hz, 1H), 3.79-3.52 (m, 14H), 3.10 (q, J=7.4 Hz, 1H), 2.99-2.87 (m, 3H), 2.76 (m, 1H), 2.54-2.44 (m, 5H), 2.40-2.32 (m, 2H), 2.24 (m, 1H), 1.48-1.39 (m, 8H), 0.94 (s, 9H).

##STR00201##

3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) ethoxy) ethoxy)-N-(2-oxo-2H-chromen-7-yl) propenamide (59)

[0117] General Procedure 2. Reaction scale: 8.0 μmol of 2 and 1.4 mg (8.8 μmol) of coumarin. Purified by pTLC (EtOAc, then 10% MeOH/CH.sub.2Cl.sub.2) to afford 59 as a glassy solid (4.3 mg, 85%). Molecular formula: C.sub.31H.sub.30N.sub.4O.sub.11 .sup.1H NMR (400 MHZ, CDCl.sub.3) δ.sub.H 2.14-2.25 (1H, m, CL1, diastereotopic proton), 2.65 (2H, t, J 5.6, C(O)CH.sub.2CH.sub.2O), 2.71-2.94 (2H, m, CL1 diastereotopic proton, other proton), 3.51-3.66 (8H, m, O(CH.sub.2).sub.2O(CH.sub.2).sub.2NH), 3.85 (2H, t, J 5.6, C(O)CH.sub.2CH.sub.2O), 4.56 (2H, s, C(O)CH.sub.2O), 4.96 (1H, dd, J 5.5, 12.4, CL1 stereocenter proton), 6.36 (1H, d, J 9.5, coumarin double bond nearest carbonyl), 7.13 (1H, d, J 8.7, ortho-linker), 7.20 (1H d, J 8.9, coumarin proton), 7.41 (1H, dd, J 2.4, 8.9, coumarin proton nearest linker), 7.51 (1H, d, J 7.4, CL1, para-linker), 7.56 (1H, br t, J 5.6, linker NH), 7.64 (1H, J 9.5, coumarin double bond farthest from carbonyl), 7.72 (1H dd, J 8.7, 7.4, CL1, meta-linker), 8.06 (1H, d, J 2.4, coumarin proton no neighbours), 8.86 (1H, br. s, NH).

##STR00202##

(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-((2-oxo-2H-chromen-7-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (60)

[0118] General Procedure 3 then 2. Reaction scale: 11.1 μmol of 2 and 2.0 mg (12.2 μmol) of coumarin. Purified by pTLC (EtOAc, then 10% MeOH/CH.sub.2Cl.sub.2) to afford 60 as a translucent glass (5.1 mg, 60%). Molecular formula: C.sub.39H.sub.47N.sub.5O.sub.9S. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.27 (s, 1H), 8.68 (s, 1H), 7.92 (d, J=2.2 Hz, 1H), 7.75-7.51 (m, 2H), 7.44 (d, J=8.7 Hz, 1H), 7.37 (q, J=8.3 Hz, 4H), 7.21-7.09 (m, 2H), 6.33 (d, J=9.6 Hz, 1H), 4.79 (t, J=8.3 Hz, 1H), 4.71-4.55 (m, 3H), 4.32 (dd, J=15.1, 5.2 Hz, 1H), 4.20 (d, J=11.3 Hz, 1H), 3.84 (td, J=9.9, 3.2 Hz, 1H), 3.79-3.44 (m, 7H), 2.78 (ddd, J=14.2, 9.9, 4.2 Hz, 1H), 2.52 (s, 3H), 2.49-2.30 (m, 2H), 2.30-2.13 (m, 3H), 1.02 (s, 9H).

##STR00203##

3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) ethoxy) ethoxy)-N-(3,3-diphenylpropyl) propenamide (61)

[0119] General Procedure 3 then 2. Molecular formula: C.sub.37H.sub.40N.sub.4O.sub.9 Yield 61:4.3 mg (78%). .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.42-7.30 (m, 4H), 7.30-7.19 (m, 11H), 7.19-7.13 (m, 2H), 6.98 (d, J=8.5 Hz, 1H), 6.51 (t, J=5.8 Hz, 1H), 4.62 (t, J=8.1 Hz, 1H), 4.56 (dd, J=15.0, 6.7 Hz, 1H), 4.50 (d, J=8.6 Hz, 2H), 4.31 (dd, J=15.0, 5.2 Hz, 1H), 4.04 (d, J=11.3 Hz, 1H), 3.94 (t, J=7.8 Hz, 1H), 3.75-3.45 (m, 10H), 3.16 (dt, J=8.1, 6.1 Hz, 2H), 2.51 (s, 3H), 2.48-2.27 (m, 4H), 2.23 (td, J=8.0, 6.3 Hz, 2H), 2.15-2.03 (m, 1H), 0.94 (s, 9H).

##STR00204##

(2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-17,17-diphenyl-7,10-dioxa-3,14-diazaheptadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (62)

[0120] General Procedure 3 then 2. Molecular formula: C.sub.45H.sub.57N.sub.5O.sub.7S. Yield

62:7.7 mg (80%).
##STR00205##
N-(2-(1H-indol-3-yl)ethyl)-3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)oxy) acetamido) ethoxy) ethoxy) propenamide (63)
[0121] General Procedure 3 then 2. Reaction scale: 17.6 μmol of 2 and 3.2 mg (20 μmol) of indole amine. Purified by pTLC (EtOAc, then 10% MeOH/CH.sub.2Cl.sub.2) to afford 63 as a yellow translucent glass (2.8 mg, 25%). Molecular formula: C.sub.32H.sub.35N.sub.5O.sub.9 .sup.1H NMR (400 MHz, CDCl.sub.3) δ.sub.H 2.04-2.12 (1H, m, CL1, diastereotopic proton), 2.41 (2H, t, J 5.9, tryptamine NHCH.sub.2CH.sub.2), 2.57-2.84 (3H, m, CL1 diastereotopic proton, imide-CH.sub.2), 2.95 (2H, t, J6.5, C(O)CH.sub.2CH.sub.2O), 3.46-3.64 (10H, m, O(CH.sub.2).sub.2O(CH.sub.2).sub.2NH, tryptamine CH.sub.2NH), 3.70 (2H, t, J5.8, C(O)CH.sub.2CH.sub.2O), 4.58 (2H, d, J2.2, C(O)CH.sub.2O), 4.90 (1H, dd, J 5.4, 12.2, CL1 stereocenter proton), 6.36 (1H, t, J 5.5, tryptamine amide NHC(O)), 7.04-7.18 (3H, m, indole), 7.13 (1H, d, J 8.7, ortho-linker), 7.32 (1H, d, J 8.1, indole), 7.52 (1H, d, J 7.4, CL1, para-linker), 7.56 (1H, d, J 7.7, indole), 7.62 (1H, br t, J 5.6, linker NH), 7.72 (1H dd, J 8.7, 7.4, CL1, meta-linker), 8.41 (1H, br s, imide NH), 8.82 (1H, br s, indole NH).
##STR00206##
(2S,4R)-1-((S)-2-(tert-butyl)-16-(1H-indol-3-yl)-4,13-dioxo-7,10-dioxa-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (64)
[0122] General Procedure 3 then 2. Molecular formula: C.sub.40H.sub.52N.sub.6O.sub.7S. Yield 64:4.3 mg (45%).
##STR00207##
(2S,4R)-1-((S)-2-(tert-butyl)-16-(3,4-dimethoxyphenyl)-4,13-dioxo-7,10-dioxa-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (65)
[0123] General Procedure 3 then 2. Molecular formula: C.sub.40H.sub.55N.sub.5O.sub.9S. Yield 65:7.9 mg (78%). .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.68 (s, 1H), 7.36 (s, 3H), 6.98 (d, J=8.4 Hz, 1H), 6.85-6.65 (m, 4H), 6.53 (d, J=5.9 Hz, 1H), 4.60 (dt, J=23.2, 7.4 Hz, 2H), 4.49 (d, J=8.5 Hz, 2H), 4.33 (dd, J=15.0, 5.1 Hz, 1H), 4.08 (d, J=11.4 Hz, 1H), 3.86 (s, 3H), 3.84 (s, 3H), 3.76-3.38 (m, 12H), 2.74 (q, J=7.4 Hz, 2H), 2.54-2.30 (m, 9H), 2.19-2.09 (m, 1H), 0.94 (s, 9H).
##STR00208##
(2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-16-(2-oxo-2,3,4,5-tetrahydro-1H-benzo[b]azepin-8-yl)-7,10-dioxa-3,14-diazahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (66)
[0124] General Procedure 3 then 2. Molecular formula: C.sub.42H.sub.56N.sub.6O.sub.8S. Yield 66:5.0 mg (50%).
##STR00209##
(2S,4R)-1-((S)-1-(1H-benzo[d]imidazol-2-yl)-14-(tert-butyl)-3,12-dioxo-6,9-dioxa-2,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (67)
[0125] General Procedure 3 then 2. Molecular formula: C.sub.38H.sub.49N.sub.7O.sub.7S. Yield 67:4.3 mg (45%).
##STR00210##
(2S,4R)-1-((S)-2-(3-(2-(3-(((3s,5s,7s)-adamantan-1-yl)amino)-3-oxopropoxy) ethoxy) propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (68)
[0126] General Procedure 3 then 2. Molecular formula: C.sub.40H.sub.57N.sub.5O.sub.7S. Yield 68:4.2 mg (43%).
##STR00211##
(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((R)-1,2,3,4-tetrahydronaphthalen-1-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)

pyrrolidine-2-carboxamide (69)

[0127] General Procedure 3 then 2. Molecular formula: C.sub.40H.sub.53N.sub.5O.sub.7S. Yield 69:5.2 mg (41%).

##STR00212##

(2S,4R)-1-((S)-3,3-dimethyl-2-(3-(2-(3-oxo-3-(((S)-1,2,3,4-tetrahydronaphthalen-1-yl)amino) propoxy) ethoxy) propanamido) butanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (70)

[0128] General Procedure 3 then 2. Molecular formula: C.sub.40H.sub.53N.sub.5O.sub.7S. Yield 70:8.2 mg (96%).

##STR00213##

(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (71)

[0129] General Procedure 1 and 3. Reaction scale: 17.0 mg (23.0 μmol, 1.00 equiv) of 14 and 11.8 mg (47.0 μmol, 2.04 equiv) of (+)-sclereolide. Purified by PTLC (50% EtOAc/hexanes) to afford 71 as a yellow oil (16.8 mg, 84%). .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.40-7.32 (m, 5H), 6.97 (d, J=8.4 Hz, 1H), 6.33 (m, 1H), 5.31 (m, 1H), 4.70 (t, J=8.1 Hz, 1H), 4.58-4.48 (m, 3H), 4.33 (dd, J=15.1, 5.3 Hz, 1H), 3.76-3.56 (m, 13H), 3.53-3.49 (m, 2H), 3.45-3.38 (m, 2H), 2.50-2.44 (m, 7H), 2.35-2.29 (m, 1H), 2.17-2.06 (m, 3H), 1.65 (m, 3H), 1.42-1.36 (m, 4H), 1.24-1.07 (m, 4H), 0.93 (s, 9H), 0.92 (s, 3H), 0.87 (s, 3H), 0.84 (s, 3H).

##STR00214##

(2S,4R)-1-((S)-17-(tert-butyl)-3,15-dioxo-1-(3-((2-((1R,4aS,8aS)-2,5,5,8a-tetramethyl-1,4,4a,5,6,7,8,8a-octahydronaphthalen-1-yl) acetamido)methyl)phenyl)-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (72)

[0130] Reaction scale: 12.3 mg (17.0 μmol, 1.00 equiv) of 12 and 10.8 mg (23.0 μmol, 1.35 equiv) of (+)-sclereolide phenyl derivative. Purified by PTLC (100% EtOAc) to afford 72 as a yellow oil (3.8 mg, 22%).

##STR00215##

(2S,4R)-1-((S)-16-(tert-butyl)-1,14-dioxo-1-((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptan-1-yl)-5,8,11-trioxa-2,15-diazaheptadecan-17-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (73)

[0131] General Procedure 1 and 3. Reaction scale: 14.4 mg (20.0 μmol, 1.00 equiv) of 14 and 7.0 mg (35.0 μmol, 1.75 equiv) of (S)-camphanic acid. Purified by PTLC (50% EtOAc/hexanes) to afford 73 as a yellow oil (13.1 mg, 80%). 1H NMR (400 MHZ, CDCl.sub.3) δ 8.67 (s, 1H), 7.45 (t, J=5.9 Hz, 1H), 7.37-7.32 (m, 4H), 7.02-6.96 (m, 2H), 4.72 (t, J=8.1 Hz, 1H), 4.55 (dd, J=15.0, 6.6 Hz, 1H), 4.51-4.45 (m, 2H), 4.33 (dd, J=15.1, 5.3 Hz, 1H), 4.11 (m, 1H), 3.71 (m, 2H), 3.63-3.44 (m, 14H), 2.52-2.47 (m, 7H), 2.16-2.11 (m, 1H), 1.95-1.84 (m, 2H), 1.68-1.62 (m, 1H), 1.08 (s, 3H), 1.08 (s, 3H), 0.94 (s, 9H), 0.88 (s, 3H).

##STR00216##

(2S,4R)-1-((S)-18-(tert-butyl)-3,16-dioxo-1-(3-(((1S,4S)-4,7,7-trimethyl-3-oxo-2-oxabicyclo[2.2.1]heptane-1-carboxamido)methyl)phenyl)-7,10,13-trioxa-2,17-diazanonadecan-19-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (74)

[0132] General Procedure 1 and 3. Reaction scale: 15.4 mg (21.0 μmol, 1.00 equiv) of 12 and 12.5 mg (30.0 μmol, 1.43 equiv) of(S)-camphanic acid phenyl derivative. Purified by PTLC (100% EtOAc) to afford 74 as a yellow oil (13.9 mg, 68%). .sup.1H NMR (400 MHz, CDCl.sub.3) δ 8.66 (s, 1H), 7.36-7.28 (m, 6H), 7.19-7.13 (m, 3H), 7.08-6.98 (m, 2H), 6.91 (d, J=8.6 Hz, 1H), 4.61 (t, J=8.1 Hz, 1H), 4.56-4.38 (m, 8H), 4.32 (dd, J=15.1, 5.3 Hz, 1H), 4.06 (d, J=11.5 Hz, 1H), 3.74 (m, 2H), 3.68-3.50 (m, 12H), 2.51-2.49 (m, 6H), 2.44-2.38 (m, 2H), 2.12-2.07 (m, 2H), 1.97-1.87 (m, 3H), 1.70-1.63 (m, 1H), 1.11 (s, 3H), 1.09 (s, 3H), 0.93 (s, 9H), 0.89 (s, 3H).

##STR00217##

(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-((R)-2,3-dihydrobenzo[b][1,4]dioxine-2-carbonyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (75)

[0133] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.1 μmol, 1.0 eq.) of 12 and 2.4 mg (10.0 μmol, 1.1 eq.) of(S)-benzodioxanpiperazine methanone. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 75 as a white solid (3.3 mg, 41%). R/=0.49 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.45H.sub.60N.sub.6O.sub.11S 892.4, found [M+H].sup.+ 893.9. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.83-7.23 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 8H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.60-2.35 (m, 9H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

##STR00218##

(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(naphthalen-2-ylmethyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (76)

[0134] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.1 μmol, 1.0 eq.) of 12 and 2.25 mg (10.0 μmol, 1.1 eq.) of naphthalene piperazine. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 76 as a slightly yellow solid (3.1 mg, 40%). R.sub.f=0.55 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.47H.sub.62N.sub.6O.sub.8S 870.4, found [M+NH.sub.4].sup.+ 888.8. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 7H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.92 (s, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 16H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).

##STR00219##

(2S,4R)-1-((S)-2-(tert-butyl)-16-(4-((4′-chloro-5,5-dimethyl-3,4,5,6-tetrahydro-[1,1′-biphenyl]-2-yl)methyl) piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (77)

[0135] General Procedure 3 then 2. Reaction scale: 6.5 mg (9.8 μmol, 1.0 eq.) of 12 and 3.4 mg (10.8 μmol, 1.1 eq.) of chlorophenyldimethylcyclohexenpiperazine. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 77 as a white solid (3.1 mg, 33%). R.sub.f=0.61 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI−) calc'd for C.sub.51H.sub.71ClN.sub.6O.sub.8S 962.5, found [M+HCOO].sup.−: 1007.3. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J 7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.62 (q, J=5.4 Hz, 4H), 3.66-3.51 (m, 8H), 3.47 (q, J=6.7, 4.9 Hz, 4H), 3.10 (s, 2H), 2.60-2.35 (m, 8H), 2.20-2.10 (m, 5H), 2.05 (s, 1H), 1.53 (m 1H), 0.93 (s, 9H), 0.88 (m, 6H).

##STR00220##

(2S,4R)-1-((S)-17-(tert-butyl)-2-cyclopentyl-1-(9-ethyl-9H-carbazol-2-yl)-3,15-dioxo-6,9,12-trioxa-2,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (78)

[0136] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μmol) of 12 and 5.29 mg (18.11 μmol) of carbazolecyclopentanamine. Purified by pTLC (8% MeOH/DCM) to afford 78 as a white solid (5.3 mg, 63%). R.sub.f=0.53 (8% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.52H.sub.68N.sub.6O.sub.8S 936.5, found [M+H].sup.+ 938.2. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 8H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 5H), 4.20 (s,

1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.66-3.51 (m, 11H), 2.60-2.35 (m, 6H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 1.60-1.87 (8H), 1.26 (t, 2H), 0.93 (s, 9H).
##STR00221##
(2S,4R)-1-((S)-19-(1-benzyl-1H-indol-3-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3,17-diazanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (79)
[0137] General Procedure 2. Reaction scale: 6.0 mg (9.05 μmol) of 12 and 4.79 mg (18.11 μmol) of benzyl indole amine. Purified by pTLC (8% MeOH/DCM) to afford 79 as a slightly orange solid (mg, %). Molecular formula: C.sub.50H.sub.64N.sub.6O.sub.8S. MS calc'd 908.5, found [M+H].sup.+ 909.7. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 9H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 2H), 5.04 (t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 3H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.66-3.51 (m, 12H), 2.60-2.35 (m, 8H), 2.20-2.10 (m, 1H), 2.13 (s, 3H), 2.05 (s, 1H), 0.93 (s, 9H).
##STR00222##
(2S,4R)-1-((2S)-16-(2-(5-(4-bromobenzyl)-1,2,4-oxadiazol-3-yl) pyrrolidin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (80)
[0138] General Procedure 3 then 2. Reaction scale: 6.0 mg (9.05 μmol) of 12 and 5.58 mg (18.11 μmol) of bromobenzyloxadiazolepyrrolidine. Purified by pTLC (8% MeOH/DCM) to afford 80 as a slightly yellow solid (mg, %). Molecular formula: C.sub.45H.sub.58BrN.sub.7O.sub.9S. MS calc'd 951.3, found [M+H].sup.+ 952.4. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.86 (br. s, 1H), 7.80-7.35 (m, 4H), 7.50 (br. t, 2H), 7.30 (s, 1H), 7.25 (br. t, 1H), 4.72 (t, 2H), 4.63 (d, J=3.1 Hz, 2H), 4.50 (s, 1H), 4.60-4.30 (m, 4H), 4.20 (s, 1H), 4.15-4.10 (d, J=7 Hz, 2H), 3.78 (ddd, J=9.1, 7.1, 2.4 Hz, 2H), 3.66-3.51 (m, 12H), 2.60-2.35 (m, 10H), 2.20-2.10 (m, 1H), 2.05 (s, 1H), 0.93 (s, 9H).
##STR00223##
(2S,4R)-1-((2S)-2-(tert-butyl)-4,16-dioxo-16-((2-oxo-5-phenyl-2,5-dihydro-1H-benzo[e][1,4]diazepin-3-yl)amino)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (81)
[0139] General Procedure 3 then 1. Reaction scale: 8 mg (12.1 μmol, 1.0 eq.) of 12 and 3.3 mg (13.3 μmol, 1.1 eq.) of benzodiazepam. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 81 as a white solid (7 mg, 68%). R/=0.57 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.47H.sub.57N.sub.7O.sub.9S 895.4, found [M+H].sup.+ 896.2. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 9.96 (s, 1H), 8.67 (d, J=6.2 Hz, 2H), 8.42 (d, J=7.8 Hz, 1H), 7.58-7.49 (m, 5H), 7.37 (s, 6H), 7.13 (t, J=8.6 Hz, 4H), 5.56-5.46 (m, 2H), 4.71-4.62 (m, 3H), 4.56-4.49 (m, 3H), 4.36-4.29 (m, 1H), 4.22 (d, J=11.5 Hz, 2H), 4.12 (q, J=7.2 Hz, 4H), 3.86 (q, J=4.4 Hz, 2H), 3.67 (d, J=5.0 Hz, 7H), 2.50 (s, 5H), 0.92 (s, 9H). Note: presence of rotamers.
##STR00224##
(2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(2′-oxo-2′,3′-dihydro-1′H-spiro[piperidine-4,4′-quinazolin]-1-yl)-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide (82)
[0140] General Procedure 3 then 1. Reaction scale: 8 mg (12.1 μmol, 1.0 eq.) of 12 and 2.88 mg (13.3 μmol, 1.1 eq.) of spiroquinazoline. Purified by pTLC (10% MeOH/CH.sub.2Cl.sub.2) to afford 82 as a yellow solid (5 mg, 48%). R.sub.f=0.40 (10% MeOH/CH.sub.2Cl.sub.2, UV-active); LC-MS (ESI+) calc'd for C.sub.44H.sub.59N.sub.7O.sub.9S 861.4, found [M+H].sup.+ 862.4. .sup.1H NMR (400 MHZ, CDCl.sub.3) δ 8.65 (d, J=16.1 Hz, 1H), 7.55 (s, 1H), 7.41-7.30 (m, 5H), 7.22-7.08 (m, 2H), 6.99 (d, J=5.5 Hz, 1H), 6.74 (d, J=7.8 Hz, 1H), 4.71 (t, J=8.6 Hz, 2H), 4.59-4.47 (m, 3H), 4.38-4.24 (m, 1H), 4.18-4.05 (m, 2H), 3.92-3.73 (m, 5H), 3.67-3.44 (m, 12H), 2.76 (dt, J=13.4, 6.3 Hz, 1H), 2.58 (dd, J=15.2, 5.5 Hz, 1H), 2.44 (s, 3H), 2.26 (dd, J=14.0, 5.5 Hz, 2H), 1.94 (q, J=14.0 Hz, 6H), 0.88 (s, 9H). Note: presence of rotamers.
TABLE-US-00004 TABLE 3 Non-limiting Examples of FragTAC Probes and Intermediates

Compound Example Compound Number Code Structure Chemical Name 1 IF-012 [00225] embedded image Tert-butyl 3-(2-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4yl)oxy)acetamido)ethoxy) propanoate 2 LPC002 [00226] embedded image Tert-butyl 3-(2-(2-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4- yl)oxy)acetamido)ethoxy)ethoxy) propanoate 3 IF-001 [00227] embedded image Tert-butyl 1-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)-2- oxo-6,9,12-trioxa-3- azapentadecan-15-oate 4 IF-092 [00228] embedded image Tert-butyl (1-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)-2- oxo-6,9, 12-trioxa-3- azatetradecan-14-yl)carbamate 5 IF-015 [00229] embedded image Tert-butyl 1-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)-2- oxo-6,9,12,15-tetraoxa-3- azaoctadecan-18-oate 6 IF-025 [00230] embedded image Tert-butyl 6-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy) acetamido)hexanoate 7 CMC-6- 194 [00231] embedded image Tert-butyl (6-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy) acetamido)hexyl)carbamate 8 IF-081 [00232] embedded image Tert-butyl 3-(3-(((S)-1-((2S,4R)- 4-hydroxy-2-((4-(4- methylthiazol-5- yl)benzyl)carbamoyl)pyrrolidin- 1-yl)-3,3-dimethyl-1- oxobutan- 2-yl)amino)-3- oxopropoxy)propanoate 9 LPC115 [00233] embedded image Tert-butyl 3-(3-(((S)-1-((2S,4S)- 4-hydroxy-2-((4-(4- methylthiazol-5- yl)benzyl)carbamoyl)pyrrolidin- 1-yl)-3,3-dimethyl-1-oxobutan- 2-yl)amino)-3- oxopropoxy)propanoate 10 LPC013 [00234] embedded image Tert-butyl 3-(2-(3-(((S)-1- ((2S,4R)-4-hydroxy-2-((4-(4- methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin- 1-yl)-3,3-dimethyl-1-oxobutan- 2-yl)amino)-3-oxopropoxy)ethoxy)propanoate 11 LPC114 [00235] embedded image Tert-butyl 3-(2-(3-(((S)-1-((2S,4S)-4-hydroxy-2-((4-(4- methylthiazol-5-yl) benzyl)carbamoyl)pyrrolidin-1- yl)-3,3-dimethyl-1-oxobutan-2- yl)amino)-3- oxopropoxy)ethoxy)propanoate 12 IF-061 [00236] embedded image Tert-butyl (S)-15-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl) benzyl) carbamoyl) pyrrolidine-1-carbonyl)-16,16- dimethyl-13-oxo-4,7,10-trioxa- 14-azaheptadecanoate 13 IF-068 [00237] embedded image Tert-butyl (S)-15-((2S,4S)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl)carbamoyl) pyrrolidine-1-carbonyl)-16,16- dimethyl-13-oxo-4,7,10-trioxa- 14-azaheptadecanoate 14 IF-069 [00238] embedded image Tert-butyl ((S)-14-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5- yl)benzyl)carbamoyl)pyrrolidine- 1-carbonyl)-15,15-dimethyl-12- oxo-3,6,9-trioxa-13- azahexadecyl)carbamate 15 IF-052 [00239] embedded image Tert-butyl (S)-18-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl)carbamoyl) pyrrolidine-1-carbonyl)-19,19- dimethyl-16-oxo-4,7,10,13- tetraoxa-17-azaicosanoate 16 IF-036 [00240] embedded image Tert-butyl 4-(((S)-1-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl) carbamoyl)pyrrolidin-1-yl)-3,3- dimethyl-1-oxobutan-2- yl)amino)-4-oxobutanoate 17 IF-084 [00241] embedded image Tert-butyl 6-(((S)-1-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl) carbamoyl)pyrrolidin-1-yl)-3,3- dimethyl-1-oxobutan-2- yl)amino)-6-oxohexanoate 18 CMC-6- 108 [00242] embedded image Tert-butyl (7-(((S)-1-((2S,4R)-4- hydroxy-2-((4-(4- methylthiazol- 5-yl) benzyl) carbamoyl)pyrrolidin-1-yl)-3,3- dimethyl-1-oxobutan-2- yl)amino)-7-oxoheptyl) carbamate 19 IF-042 [00243] embedded image Methyl 8-(((S)-1-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl)carbamoyl) pyrrolidin-1-yl)-3,3-dimethyl-1- oxobutan-2-yl)amino)-8- oxooctanoate 20 IF-045 [00244] embedded image Methyl 10-(((S)-1-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl) benzyl) carbamoyl)pyrrolidin-1-yl)-3,3- dimethyl-1-oxobutan-2- yl)amino)-10-oxodecanoate 21 IF-014 [00245] embedded image N-(2-(3-(4- benzhydrylpiperazin- 1-yl)-3-oxopropoxy)ethyl)-2- ((2-(2,6-dioxopiperidin-3-yl)- 1,3- dioxoisoindolin-4- yl)oxy)acetamide 22 LPC011 [00246] embedded image N-(2-(2-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy)ethoxy)ethyl)-2- ((2-(2,6-dioxopiperidin-3-yl)- 1,3- dioxoisoindolin-4- yl)oxy)acetamide 23 IF-003 [00247] embedded image N-(2-(2-(2-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy)ethoxy)ethoxy) ethyl)-2-((2-(2,6- dioxopiperidin-3- yl)-1,3-dioxoisoindolin-4- yl)oxy)acetamide 24 IF-017 [00248] embedded image N-(15-(4- benzhydrylpiperazin- 1-yl)-15-oxo-3,6,9,12- tetraoxapentadecyl)-2-((2-(2,6- dioxopiperidin-3- yl)-1,3- dioxoisoindolin-4- yl)oxy)acetamide 25 IF-027 [00249] embedded image N-(6-(4-

benzhydrylpiperazin-1- yl)-6-oxohexyl)-2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)acetamide 26 IF-049 [00250]embedded image (2S,4R)-1-((S)-2-(3-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy)propanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 27 LPC118 [00251]embedded image (2S,4S)-1-((S)-2-(3-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy) propanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 28 LPC020 [00252]embedded image (2S,4R)-1-((S)-2-(3-(2-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy) ethoxy)propanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2- carboxamide 29 LPC117 [00253]embedded image (2S,4S)-1-((S)-2-(3-(2-(3-(4- benzhydrylpiperazin-1-yl)-3- oxopropoxy) ethoxy)propanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 30 IF-051 [00254]embedded image (2S,4R)-1-((S)-16-(4- benzhydrylpiperazin-1-yl)-2- (tert-butyl)-4,16-dioxo-7,10,13- trioxa-3-azahexadecanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 31 LPC116 [00255]embedded image (2S,4S)-1-((S)-16-(4-benzhydrylpiperazin-1-yl)-2- (tert-butyl)-4,16-dioxo-7,10,13- trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 32 IF-053 [00256] embedded image (2S,4R)-1-((S)-19-(4- benzhydrylpiperazin-1-yl)-2- (tert-butyl)-4,19-dioxo-7,10,13,16-tetraoxa-3- azanonadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 33 IF-038 [00257]embedded image (2S,4R)-1-((S)-2-(4-(4-benzhydrylpiperazin-1-yl)-4- oxobutanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 34 IF-041 [00258]embedded image (2S,4R)-1-((S)-2-(6-(4- benzhydrylpiperazin-1-yl)-6- oxohexanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 35 IF-044 [00259] embedded image (2S,4R)-1-((S)-2-(8-(4- benzhydrylpiperazin-1-yl)-8- oxooctanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 36 IF-047 [00260]embedded image (2S,4R)-1-((S)-2-(10-(4- benzhydrylpiperazin-1-yl)-10- oxodecanamido)-3,3- dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 37 IF-085 [00261]embedded image (2S,4R)-1-((S)-16-((R)-4- benzhydryl-2-phenylpiperazin- 1-yl)-2-(tert-butyl)-4,16-dioxo- 7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 38 CMC-6- 123 [00262]embedded image (2S,4R)-1-((S)-16-((S)-4- benzhydryl-2-phenylpiperazin- 1-yl)-2-(tert-butyl)-4,16-dioxo- 7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 39 IF-088 [00263]embedded image (2S,4R)-1-((2S)-2-(tert-butyl)- 4,16-dioxo-16-(4- (phenyl(pyridin-2-yl)methyl) piperazin-1-yl)-7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2- carboxamide 40 CMC-6- 125 [00264]embedded image (2S,4R)-1-((S)-16-((S)-4- benzhydryl-3-methylpiperazin- 1-yl)-2-(tert-butyl)-4,16-dioxo- 7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 41 IF-082 [00265]embedded image 2-((2-(2,6-dioxopiperidin-3-yl)- 1,3-dioxoisoindolin-4-yl)oxy)-N-(2-(2-(2-(3-oxo-3-(4- tosylpiperazin-1-yl) propoxy) ethoxy)ethoxy)ethyl)acetamide 42 IF-079 [00266]embedded image 2-((2-(2,6-dioxopiperidin-3-yl)- 1,3-dioxoisoindolin-4-yl)oxy)- N-(6-oxo-6-(4-tosylpiperazin-1- yl)hexyl)acetamide 43 IF-058 [00267]embedded image (2S,4R)-1-((S)-2-(tert-butyl)- 4,16-dioxo-16-(4-tosylpiperazin- 1-yl)-7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 44 IF-080 [00268] embedded image (2S,4R)-1-((S)-3,3-dimethyl-2- (6-oxo-6-(4-tosylpiperazin-1-yl)hexanamido)butanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 45 CMC-6- 109 [00269]embedded image N-(1-((2-(2,6-dioxopiperidin-3- yl)-1,3-dioxoisoindolin-4- yl)oxy)-2-oxo-6,9,12-trioxa-3- azatetradecan-14-yl)-3-(4,5- diphenyloxazol-2-yl)propanamide 46 CMC-6- 100 [00270]embedded image N-(6-(2-((2-(2,6-dioxopiperidin- 3-yl)-1,3-dioxoisoindolin-4-yl) oxy)acetamido)hexyl)-3-(4,5- diphenyloxazol-2-yl) propanamide 47

IF-072 [00271]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-19- (4,5-diphenyloxazol-2-yl)-4,17- dioxo-7,10,13-trioxa-3,16- diazanonadecanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 48 CMC-6- 111 [00272]embedded image (2S,4R)-1-((S)-2-(7-(3-(4,5- diphenyloxazol-2- yl)propanamido)heptanamido)- 3,3-dimethylbutanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 49 CMC-6- 110 [00273] embedded image 5-chloro-N-(1-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)-2-oxo-6,9,12-trioxa-3- azatetradecan-14-yl)-3-phenyl- 1H-indole-2-carboxamide 50 CMC-6- 102 [00274]embedded image 5-chloro-N-(6-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy) acetamido)hexyl)-3-phenyl-1H- indole-2-carboxamide 51 IF-073 [00275] embedded image 5-chloro-N-((S)-14-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl) carbamoyl)pyrrolidine-1- carbonyl)-15,15-dimethyl-12- oxo-3,6,9-trioxa-13- azahexadecyl)-3-phenyl-1H- indole-2-carboxamide 52 CMC-6- 112 [00276]embedded image 5-chloro-N-(7-(((S)-1-((2S,4R)- 4-hydroxy-2-((4-(4- methylthiazol-5-yl) benzyl) carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2- yl)amino)-7-oxoheptyl)-3- phenyl-1H-indole-2- carboxamide 53 CMC-6-27 [00277]embedded image (2S,4R)-1-((S)-2-(tert-butyl)- 4,16-dioxo-16-(4-oxo-3-(3,4,5-trifluorobenzyl)-3,5,7,8- tetrahydropyrido[4,3- d]pyrimidin-6(4H)-yl)-7,10,13- trioxa-3-azahexadecanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 54 CMC-6- 35 [00278]embedded image (2S,4R)-1-((S)-1-(4- (benzyloxy)phenyl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxa- 3,16-diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2- carboxamide 55 CMC-6- 40 [00279]embedded image (2S,4R)-1-((17S)-17-(tert-butyl)- 2,15-dioxo-1-(4-oxo-2,3,4,5- tetrahydrobenzo[b][1,4] thiazepin- 3-yl)-6,9,12-trioxa-3,16- diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 56 CMC-6- 46 [00280]embedded image 1-benzyl-N-((S)-14-((2S,4R)-4- hydroxy-2-((4-(4-methylthiazol- 5-yl)benzyl) carbamoyl)pyrrolidine-1-carbonyl)-15,15-dimethyl-12- oxo-3,6,9-trioxa-13- azahexadecyl)-6-oxo-1,6- dihydropyridazine-3-carboxamide 57 CMC-6- 73 [00281]embedded image (2S,4R)-1-((2S)-2-(tert-butyl)- 16-(3-(2-methoxy-5-(3- methylisoxazol-5-yl) pyrimidin- 4-yl)piperidin-1-yl)-4,16-dioxo- 7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl) benzyl)pyrrolidine-2- carboxamide 58 CMC-6- 74 [00282]embedded image (2S,4R)-1-((S)-17-(tert-butyl)- 3,15-dioxo-1-(4-(4-(pyrimidin- 2-ylmethyl)piperidine-1- carbonyl)phenyl)-6,9,12-trioxa- 2,16-diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 59 LPC004 [00283]embedded image 3-(2-(2-(2-((2-(2,6- dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4-yl)oxy)acetamido) ethoxy)ethoxy)-N-(2-oxo-2H- chromen-7-yl)propanamide 60 LPC019 [00284] embedded image (2S,4R)-1-((S)-3,3-dimethyl-2- (3-(2-(3-ox0-3-((2-oxo-2H- chromen-7-yl)amino) propoxy)ethoxy)propanamido) butanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2- carboxamide 61 LPC006 [00285]embedded image 3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3- dioxoisoindolin-4- yl)oxy)acetamido) ethoxy)ethoxy)-N-(3,3-diphenylpropyl)propanamide 62 LPC022 [00286]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-4,13-dioxo-17,17-diphenyl-7,10- dioxa-3,14-diazaheptadecanoyl)- 4-hydroxy-N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 63 LPC015 [00287]embedded image N-(2-(1H-indol-3-yl)ethyl)-3-(2- (2-(2-((2-(2,6-dioxopiperidin-3- yl)-1,3-dioxoisoindolin-4-yl)oxy)acetamido) ethoxy)ethoxy)propanamide 64 LPC024 [00288]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-16- (1H-indol-3-yl)-4,13-dioxo- 7,10-dioxa-3,14- diazahexadecanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 65 LPC033 [00289] embedded image (2S,4R)-1-((S)-2-(tert-butyl)-16- (3,4-dimethoxyphenyl)-4,13- dioxo-7,10-dioxa-3,14- diazahexadecanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2-carboxamide 66 LPC034 [00290]embedded image (2S,4R)-1-((S)-2-(tert-butyl)- 4,13-dioxo-16-(2-oxo-2,3,4,5- tetrahydro-1H-benzo[b]azepin- 8-yl)-7,10-dioxa-3,14- diazahexadecanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 67 LPC036 [00291] embedded image (2S,4R)-1-((S)-1-(1H- benzo[d]imidazol-2-yl)-14-(tert- butyl)-3,12-dioxo-6,9-

dioxa- 2,13-diazapentadecan-15-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2-carboxamide 68 LPC042 [00292]embedded image (2S,4R)-1-((S)-2-(3-(2-(3- (((3s,5s,7s)-adamantan-1- yl)amino)-3-oxopropoxy) ethoxy)propanamido)-3,3- dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 69 LPC039 [00293] embedded image (2S,4R)-1-((S)-3,3-dimethyl-2- (3-(2-(3-oxo-3-(((R)-1,2,3,4-tetrahydronaphthalen-1- yl)amino)propoxy)ethoxy) propanamido) butanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 70 LPC049 [00294]embedded image (2S,4R)-1-((S)-3,3-dimethyl-2- (3-(2-(3-oxo-3-(((S)-1,2,3,4- tetrahydronaphthalen-1-yl)amino)propoxy)ethoxy) propanamido) butanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2- carboxamide 71 CMC-6- 175 [00295]embedded image (2S,4R)-1-((S)-16-(tert-butyl)- 1,14-dioxo-1-((1S,4S)-4,7,7- trimethyl-3-oxo-2- oxabicyclo[2.2.1]heptan-1-yl)- 5,8,11-trioxa-2,15- diazaheptadecan-17-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 72 CMC-6- 182 [00296]embedded image (2S,4R)-1-((S)-17-(tert-butyl)- 3,15-dioxo-1-(3-((2- ((1R,4aS,8aS)-2,5,5,8a- tetramethyl-1,4,4a,5,6,7,8,8a-octahydronaphthalen-1- yl)acetamido)methyl)phenyl)- 6,9,12-trioxa-2,16- diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 73 CMC-6- 177 [00297]embedded image (2S,4R)-1-((S)-16-(tert-butyl)- 1,14-dioxo-1-((1S,4S)-4,7,7- trimethyl-3-oxo-2- oxabicyclo[2.2.1]heptan-1-yl)- 5,8,11-trioxa-2,15- diazaheptadecan-17-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 74 CMC-6- 181 [00298] embedded image (2S,4R)-1-((S)-18-(tert-butyl)- 3,16-dioxo-1-(3-(((1S,4S)-4,7,7- trimethyl-3-oxo-2- oxabicyclo[2.2.1]heptane-1- carboxamido)methyl)phenyl)- 7,10,13-trioxa-2,17-diazanonadecan-19-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2-carboxamide 75 IF-057 [00299]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-16- (4-((R)-2,3-dihydrobenzo[b][1,4]dioxine-2- carbonyl)piperazin-1-yl)-4,16- dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 76 IF-059 [00300]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-16- (4-(naphthalen-2-ylmethyl)piperazin-1-yl)-4,16- dioxo-7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 77 IF-060 [00301]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-16- (4-((4′-chloro-5,5-dimethyl- 3,4,5,6-tetrahydro-[1,1′- biphenyl]-2-yl)methyl)piperazin-1-yl)-4,16- dioxo-7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 78 IF-062 [00302]embedded image (2S,4R)-1-((S)-17-(tert-butyl)-2- cyclopentyl-1-(9-ethyl-9H- carbazol-2-yl)-3,15-dioxo- 6,9,12-trioxa-2,16- diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2-carboxamide 79 IF-063 [00303]embedded image (2S,4R)-1-((S)-19-(1-benzyl-1H- indol-3-yl)-2-(tert-butyl)-4,16- dioxo-7,10,13-trioxa-3,17- diazanonadecanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5- yl)benzyl)pyrrolidine-2- carboxamide 80 IF-064 [00304]embedded image (2S,4R)-1-((2S)-16-(2-(5-(4- bromobenzyl)-1,2,4-oxadiazol- 3-yl)pyrrolidin-1-yl)-2-(tert-butyl)-4,16-dioxo-7,10,13- trioxa-3-azahexadecanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide 81 IF-065 [00305]embedded image (2S,4R)-1-((2S)-2-(tert-butyl)- 4,16-dioxo-16-((2-oxo-5- phenyl-2,5-dihydro-1H- benzo[e][1,4]diazepin-3-yl)amino)-7,10,13-trioxa-3- azahexadecanoyl)-4-hydroxy-N- (4-(4-methylthiazol-5-yl)benzyl) pyrrolidine-2-carboxamide 82 IF-066 [00306]embedded image (2S,4R)-1-((S)-2-(tert-butyl)-4,16-dioxo-16-(2′-oxo-2′,3′- dihydro-1′H-spiro[piperidine- 4,4′-quinazolin]-1-yl)-7,10,13- trioxa-3-azahexadecanoyl)-4- hydroxy-N-(4-(4-methylthiazol- 5-yl)benzyl)pyrrolidine-2- carboxamide

[0141] The present invention is directed to designed bifunctional small molecules that modulate protein modifications via proximity-induced effects, such as ubiquitination-inducing small molecules. These represent a transformative therapeutic strategy. The strategy formulates FragTAC heterobifunctional compositions which integrate powerful chemical proteomic platforms with robust chemical biology tools to expedite the discovery of proteins amendable to these approaches as well as their corresponding bifunctional chemical probes. The techniques and chemical tools of the present invention provide a broad utility of powerful targeted protein degradation (TPD) methods. The new new E3 ligase systems and corresponding ligands that are capable of supporting TPD enable modification and treatment of such maladies as autoimmune disease and neoplastic disease both benign and malignant, and especially malignant such as cancer.

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SUMMARY STATEMENTS

[0158] The inventions, examples, biological assays and results described and claimed herein have may attributes and embodiments include, but not limited to, those set forth or described or referenced in this application.

[0159] All patents, publications, scientific articles, web sites and other documents and material references or mentioned herein are indicative of the levels of skill of those skilled in the art to which the invention pertains, and each such referenced document and material is hereby incorporated by reference to the same extent as if it had been incorporated verbatim and set forth in its entirety herein. The right is reserved to physically incorporate into this specification any and all materials and information from any such patent, publication, scientific article, web site, electronically available information, textbook or other referenced material or document.

[0160] The written description of this patent application includes all claims. All claims including all original claims are hereby incorporated by reference in their entirety into the written description portion of the specification and the right is reserved to physically incorporated into the written description or any other portion of the application any and all such claims. Thus, for example, under no circumstances may the patent be interpreted as allegedly not providing a written description for a claim on the assertion that the precise wording of the claim is not set forth in haec verba in written description portion of the patent.
[0161] While the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Thus, from the foregoing, it will be appreciated that, although specific nonlimiting embodiments of the invention have been described herein for the purpose of illustration, various modifications may be made without deviating from the spirit and scope of the invention. Other aspects, advantages, and modifications are within the scope of the following claims and the present invention is not limited except as by the appended claims.
[0162] The specific methods and compositions described herein are representative of preferred nonlimiting embodiments and are exemplary and not intended as limitations on the scope of the invention. Other objects, aspects, and embodiments will occur to those skilled in the art upon consideration of this specification and are encompassed within the spirit of the invention as defined by the scope of the claims. It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably may be practiced in the absence of any element or elements, or limitation or limitations, which is not specifically disclosed herein as essential. Thus, for example, in each instance herein, in nonlimiting embodiments or examples of the present invention, the terms “comprising”, “including”, “containing”, etc. are to be read expansively and without limitation. The methods and processes illustratively described herein suitably may be practiced in differing orders of steps, and that they are not necessarily restricted to the orders of steps indicated herein or in the claims.
[0163] The terms and expressions that have been employed are used as terms of description and not of limitation, and there is no intent in the use of such terms and expressions to exclude any equivalent of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention as claimed. Thus, it will be understood that although the present invention has been specifically disclosed by various nonlimiting embodiments and/or preferred nonlimiting embodiments and optional features, any and all modifications and variations of the concepts herein disclosed that may be resorted to by those skilled in the art are considered to be within the scope of this invention as defined by the appended claims.
[0164] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skill in the art.
[0165] The term “about” as used herein, when referring to a numerical value or range, allows for a degree of variability in the value or range, for example, within 10%, or within 5% of a stated value or of a stated limit of a range.
[0166] All percent compositions are given as weight-percentages, unless otherwise stated.
[0167] All average molecular weights of polymers are weight-average molecular weights, unless otherwise specified.
[0168] The term “may” in the context of this application means “is permitted to” or “is able to” and is a synonym for the term “can.” The term “may” as used herein does not mean possibility or chance.
[0169] It is also to be understood that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise, for example, the term “X and/or Y” means “X” or “Y” or both “X” and “Y”, and the letter “s”

following a noun designates both the plural and singular forms of that noun. In addition, where features or aspects of the invention are described in terms of Markush groups, it is intended, and those skilled in the art will recognize, that the invention embraces and is also thereby described in terms of any individual member and any subgroup of members of the Markush group, and the right is reserved to revise the application or claims to refer specifically to any individual member or any subgroup of members of the Markush group.

[0170] The term and/or means both as well as one or the other as in A and/or B means A alone, B alone and A and B together.

Claims

1. A heterobifunctional FragTAC composition comprising a compound of Formula I,
PBF-L-RBF (Formula I) wherein PBF is a protein binding fragment of a hetero-organic molecule which is capable of binding an endogenous protein, L is an organic linker and RBF is a recruiter binding fragment of a hetero-organic molecule which is capable of binding a ligase of the polyubiquitin system.

2. A FragTAC composition according to claim 1, wherein the protein binding fragment is selected from a group consisting of ##STR00307##

3. A FragTAC composition according to claim 1, where in the recruiter binding fragment is a fragment of thalidomide, a thalidomide derivative or a VHL ligand.

4. A FragTAC composition according to claim 1, wherein Formula I is a cereblon ligase binding molecule or a VHL ligase binding molecule of the Cereblon FragTAC structure or the VHL FragTAC structure wherein n is an integer of 1 to 10: ##STR00308##

5. A composition according to claim 1, wherein the linker is a dimer, trimer, tetramer, or oligomer of an multiether, multi-PEG, multi-amide, or alkyl moiety.

6. A method for cleaving an endogenous protein comprising contacting the protein with a heterobifunctional composition of claim 1 and an E3 ligase.

7. A method according to claim 6, comprising conducting the contacting step in an aqueous medium.

8. A method according to claim 7, wherein the aqueous medium is cytoplasm.

9. A method according to claim 8, wherein the contacting is conducted in a viable cell.

10. A method according to claim 9, wherein the viable cell is a cell culture.

11. A method according to claim 9, wherein the viable cell is within a living organism.